SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

FACULTY OF ENGINEERING AND TECHNOLOGY

SCHOOL OF COMPUTING



SRM Institute of Science and Technology
School of Computing



LAB MANUAL

21CSC302J COMPUTER NETWORKS

JULY – NOVEMBER 2025

Revision History:

Date	Version	Modification Done	Prepared By	Reviewed by	Signature
	1.0	Initial Release	Dr. Nadana Ravishankar T	Dr. D. Malathi	

Course Code	21CSC302J	Course Name	COMPUTER NETWORKS	Course Category	C	PROFESSIONAL CORE	L	T	P	C
							3	0	2	4

Pre-requisite Courses	Nil	Co- requisite Courses	Nil	Progressive Courses	Nil
Course Offering Department	School of Computing	Data Book / Codes / Standards	Nil		

Course Learning Rationale (CLR):	The purpose of learning this course is to:
CLR-1:	define the layered network architecture
CLR-2:	produce knowledge in IP addressing
CLR-3:	identify suitable routing algorithms based on geographical location of the devices
CLR-4:	apply the concept of Error detection to identify the errors in data
CLR-5:	exploring reliable and unreliable protocols

Course Outcomes (CO):	At the end of this course, learners will be able to:	Program Outcomes (PO)												Program Specific Outcomes		
		1	2	3	4	5	6	7	8	9	10	11	12			
		Engineering Knowledge	Problem Analysis	Design/development of solutions	Conduct investigations of complex problems	Modern Tool Usage	The engineer and society	Environment & Sustainability	Ethics	Individual & Team Work	Communication	Project Mgt. & Finance	Life Long Learning	PSO-1	PSO-2	PSO-3
CO-1:	apply the knowledge of communication	3	-	-	-	3	-	-	-	-	-	-	-	1	-	-
CO-2:	construct the network using addressing schemes	3	-	-	2	-	-	-	-	-	-	-	-	1	-	-
CO-3:	design and implement the various Routing Protocols	3	-	-	2	3	-	-	-	-	-	-	-	1	-	-
CO-4:	identify and correct the errors in transmission	3	-	-	-	-	-	-	-	-	-	-	-	1	-	-
CO-5:	analyze the services provided by Transport and Application layers	3	-	-	-	-	-	-	-	-	-	-	-	1	-	-

Unit-1 - Introduction to Networks — **15 Hour**

Network Types: LAN, MAN, PAN, WAN - Network Topology : BUS, STAR, RING, MESH, HYBRID - Switching : Circuit Switching, Packet Switching - OSI Layered Architecture - TCP/IP Model - Physical Layer Overview - Latency, Bandwidth, Delay - Guided Media : Twisted pair, Coaxial cable, Fiber optic cable - Unguided Media : Radio waves, Microwaves, Infrared.
Lab 1: Introduction to Packet Tracer, Peer to Peer communication, study of cables and its colour codes
Lab 2: Implementation of Network Topologies
Lab 3: Router Configuration (Creating Passwords, Configuring Interfaces)

Unit-2 - Addressing — **15 Hour**

IPV4 Addressing - Address space - Classful addressing - Subnet mask - FLSM - Classless Addressing - VLSM – NAT – Super netting - Network Devices: Hub, Repeaters, Switch, Bridge, Router
Lab 4: IP addressing and Sub netting (VLSM)
Lab 5: Static and Default Routing
Lab 6: NAT Configuration

Unit-3 - Routing — **15 Hour**

Forwarding of IP Packets — Static and Default Routing — Unicast Routing Algorithms: Distance Vector Routing, Link State Routing, Path Vector Routing — Protocols: RIP V1, RIP V2, OSPF, BGP, EIGRP — Multicasting Basics — IPV6 Addressing Basics
Lab 7: Implementation of RIP version 1
Lab 8: Implementation of RIP version 2
Lab 9: Implementation of Single Area OSPF

Unit-4 – Medium Access Control — **15 Hour**

ALOHA ,CSMA/CD, CSMA/CA, Ethernet, Token Ring - Flow Control :Stop and Wait, Sliding Window - Error Control: Stop and Wait ARQ, Sliding Window ARQ - Error Detection : Parity Check, Checksum, CRC - Error Correction: Hamming codes - Data-Link Layer Protocols : HDLC, PPP.
Lab 10: Implementation of Multi Area OSPF
Lab 11: PPP Configuration
Lab 12: HDLC Configuration

Unit-5 – Transport and Application Layer Protocols — **15 Hour**

Port Numbers — User Datagram Protocol — Transmission Control Protocol — WWW and HTTP — FTP — Email –Telnet – DNS.
Lab 13: Implementation of BGP
Lab 14: Implementation of EIGRP
Lab 15: Telnet Configuration

Learning Resources

1. Behrouz A. Forouzan, "Data Communication and Networking",5th ed.,2010
2. Bhushan Trivedi, " Data Communication and Networks" 2016
3. William Stallings, " Data and Computer Communications",9th ed.,2010
4. Todd Lammle, "CCNA Study Guide",7th ed.,2011

21CSC302J – COMPUTER NETWORKS

LIST OF EXPERIMENTS

Sl. No	**Date of Experiment**	**Title of the Experiment**	Executed Date & Marks **(6)**	Viva Marks **(4)**	Total Marks (10)	Faculty Sign with Date
1		**Exp-1:** 1a-Introduction to Packet tracer 1b- Networking Commands 1c-Study of Cables				
2		**Exp-2:** 2a-Demonstration of P2P 2b-Demonstration of simple LAN 2c-Implementation of Topologies				
3		**Exp-3:** Router configuration through a console				
4		**Exp-4:** 4a-Configuration of IP address in a Router 4b-Subnetting in WAN				
5		**Exp-5:** Demonstration of static & default routing				
6		**Exp-6:** Configuration of NAT				
7		**Exp-7:** Configuration of RIP v1 and RIP v2				
8		**Exp-8:** Configuration of 8a – Single area OSPF 8b – Multi-area OSPF				
9		**Exp-9:** Configuration of 9a – Point-to-point Protocol (PPP) 9b – High Level Data Link Control				
10		**Exp-10:** Configuration of Border Gateway Protocol (BGP)				
11		**Exp-11:** Configuration of Enhanced Interior Gateway Routing Protocol (EIGRP)				
12		**Exp-12:** Configuration of TELNET & SSH protocols.				

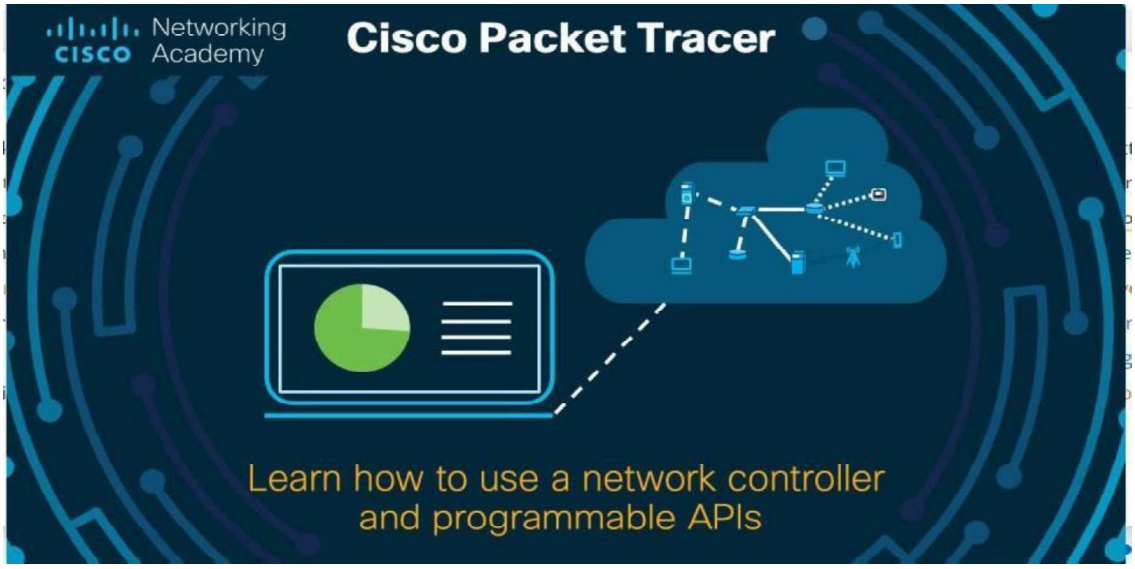
EXERCISE 1

AIM:

To understand the basics of Packet Tracer, Networking commands and Study of different types of cables

1.1 INTRODUCTION TO PACKET TRACER

Cisco Packet Tracer is a free application that enables you to practice network configuration and troubleshooting on your desktop or laptop computer. It enables you to mimic networks without having physical access to the underlying hardware. Along with networking, you may improve your Internet of Things (IoT) and cybersecurity skills through education and practice. You have the option of creating a network from scratch, using a pre-built sample network, or completing lab projects. While Packet Tracer is not a substitute for practising on physical routers, switches, firewalls, and servers, it does offer a number of advantages.

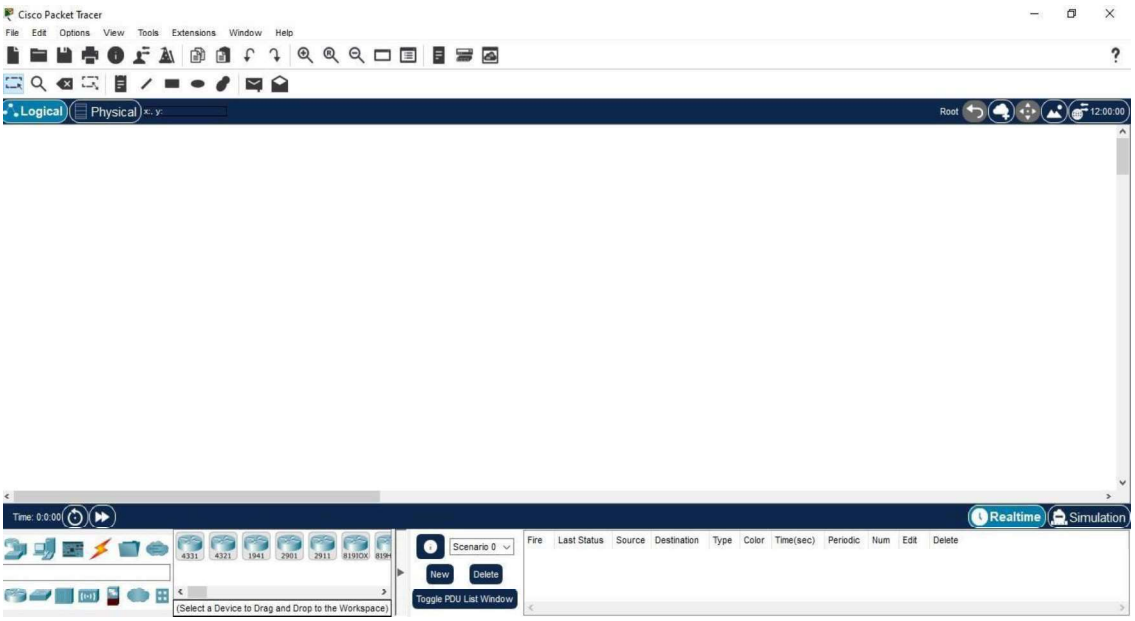


1.a What are the Benefits of Using Packet Tracer?

Imagine being able to peer inside a small business network or the internet. Have you ever wished to create an Internet of Things system that would notify you through the phone if there was an issue in your home environment? Welcome to Cisco Packet Tracer, the simulation environment that may assist you in doing all of these tasks and more. It is intended to familiarize you with the Cisco Packet Tracer network simulation and visualization tool.

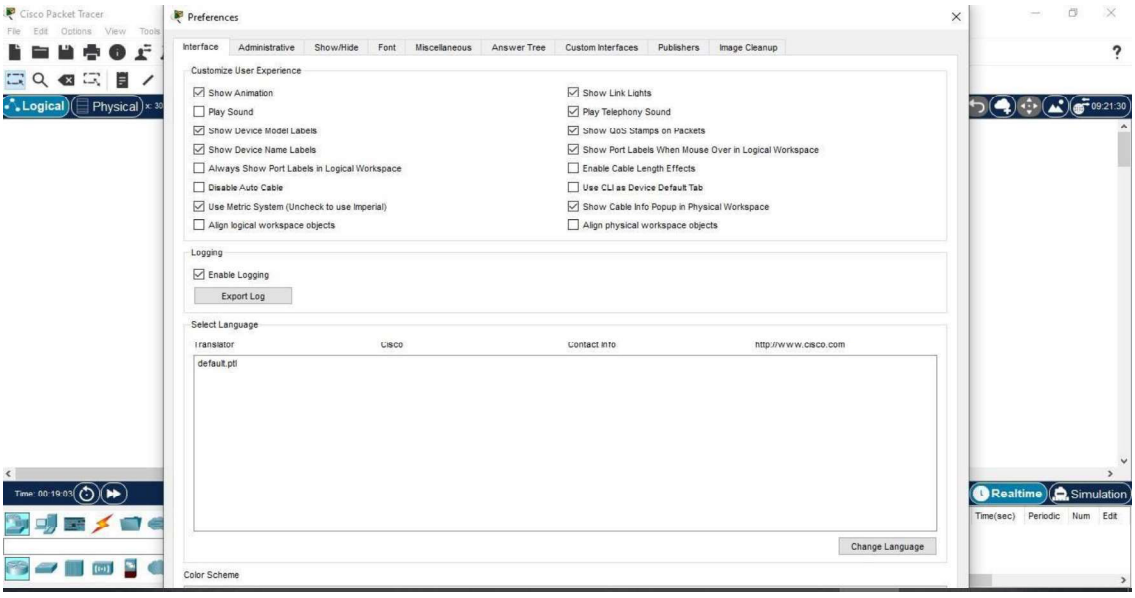
In Packet Tracer, you will design your own network (PT). Additionally, you will learn about the many sorts of PT files.

1.b Packet Tracer UI:



Packet Tracer is a tool that allows you to simulate real networks. It provides three main menus that you can use for the following:

- Add devices and connect them via cables or wireless.
- Select, delete, inspect, label, and group components within your network.
- Manage your network.



The network management menu lets you do the following:

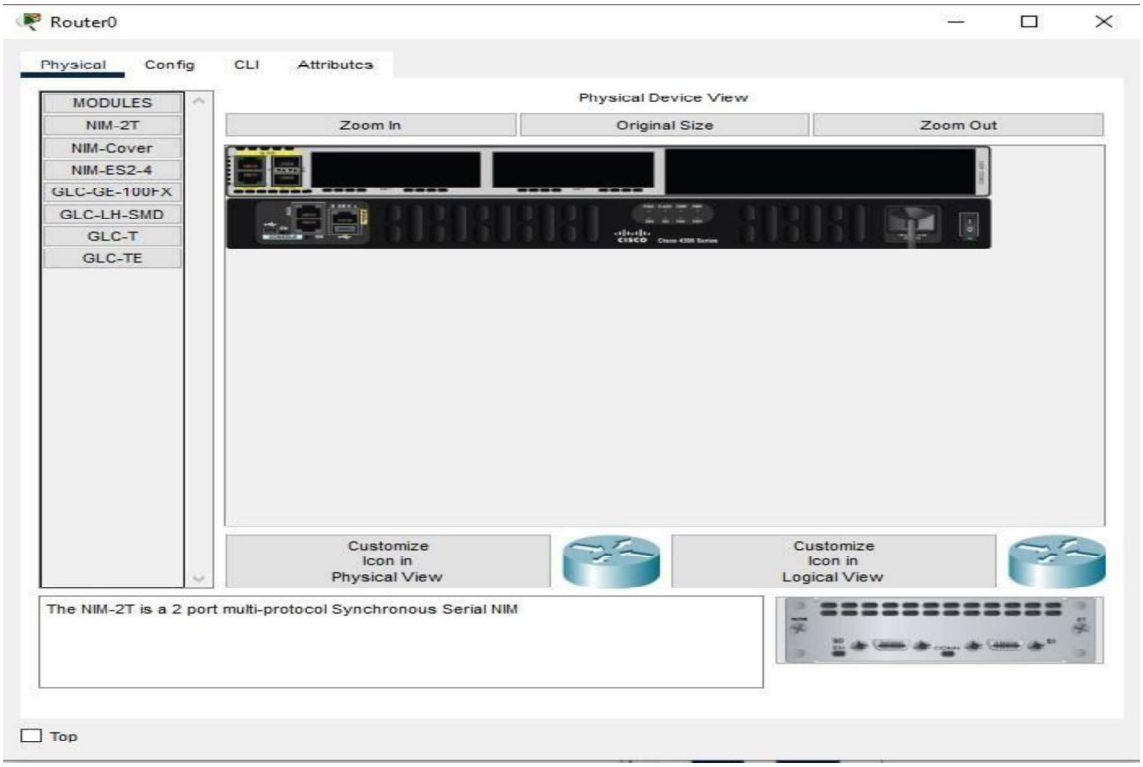
- Open an existing/sample network.
- Save your current network.
- Modify your user profile or your preferences.

Packet Tracer also provides a variety of tabs for device configuration including the following:

- Physical
- Config
- CLI
- Desktop
- Services

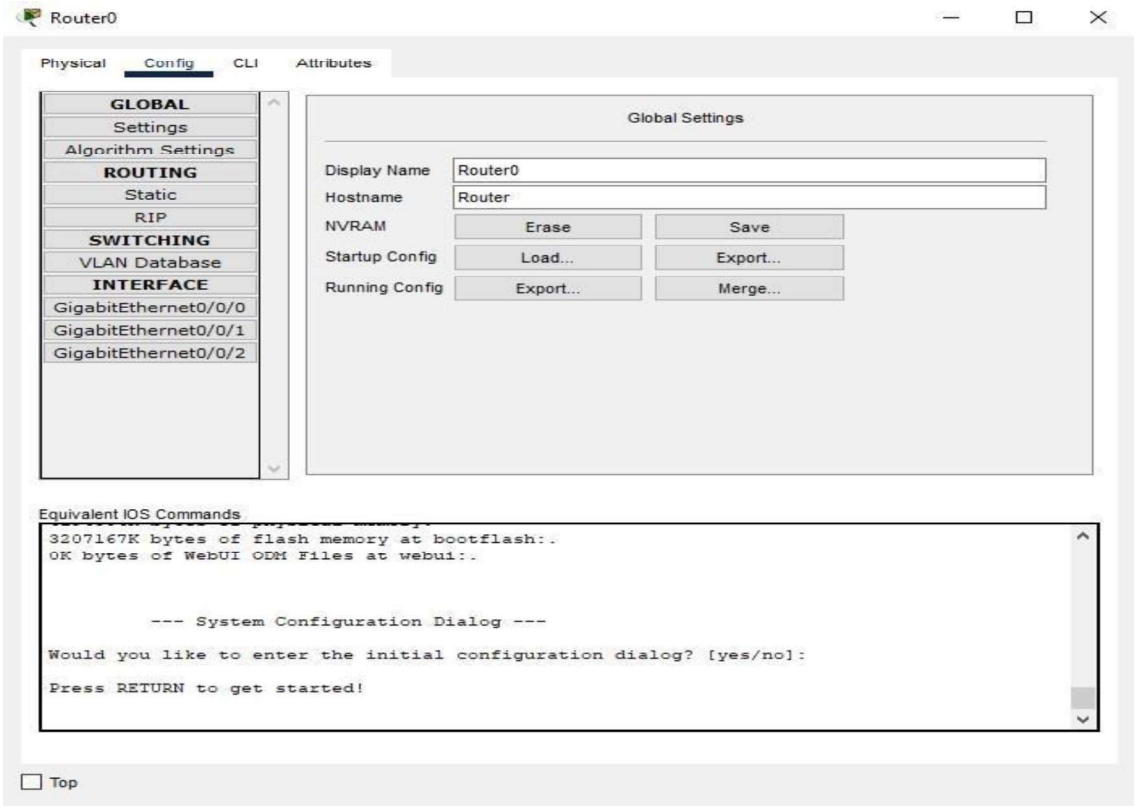
The tabs that are shown depend on the device you are currently configuring.

Physical Tab



The Physical tab provides an interface for interacting with the device including powering it on or off or installing different modules, such as a wireless network interface card (NIC).

Config Tab

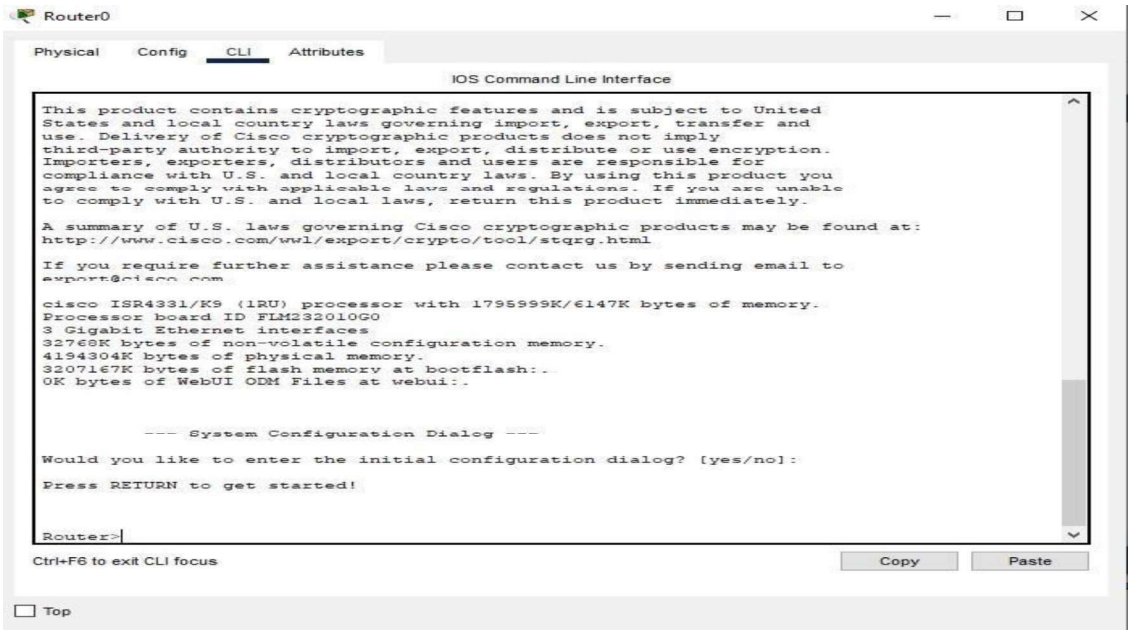


For intermediate devices such as routers and switches, there are two ways to access device configurations. Configurations can be accessed via a Config tab, which is a Graphical User Interface (GUI). Configurations can also be accessed using a command line interface (CLI).

The Config tab does not simulate the functionality of a device. This tab is unique to Packet Tracer. If you don't know how to use the command line interface, this tab provides a way to use a Packet Tracer-only GUI to configure basic settings. As settings are changed in the GUI, the equivalent CLI commands appear in the Equivalent IOS Commands window. This helps you to learn the CLI commands and the Cisco Internetwork Operating System (IOS) while you are using the Config tab.

For example, in the figure, the user has configured MyRouter as the name of the device. The Equivalent IOS Commands window shows the IOS command that achieves the same results in the CLI. In addition, device configuration files can be saved, loaded, erased, and exported here.

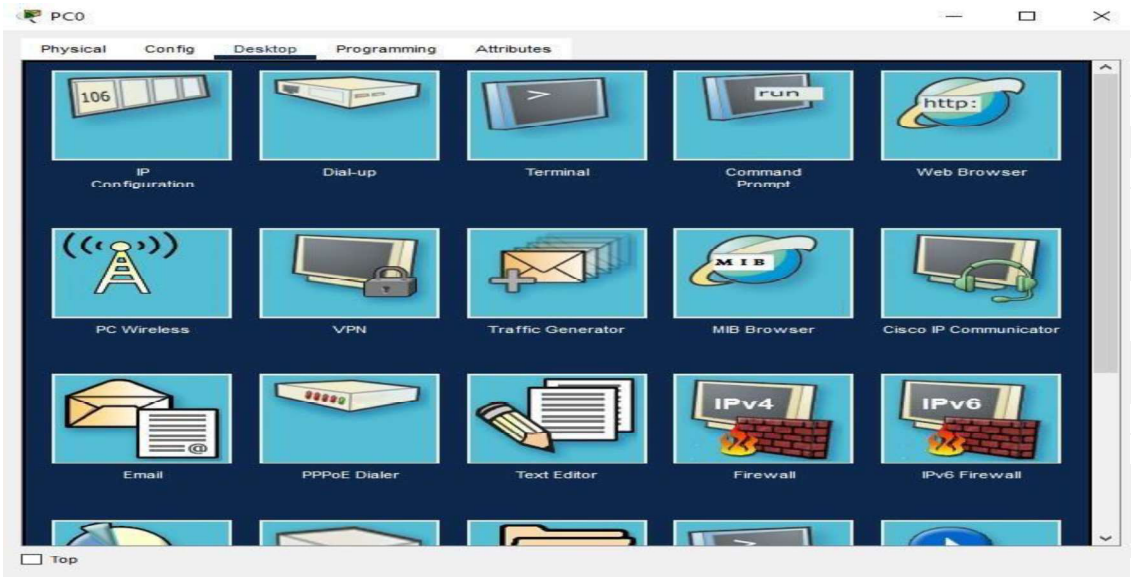
CLI Tab



The CLI tab provides access to the command line interface of a Cisco device. Using the CLI tab requires knowledge of device configuration with IOS. Here, you can practice configuring Cisco devices at the command line. CLI configuration is a necessary skill for more advanced networking implementations.

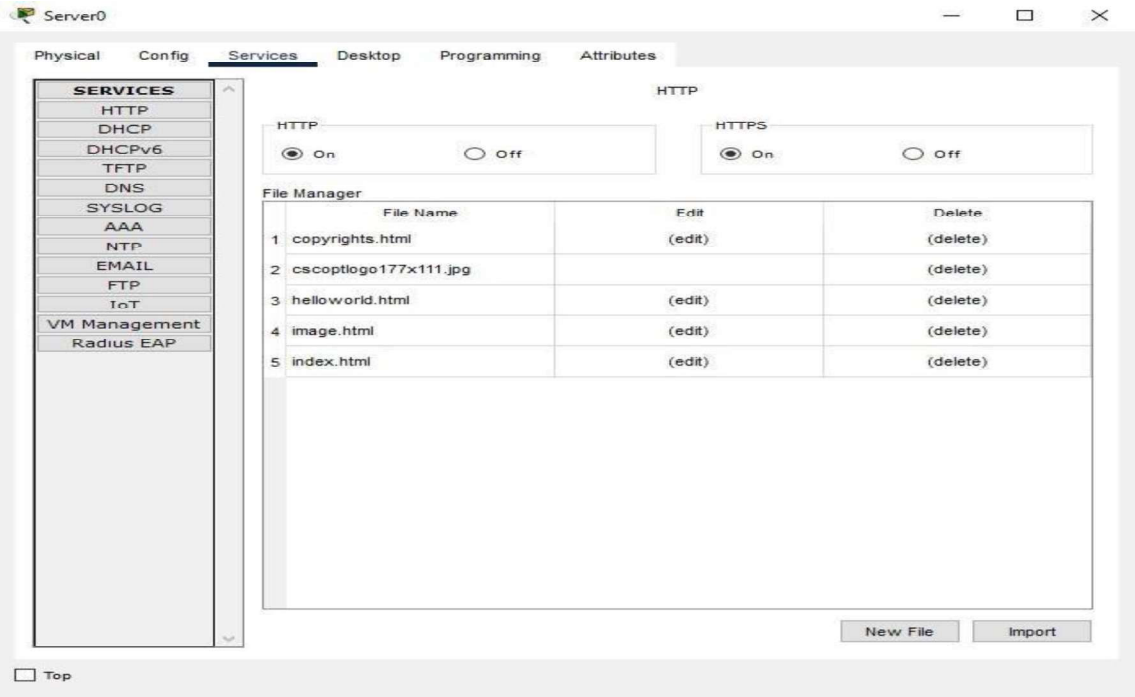
Note: Any commands that were entered from the Config tab are also shown in the CLI tab.

Desktop Tab



For some end devices, such as PCs and laptops, Packet Tracer provides a desktop interface that gives you access to IP configuration, wireless configuration, a command prompt, a web browser, and other applications.

Services Tab



A server has all of the functions of a host with the addition of one more tab, the Services tab. This tab allows a server to be configured with common server processes such as HTTP, DHCP, DNS, or other services, as shown in the figure.

1.b) NETWORKING COMMANDS

Ping (Unix/Windows)

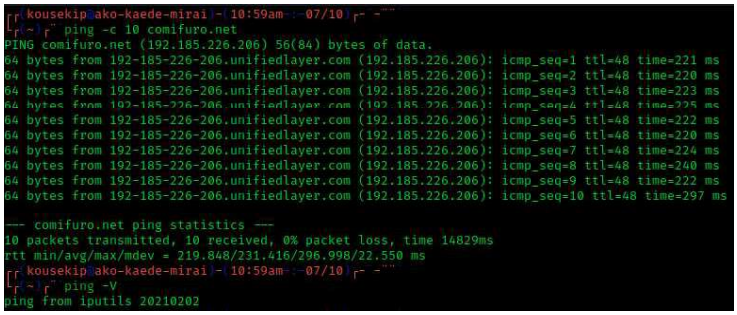
Ping dates from the 70s and is known for being one of the most basic network commands. However, it is not as simple as we believe and has many more uses than those we already know. It is based on the ICMP protocol and is used to determine:

- If there is connectivity between your machine and another machine on the network.
- It's used to measure the "speed" or latency time.

It is a command that exists on all operating systems that support TCP/IP, and it is a basic command that you should know.

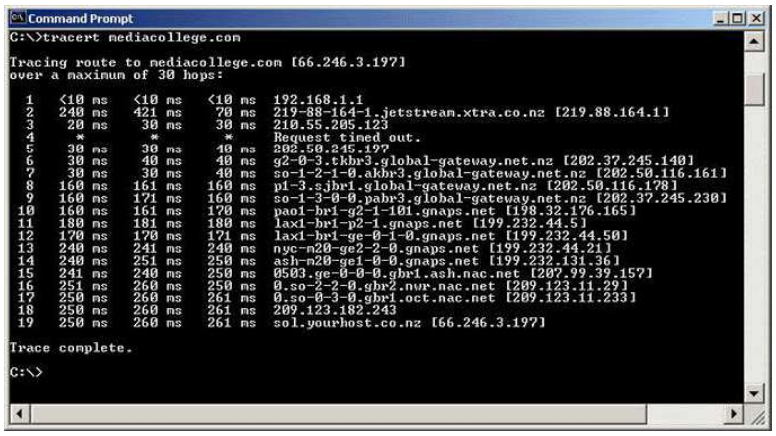
Ping is known for having dozens of parameters and the one that we find more useful is the one responsible for monitoring "the number of packages to send." There are networks that undo the first package, so it is essential to send at least three so we can check that at least one has arrived without being discarded. For this, we use the -c parameter.

The same technique can be used to determine the loss percentage of packages in our network, sending ten packages and seeing if any gets lost. The number of packages that usually get lost in the network will surprise you. (This tool is included in Pandora FMS)
Execution: Ping name/System IP



Traceroute (Unix/Windows)

The main objective of this tool is to know the traveling path of a package through our network. This network command will tell us where the package is going through (machines, switches, routers) and check that our network is working properly. If you encounter any problems, it will allow us to have a rough idea about where the fault lies.



Execution:

traceroute –n (on Unix / Linux)

tracert –d (on Windows)

Arp (Unix/Windows)

This network command is used to change and view the ARP table, which contains the mappings between the IP address and the MAC address. It only sees the connections in our local area network segment (LAN), so it could be called “low level”. However, it’s used to discover what machines are directly connected to our host or what machines we are connected to. It is a diagnostic tool, and sometimes it can be interesting to monitor it in order to discard ARP Poisoning attacks, which are one of the most common forms of phishing attacks in local networks.

Execution: arp –a

Curl and wget (Unix/ Windows)

These are essential commands to do HTTP, HTTPS or FTP requests to remote servers. It allows you to download files or whole web pages, even recursively (it literally allows us to make a “copy” of a website, including images). It supports cookies and allows you to send POST requests, in addition to “simulate a” user agent, use a http proxy or even a SOCKS4/5 proxy.

One of the most common utilities in integration with Pandora FMS, is to verify the contents of a specific web page. Because wget / curl allows us to download the entire contents of a web, it is easy to compare the MD5 of that content with a value previously verified. If it changes, it means that the Web has been altered.

Netstat (Unix/Windows)

Network command identifies all TCP connections and UDP open on a machine. Besides this, it allows us to know the following information:

- Routing tables to meet our network interfaces and its outputs.
- Ethernet statistics that show sent and received packages and possible errors.
- To know the id of the process that is being used by the connection.
- Netstat is another basic command as Ping that meets many elementary functions.

Whois (Unix/ Windows)

This network command is used to query data domains: to find out who owns the domain, when that domain expires, to view the configured logs, contact details, etc. Its use is highly recommended to contact the administrators of the domains or when incidents of migration of services such as mail and web happen.

To use 'whois' on Windows you need to download the software from this url: https://technet.microsoft.com/en-us/sysinternals/whois.aspx

VNStat

It is one of the most complete network commands. It works on all Linux and BSD systems, and allows us to monitor network traffic from the console.

- Installation is simple and fairly quick, allowing monitoring of all network interfaces.
- With VNStat we can collect all traffic needed from any configured interface.
- One of the big differences between VNStat and other tools is that VNStat collects kernel data instead of the interface itself, which means a lighter execution for the system.
- It will not require administrator permissions to run.
- It has the ability to store gathered information so your information never goes missing, even if the system crashes or reboots itself.
- You can set Vnstat to listen to traffic, daily or by billing period, as well as many other options.
- It stands out for its flexibility when configuring the reading of traffic.

Finally, it is possible to set Vnstat output to generate console graphics and even customize them with colours.

SSH (Unix/Linux/Windows)

Command to run terminals on remote machines safely. SSH allows any user to run a console just by registering and entering his credentials. So you can run the commands you want as if you were in local.

More details you need to know about SSH:

Putty is recommended when using SSH in Windows. You can find it here: http://www.putty.org/

TCPDump (Unix/Linux/Windows)

It is one of the "basic" tools of network commands, and when used right, goes on to become a great ally for network administrators, system administrators or programmers.

TCPDump is an advanced command used to inspect traffic from different interfaces of a machine so you can get the exchanged packages. You can dump output to file so then you can analyse it with more powerful sniffers and graphical interfaces such as Wireshark. For Windows, you must use WinDump.

Ngrep (Unix/Linux/Windows)

- The grep command power is taken to the network.
- It is a TCPDump with a substring text filter in real time.
- It has a very powerful filtering system for regular expressions and it is typically used to process files generated by tcpdump, wireshark, etc.
- It is a communication package filter over HTTP, SMTP, FTP, DNS and other protocols.

NMAP (Unix/Windows)

NMAP is considered the father of the general network scanners. Although today there are more reliable tools for some tasks (like Fping), NMAP is a very versatile tool for scanning networks. It is used to determine which hosts are alive in a network and to do different ways of scanning.

Netcat (Windows/Unix)

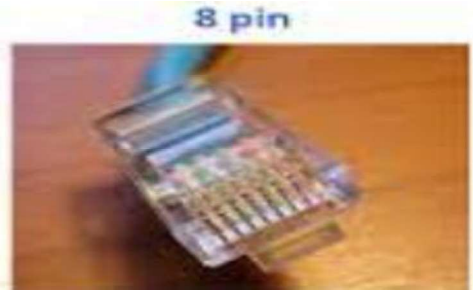
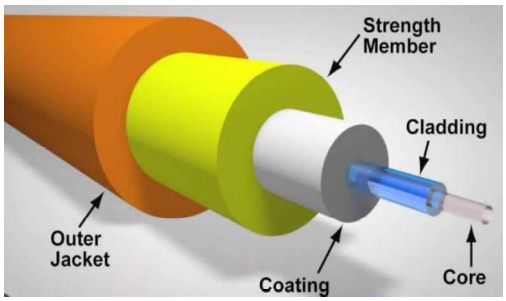
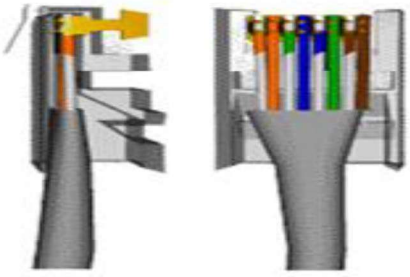
NetCat, or NC, is the network command most versatile that exists nowadays and one of the lightest. However, its use requires some imagination. Only if you've played with scripting, you will understand the subtlety of its name: NetCat. It is a tool designed to be used as a destination of a redirect (one pipe or |). It is used to send or receive information about a connection. For example, a WEB request to service would be something as simple as:

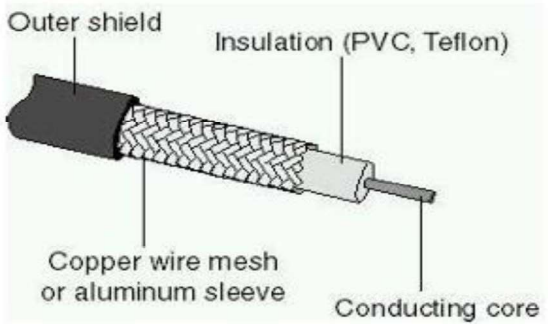
echo -e "GET http://pandorafms.com HTTP/1.0\n\n" | nc pandorafms.com 80

IPtraf (Linux)

Special command to obtain traffic statistics. It has a ncurses interface (text) to analyze real-time traffic passing through an interface. It allows you to work at low-level and to see what pairs of connections are established on each machine, and to see in detail the traffic connection of every pair, all in real-time.

1.C – Study of Cables & it's Color codes

Term	Description	Cross Sectional Diagram
BNC	British Naval Connector is the connector used with coaxial cables.	
RJ – 11	A registered jack 11 is a telephone connector used on modern telephone lines.	
RJ – 45	A registered jack 45 is an eight-wire connector used to connect computers to category 5 unshielded twisted pair (UTP) cables in a network.	
Fiber Optic Cable	Fiber optic cable uses light to transmit information across a network. The core of the cable is made of glass, which is protected by a layer of gel or plastic. A plastic cover surrounds the entire cable.	
UTP	Unshielded Twisted-Pair Cable is network cable that consists of up to 4 pairs of wires. Each pair is twisted around each other at a different rate and the entire cable is encased in a protective plastic covering. The twisting of the wires in cables is to help prevent EMI (Electro- Magnetic Interference).	

Coaxial Cable	Thin coaxial cable is often referred to as ThinNet. It consists of a copper wire surrounded first by a layer of plastic, then a layer of metal mesh and a final layer of protective plastic. It is used for peer-to-peer networking	

NETWORK DEVICES:

1. **Repeater:** Functioning at Physical Layer. A**repeater** is an electronic device that receives a signal and retransmits it at a higher level and/or higher power, or onto the other side of an obstruction, so that the signal can cover longer distances. Repeater have two ports, so cannot be use to connect for more than two devices.

2. **Hub:** An **Ethernet hub**, **active hub**, **network hub**, **repeater hub**, **hub** or **concentrator** is a device for connecting multiple twisted pair or fiber optic Ethernet devices together and making them act as a single network segment. Hubs work at the physical layer (layer 1) of the OSI model. The device is a form of multiport repeater. Repeater hubs also participate in collision detection, forwarding a jam signal to all ports if it detects a collision.

3. **Switch:** A**network switch** or **switching hub** is a computer networking device that connects network segments. The term commonly refers to a network bridge that processes and routes data at the data link layer (layer 2) of the OSI model. Switches that additionally process data at the network layer (layer 3 and above) are often referred to as Layer 3 switches or multilayer switches.

4. **Bridge:** A **network bridge** connects multiple network segments at the data link layer (Layer 2) of the OSI model. In Ethernet networks, the term *bridge* formally means a device that behaves according to the IEEE 802.1D standard. A bridge and switch are very much alike; a switch being a bridge with numerous ports. *Switch* or *Layer 2 switch* is often used interchangeably with *bridge* .Bridges can analyze incoming data packets to determine if the bridge is able to send the given packet to another segment of the network.

5. **Router:** A **router** is an electronic device that interconnects two or more computer networks, and selectively interchanges packets of data between them. Each data packet contains address information that a router can use to determine if the source and destination are on the same network, or if the data packet must be transferred from one network to another. Where multiple routers are used in a large collection of interconnected networks, the routers exchange information about target system addresses, so that each router can build up a table showing the preferred paths between any two systems on the interconnected networks.

6. **Gate Way:** In a communications network, a network node equipped for interfacing with another network that uses different protocols.

• A gateway may contain devices such as protocol translators, impedance matching devices, rate converters, fault isolators, or signal translators as necessary to provide system interoperability. It also requires the establishment of mutually acceptable administrative procedures between both networks.

• A protocol translation/mapping gateway interconnects networks with different network protocol technologies by performing the required protocol conversions.

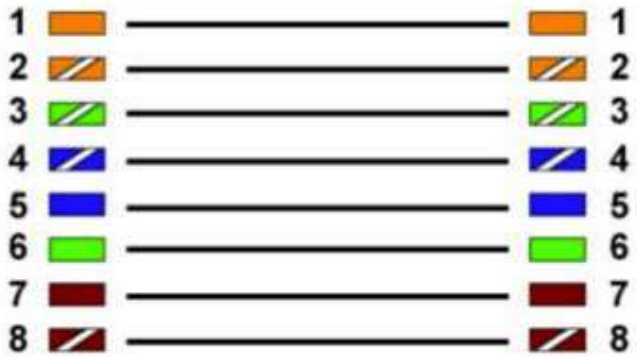
Exercise 2: Cabling – Straight Through and Cross-over Cabling

Ethernet cable:

An Ethernet cable is a network cable used for high-speed wired network connections between two devices. This network cable is made of four-pair cable, which is consists of twisted pair conductors. It is used for data transmission at both ends of the cable, which is called RJ45 connector.

The Ethernet cables are categorized as Cat 5, Cat 5e, Cat 6, and UTP cable. Cat 5 cable can support a 10/100 Mbps Ethernet network while Cat 5e and Cat 6 cable to support Ethernet network running at 10/100/1000 Mbps.

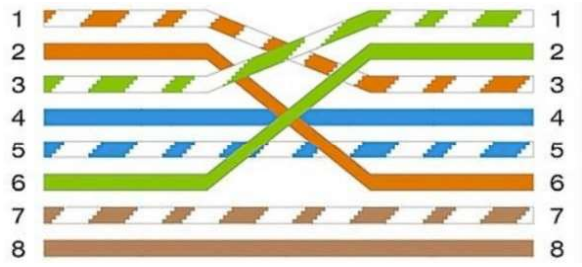
Straight Through Cable:



Straight Through Cable

Straight-through cable is a type of CAT5 with RJ-45 connectors at each end, and each has the same pin out. It is in accordance with either the T568A or T568B standards. It uses the same color code throughout the LAN for consistency. This type of twisted-pair cable is used in LAN to connect a computer or a network hub such as a router. It is one of the most common types of network cable.

Crossover Cable:



Crossover Cable

A Crossover cable is a type of CAT 5 where one end isT568A configuration and the other end as T568BConfiguration. In this type of cable connection, Pin 1 is crossed with Pin 3, and Pin 2 is crossed with Pin 6.

Crossover cable is used to connect two or more computing devices. The internal wiring of crossover cables reverses the transmission and receive signals. It is widely used to connect two devices of the same type: e.g., two computers or two switches to each other.

In regard to physical appearance, Crossover Ethernet cables are very much similar to regular Ethernet cables. Still, they are different with regard to the order with which the wires are arranged. This type of Ethernet cable is made to connect to network devices of the same kind over Ethernet directly. Crossover cables are mostly used to connect two hosts directly.

Devices Connectivity:

DEVICES	HUB	SWITCH	ROUTER	PC
HUB	**CO**	**CO**	**ST**	**ST**
SWITCH	**CO**	**CO**	**ST**	**ST**
ROUTER	**ST**	**ST**	**CO**	**CO**
PC	**ST**	**ST**	**CO**	**CO**

Exercise 2.a Demonstration of P2P Network

Objective: To demonstrate the Copper Cross-over cabling by designing a Peer to Peer Network

Components:

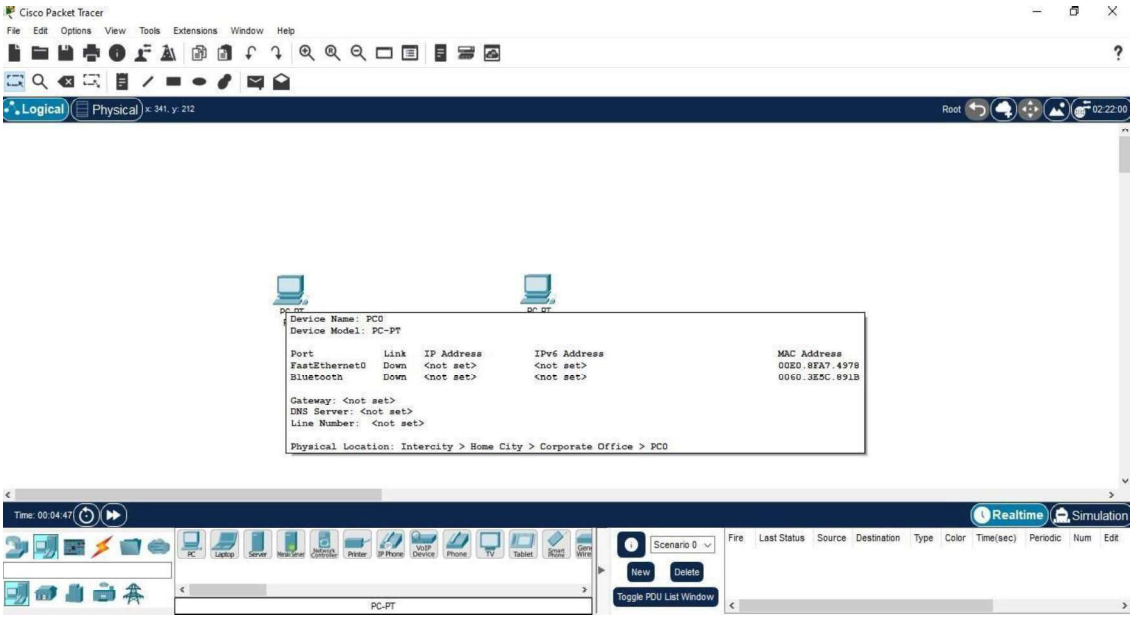
Devices	Required Nos
PCs	2
Copper Cross – Over Cable	1

Addressing Table:

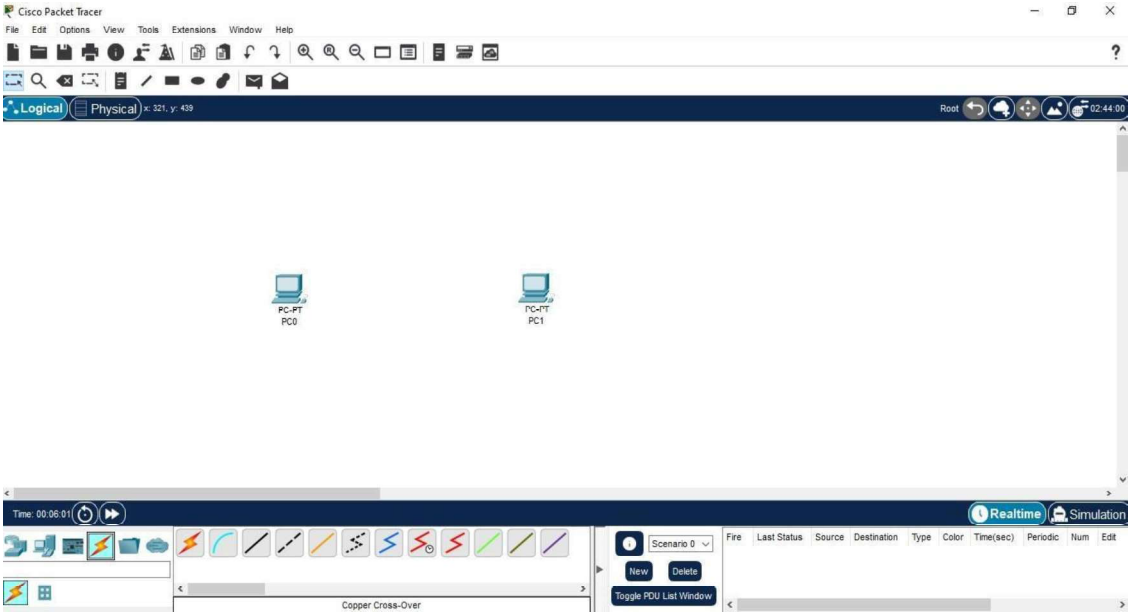
Device	Interface	IP Address	Subnet Mask
PC0	Fa0/0	192.168.10.1	255.255.255.0
PC1	Fa0/0	192.168.10.2	255.255.255.0

Procedure:

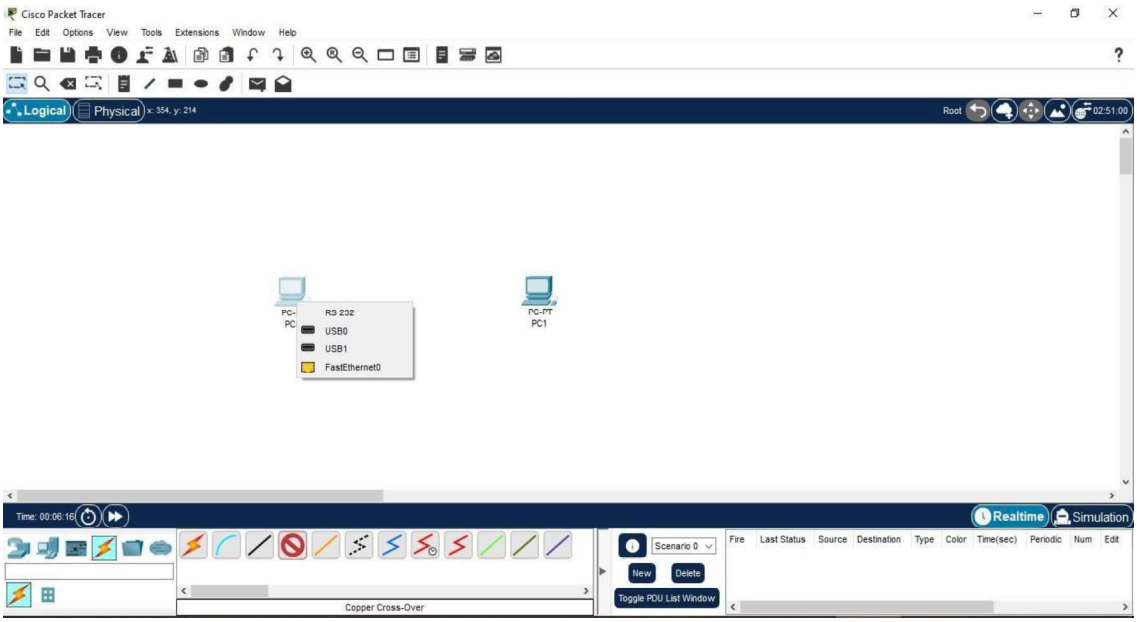
Step 1: Drag 2 PCs in the console area. Each PC will have interfaces as shown in the figure.



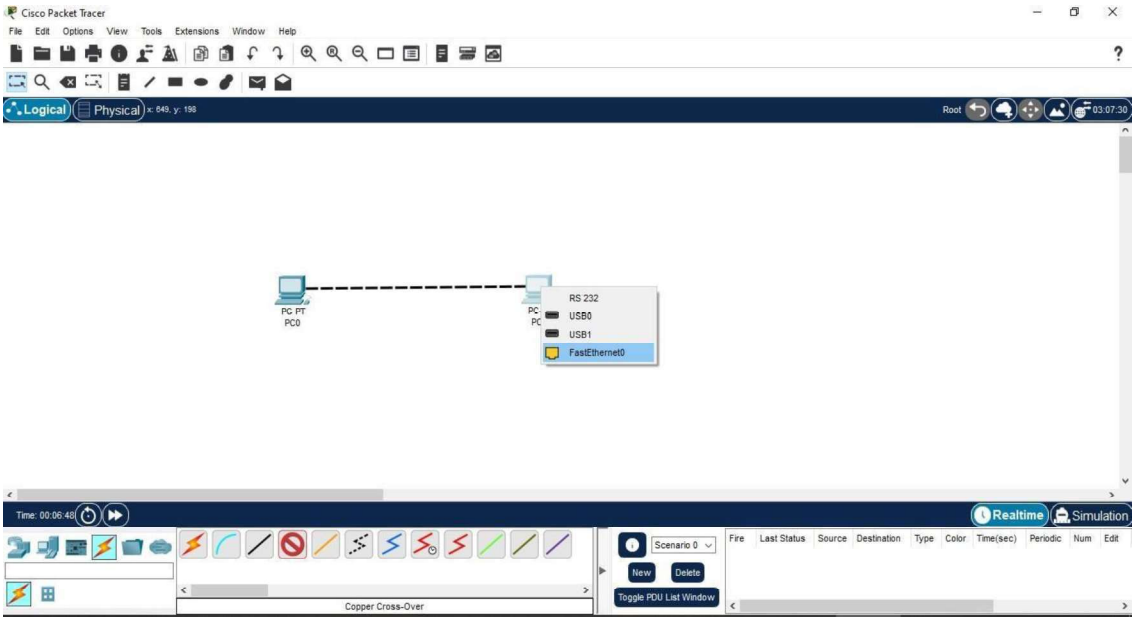
Step 2: Select Connectivity & Copper cross-over cable.



Step 3: Click on PC0 to get the interface options. Select Fa0/0

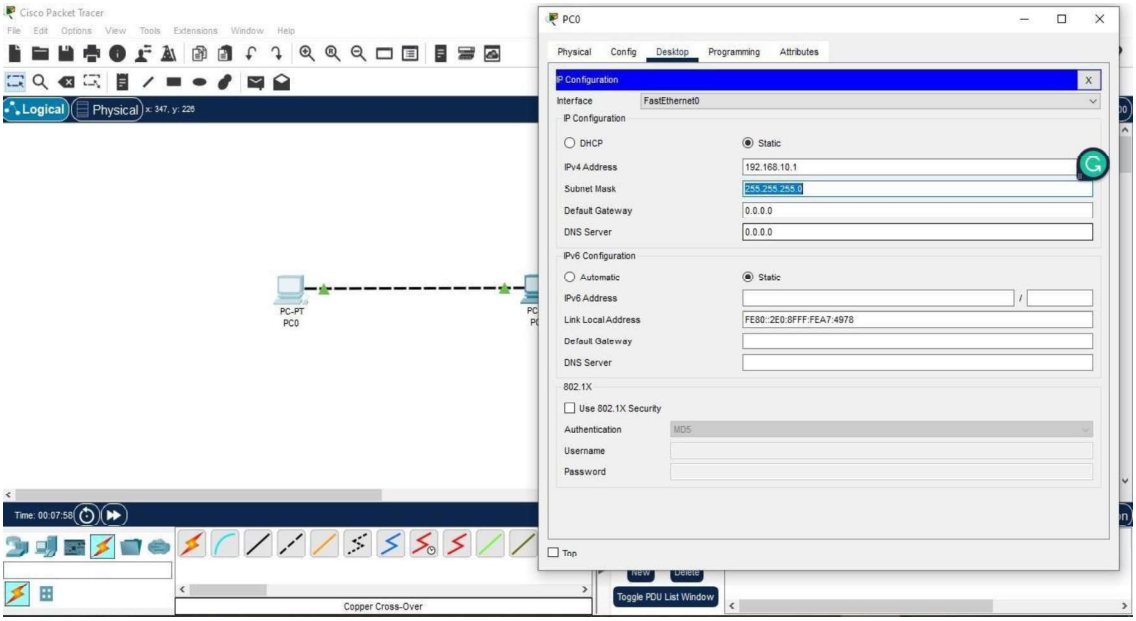


Step 4: Click on PC1 to get the interface options and select Fa0/0.

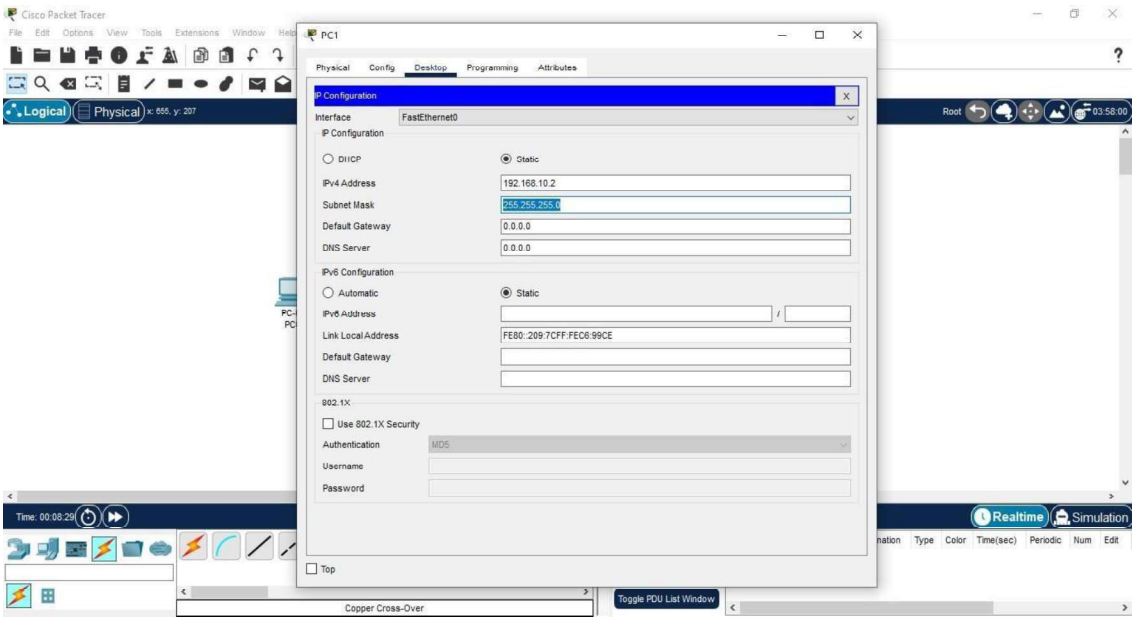


Step 5: Now the PCs are physically connected. To establish logical connectivity,

- Click on PC0.
- Select Desktop tab.
- Click on IP Configuration icon.
- Configure as in the following figure

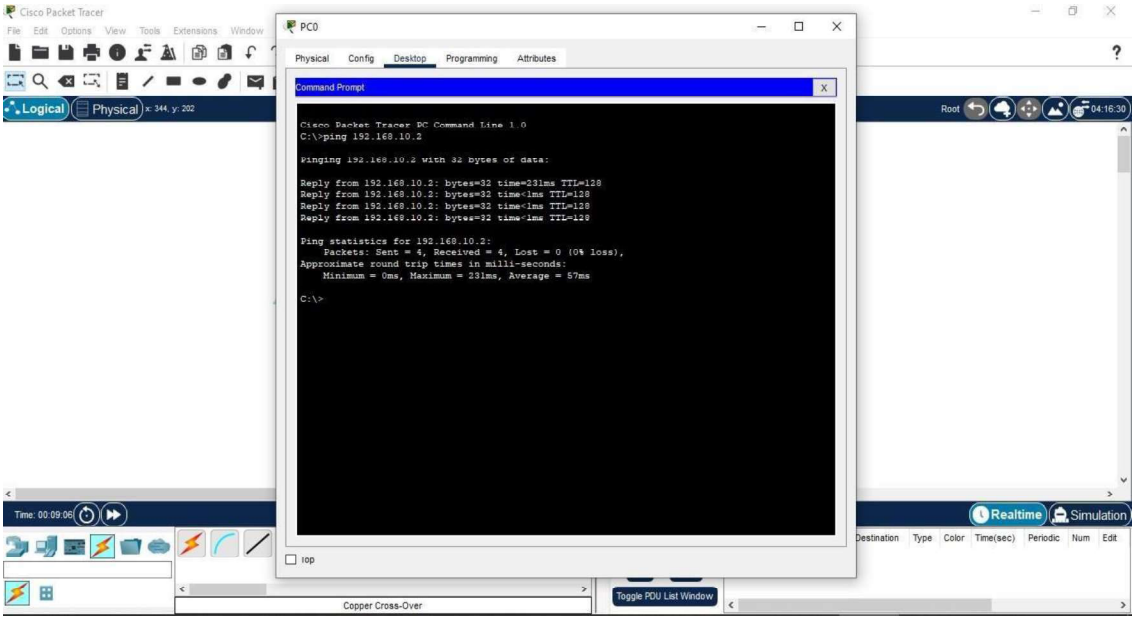


Step 6: Configure IP address for PC1 with the same procedure.



Step 7: Now both the PCs are physically and logically connected. To check the logical connectivity,

- Click on PC0.
- Select Desktop tab.
- Click on Command Prompt icon.
- Type ping 192.168.10.2 to fetch the output as follows



Exercise 2.b Demonstration of Simple LAN

Objective: To demonstrate the straight through cabling by designing a Local Area Network

Components:

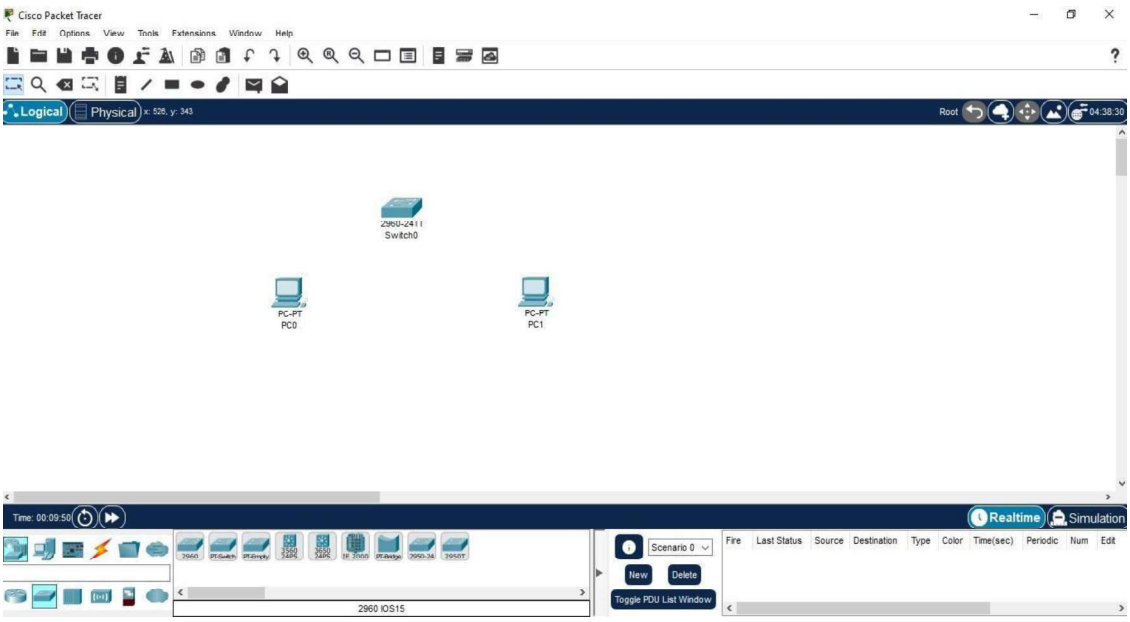
Devices	Required Nos
PCs	2
Copper Straight – Through Cables	2
Switch	1

Addressing Table:

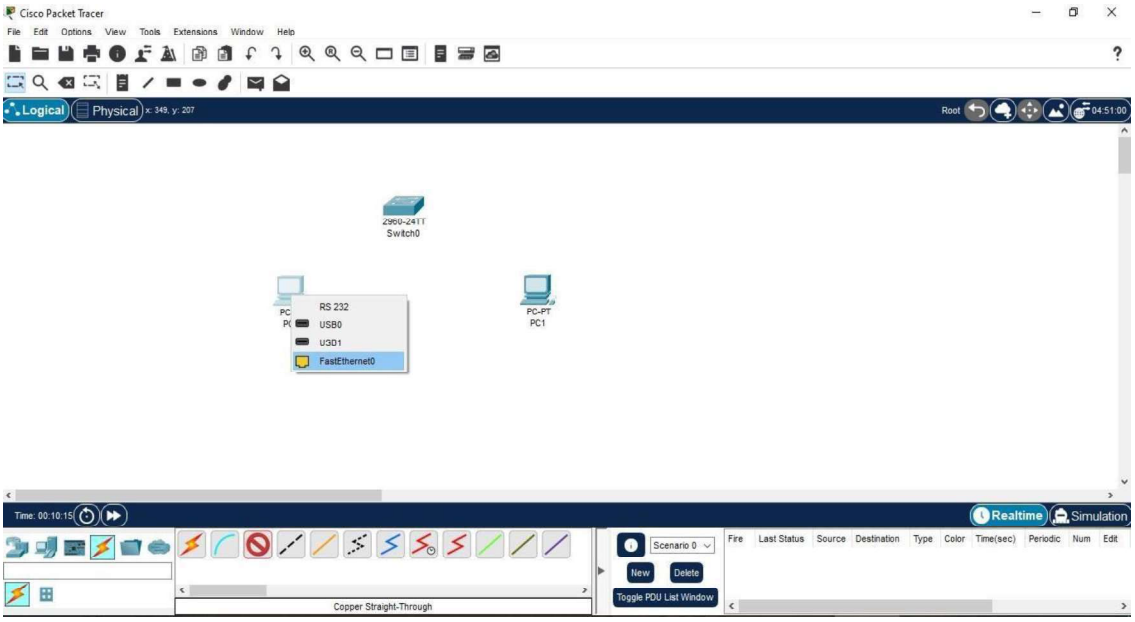
Device	Interface	IP Address	Subnet Mask
PC0	Fa0/0	192.168.10.1	255.255.255.0
PC1	Fa0/0	192.168.10.2	255.255.255.0

Procedure:

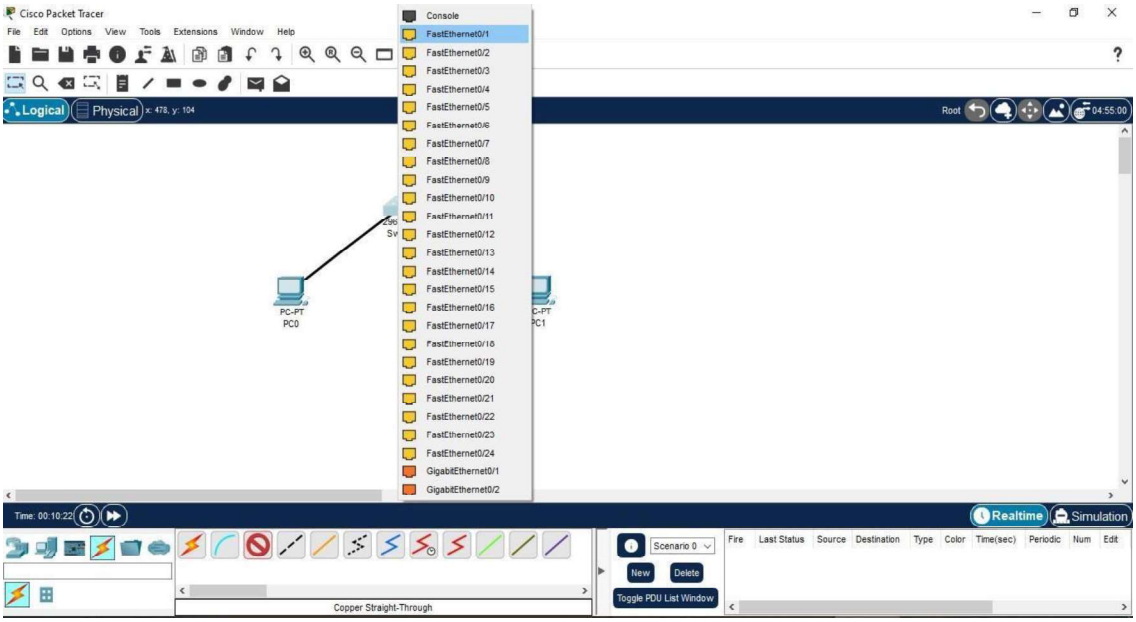
Step 1: Drag 2 PCs and a switch in the console area. Each PC will have interfaces as shown in the figure.



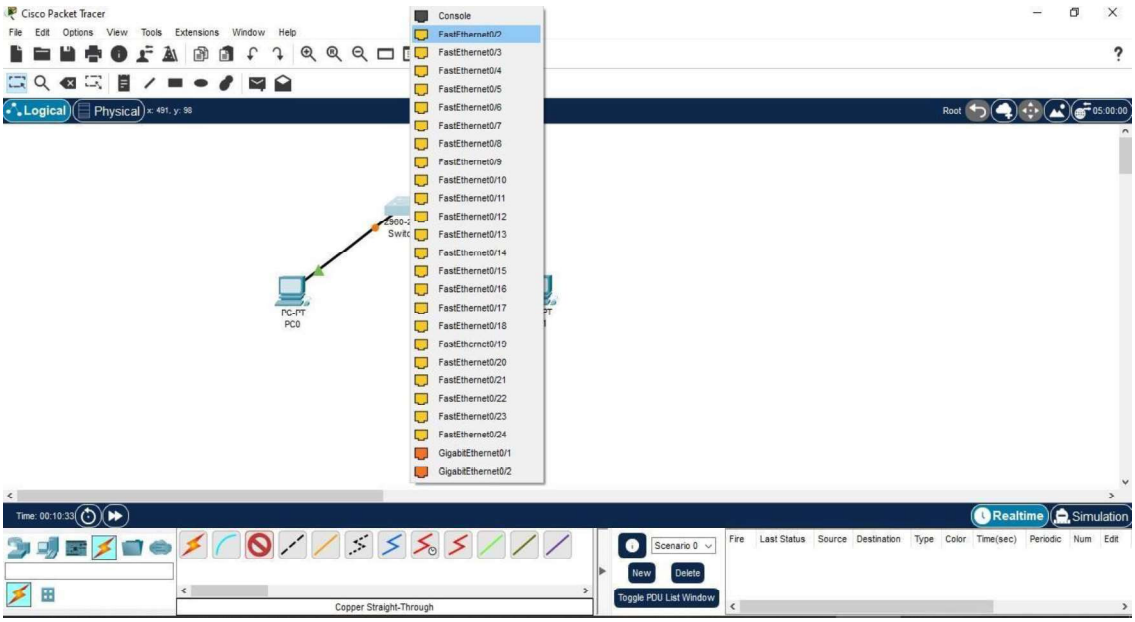
Step 2: Select Connectivity & Copper Straight-Through cable. Click on PC0 to get the interface options. Select Fa0/0



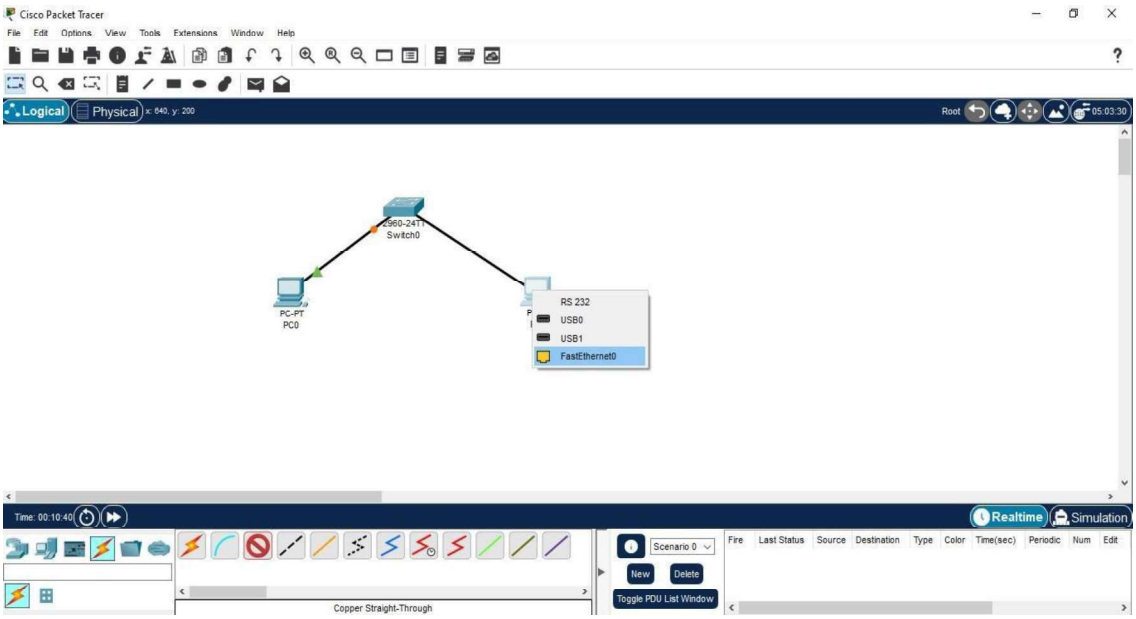
Step 3: Click on Switch to get the interface options and select Fa0/0.



Step 4: Now PC0 and Switch are physically connected. Again select copper straight-through cable and again click on Switch to get the interface options and select Fa0/1.



Step 5: Click on PC1 to get the interface options and select Fa0/0.

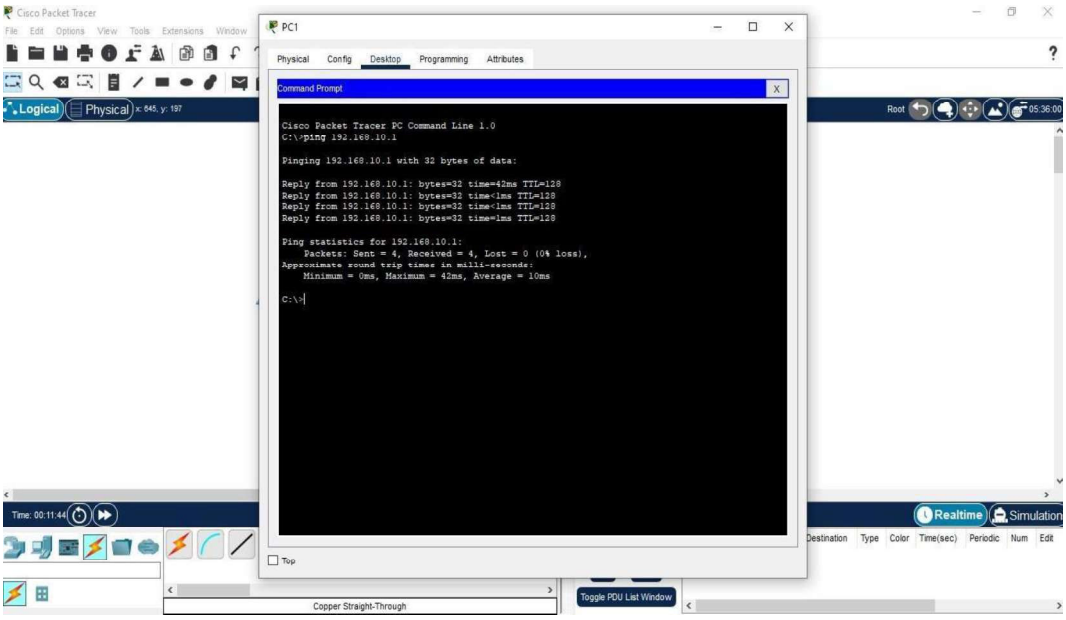


Step 6: Now the PCs are physically connected through switch. To establish logical connectivity,

- Click on PC0.
- Select Desktop tab.
- Click on IP Configuration icon.
- Configure the ip address 192.168.10.1 and subnet mask 255.255.255.0
- Repeat the same procedure for PC1 and configure with the ip address 192.168.10.2 and subnet mask 255.255.255.0

Step 7: Now both the PCs are physically and logically connected. To check the logical connectivity,

- Click on PC1.
- Select Desktop tab.
- Click on Command Prompt icon.
- Type ping 192.168.10.1 to fetch the output as follows



Result:

Thus the implementation of P2P and LAN is done successfully.

EXERCISE 3: ROUTER CONFIGURATION THROUGH A CONSOLE

Objective:

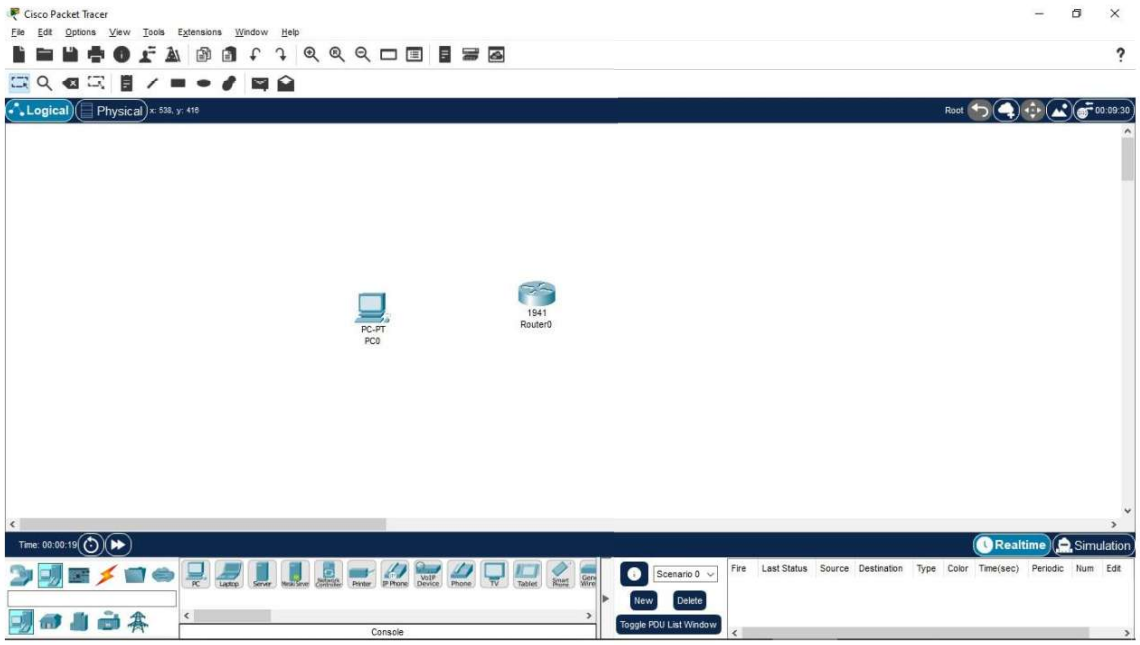
To demonstrate the configuration of Router with authentication through a console

Components:

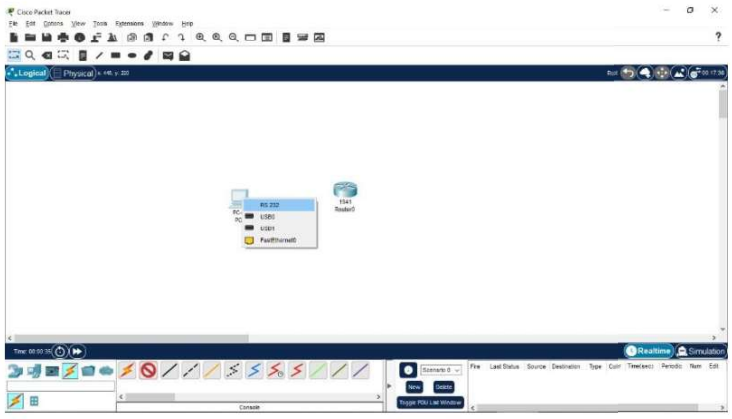
Devices	Required Nos
PCs	1
Console Cables	1
Router 1941	1

Procedure:

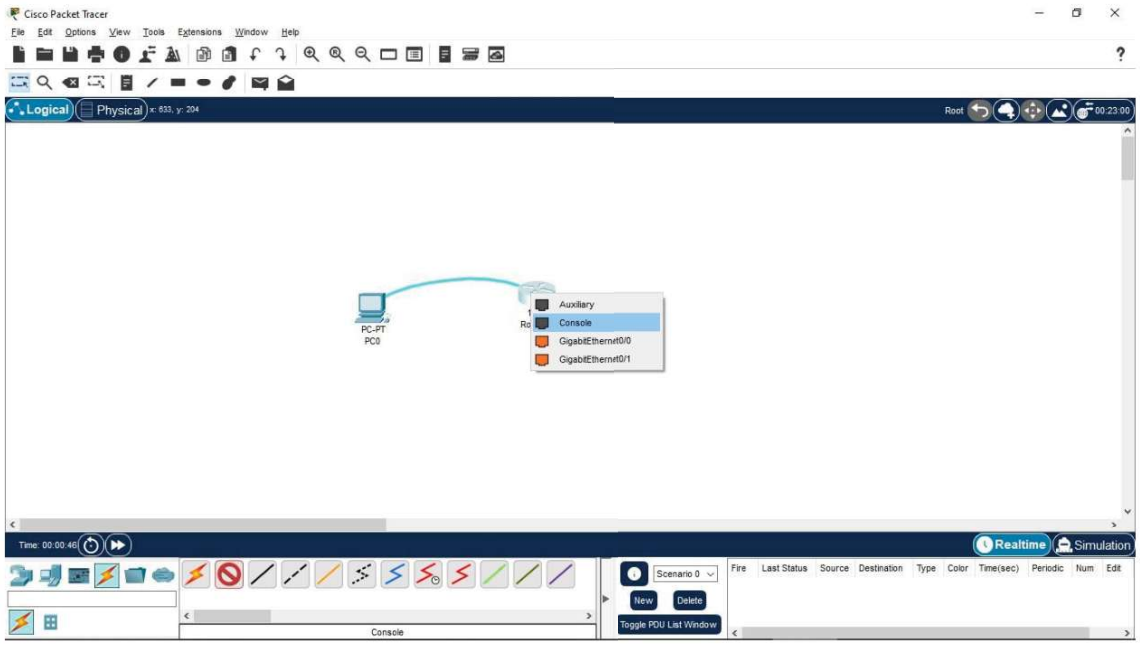
Step 1: Drag 1 PC and 1 number of 1941 Router in the console area



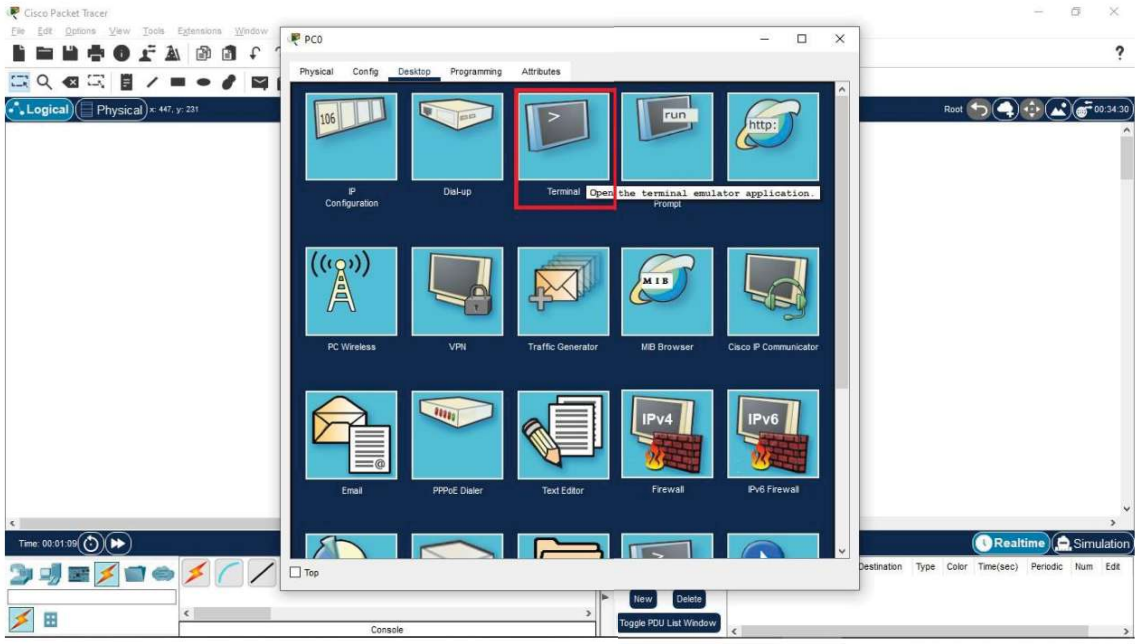
Step 2: Click on Connectivity and select Console cable. Click on PC and select RS232 interface.



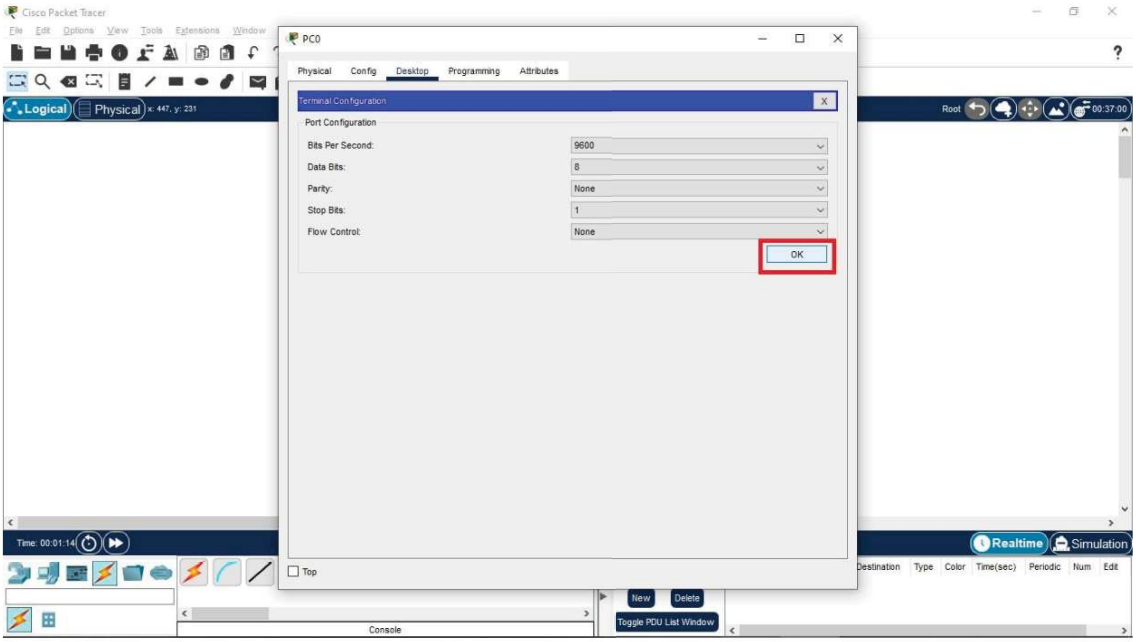
Step 3: Now click on Router and select Console interface to connect PC and Router with Console cable



Step 4: Now click on PC0 and go to Desktop => Terminal



Step 5: Click on OK for default Terminal parameters.



Step 6:

- Enter "no" when prompted, and type the following to configure "line console password" as "cisco" and privilege mode password as "cisco123"
- The "copy run start" command copies the running configuration parameters to start-up configuration parameters such that when you reload or reboot the router the configured parameters will not be erased from the Router memory RAM.

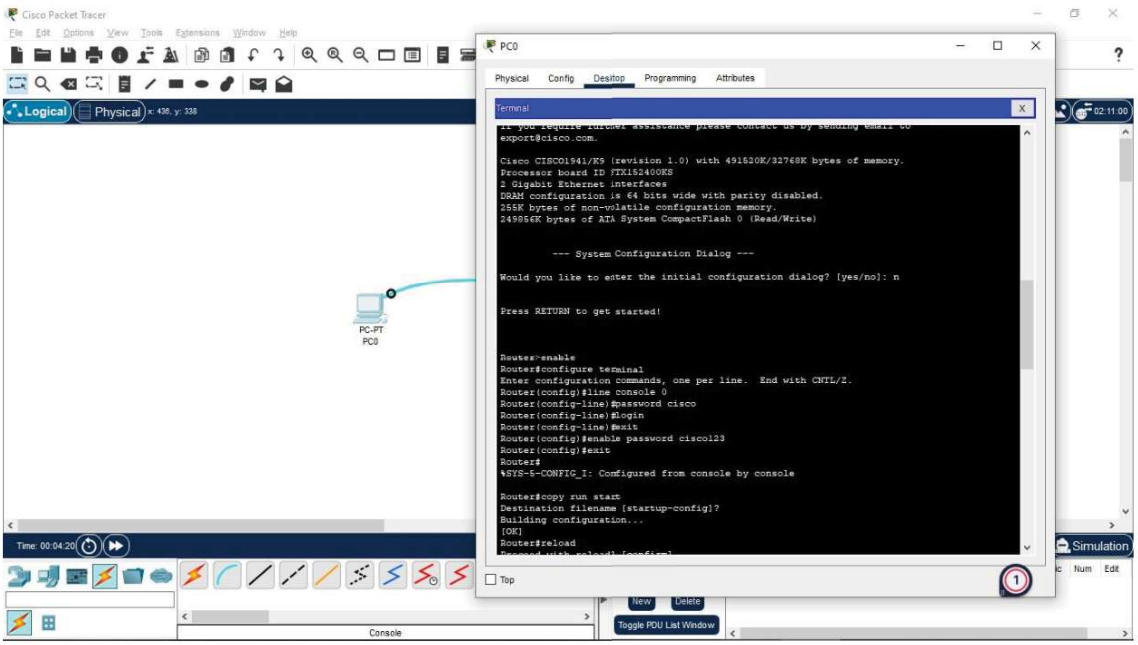
```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#line console 0
Router(config-line)#password cisco
Router(config-line)#login
Router(config-line)#exit
Router(config)#enable password cisco123
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#copy run start
Destination filename [startup-config]?
```

Building configuration...

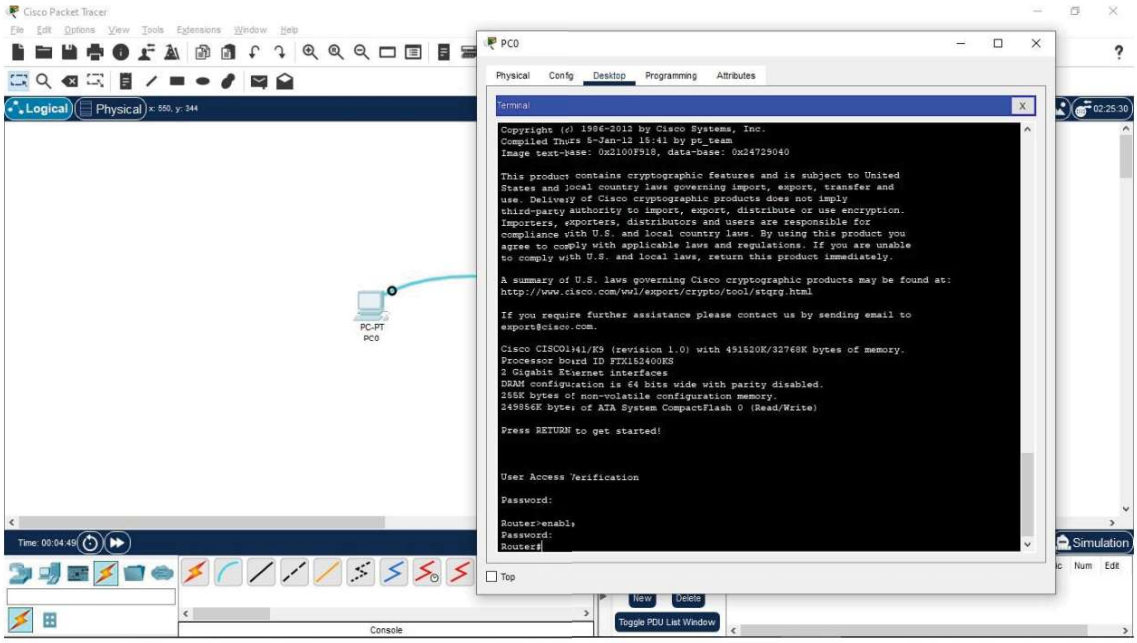
[OK]

Router#reload

Proceed with reload? [confirm]



Step 7: After reloading enter “cisco” as the password, when prompted to remote access and the password “cisco123” when you type “enable” to use Privilege mode access of Router from PC.



Configure Router Interfaces

```
R1(config)# interface gigabitethernet 0/0/0
R1(config-if)# ip address 192.168.10.1 255.255.255.0
R1(config-if)# ipv6 address 2001:db8:acad:1::1/64
R1(config-if)# description Link to LAN 1
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)# interface gigabitethernet 0/0/1
R1(config-if)# ip address 192.168.11.1 255.255.255.0
R1(config-if)# ipv6 address 2001:db8:acad:2::1/64
R1(config-if)# description Link to LAN 2
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)# interface serial 0/0/0
R1(config-if)# ip address 209.165.200.225 255.255.255.252
R1(config-if)# ipv6 address 2001:db8:acad:3::225/64
R1(config-if)# description Link to R2
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)#
```

Result:

Thus, the initial configuration of a router is done successfully.

EXERCISE 4 - A: CONFIGURATION OF IP ADDRESS IN ROUTER

Objective: To demonstrate the configuration of IP Address in router

Pre-requisite: IP Address, Range of IP Address and Classes of IP Address

Components:

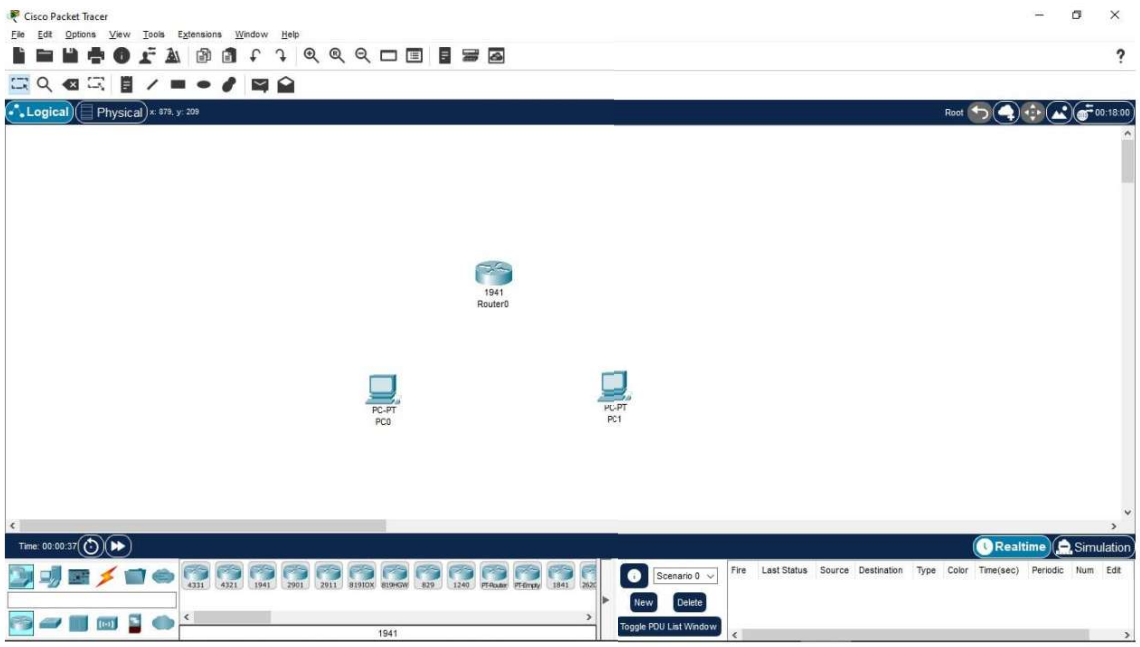
Devices	Required Nos
PCs	2
Copper cross-over Cables	2
Router	1

Addressing Table:

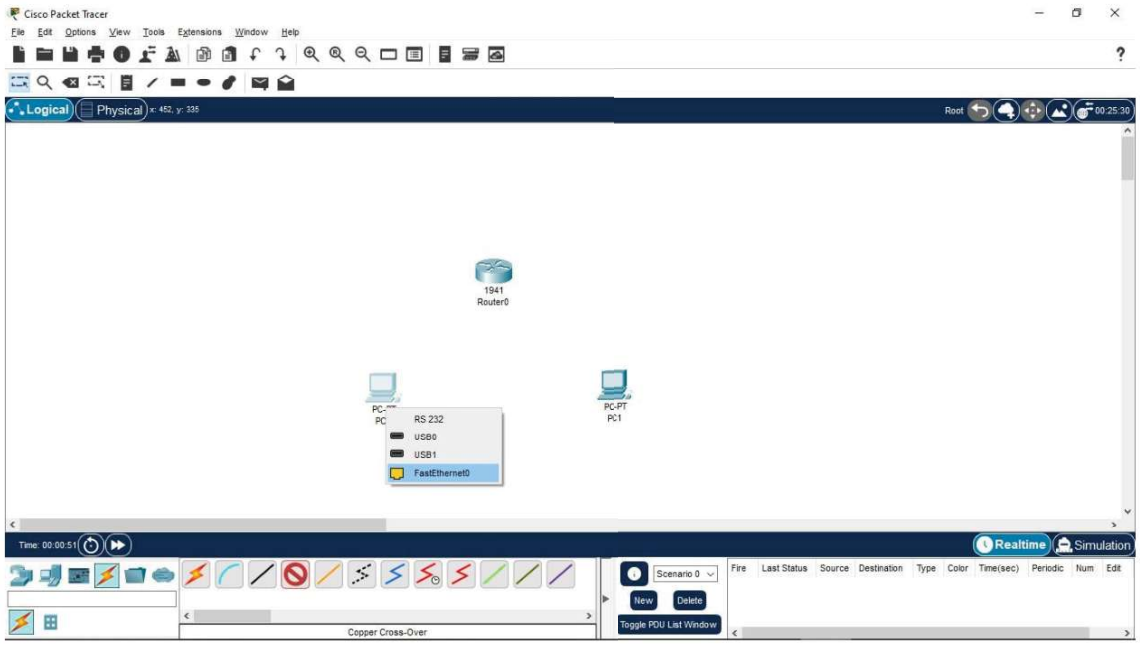
Device	Interface	IP Address	Subnet Mask	Gateway
PC0	Fa0/0	192.168.10.2	255.255.255.0	192.168.10.1
PC1	Fa0/0	192.168.11.2	255.255.255.0	192.168.11.1
Router0	Gigabit 0/0	192.168.10.1	255.255.255.0	-
Router0	Gigabit 0/1	192.168.11.1	255.255.255.0	-

Procedure:

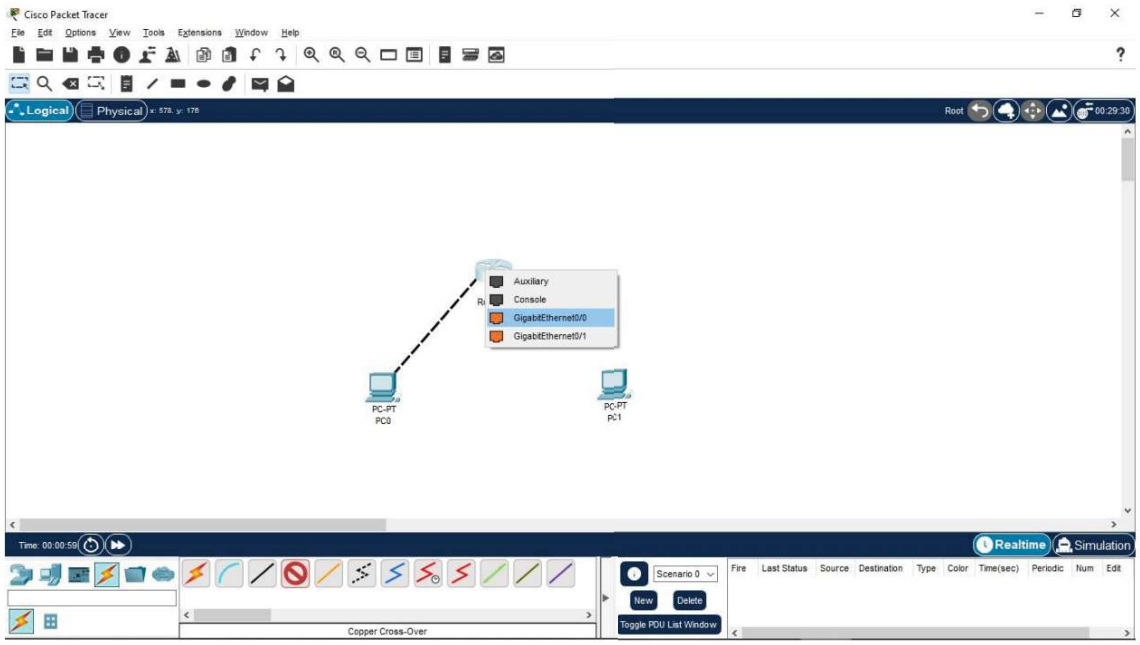
Step 1: Drag 2 PCs and a router in the console area.



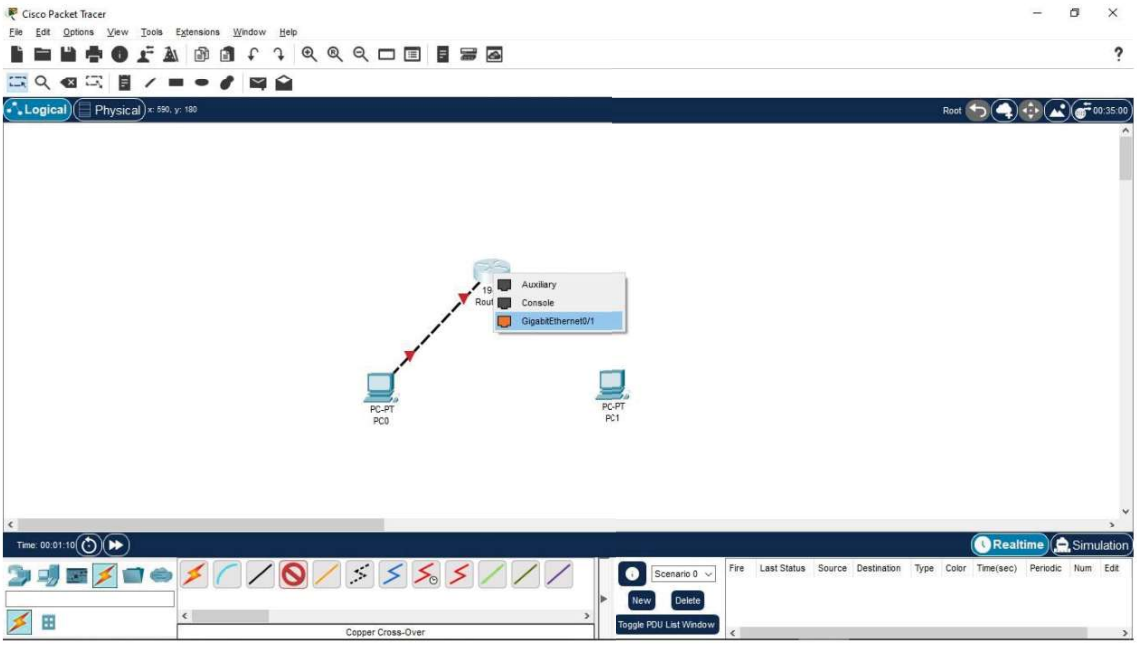
Step 2: Select Connectivity & Copper cross-over cable. Click on PC0 to get the interface options. Select Fa0/0



Step 3: Click on router0 to get the interface options and select GigabitEthernet0/0.



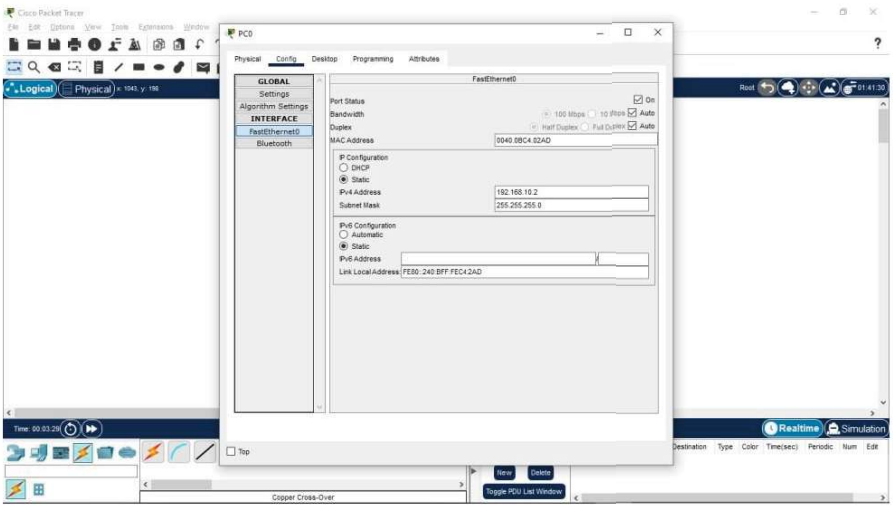
Step 4: Now PC0 and Router0 are physically connected. Again select copper cross-over cable and again click on Router0 to get the interface options and select GigabitEthernet0/1.



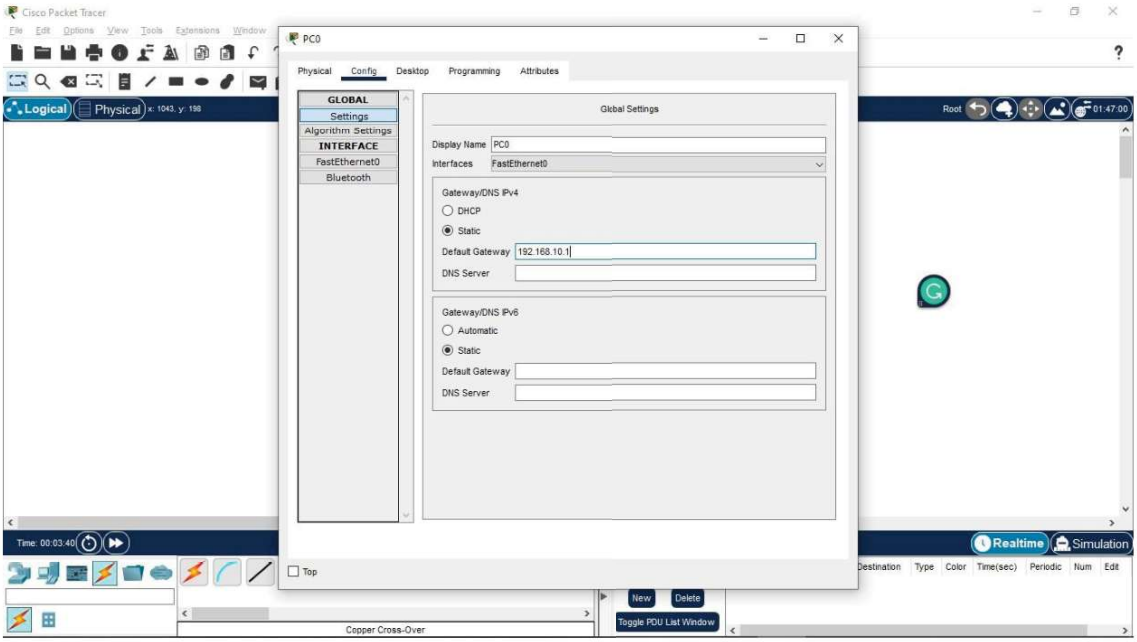
Step 5: Click on PC1 to get the interface options and select Fa0/0.

Step 6: Now the PCs are physically connected through Router. To establish logical connectivity,

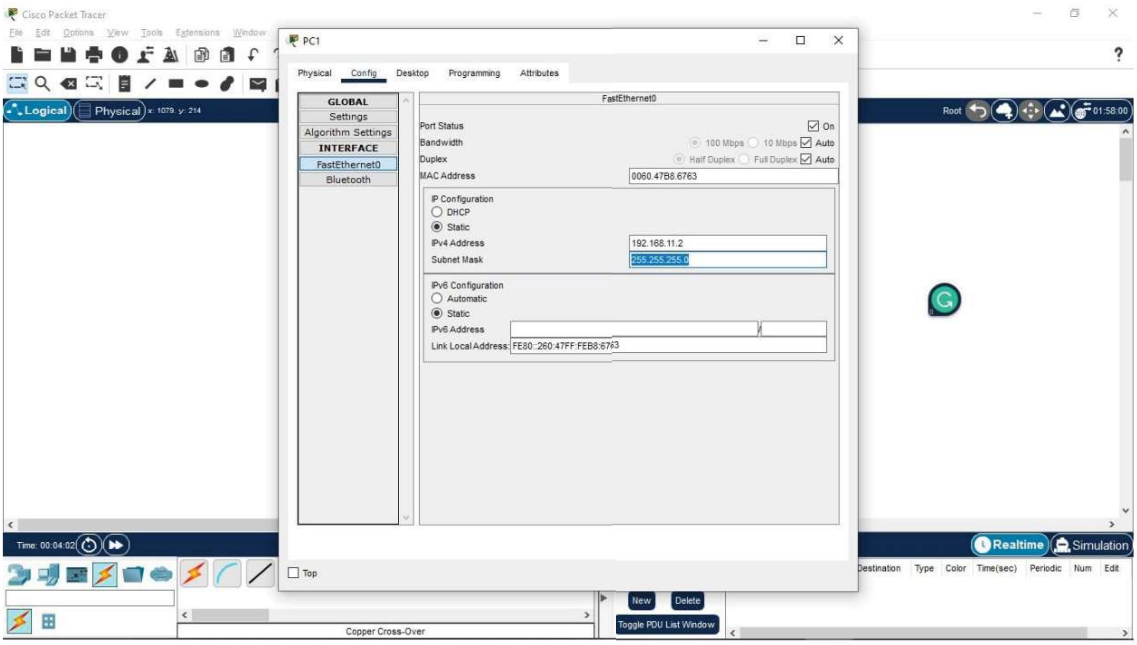
- Click on PC0.
- Select Config tab.
- Click on FastEhternet0/0 in the left pane.
- Configure the ip address 192.168.10.2 and subnet mask 255.255.255.0



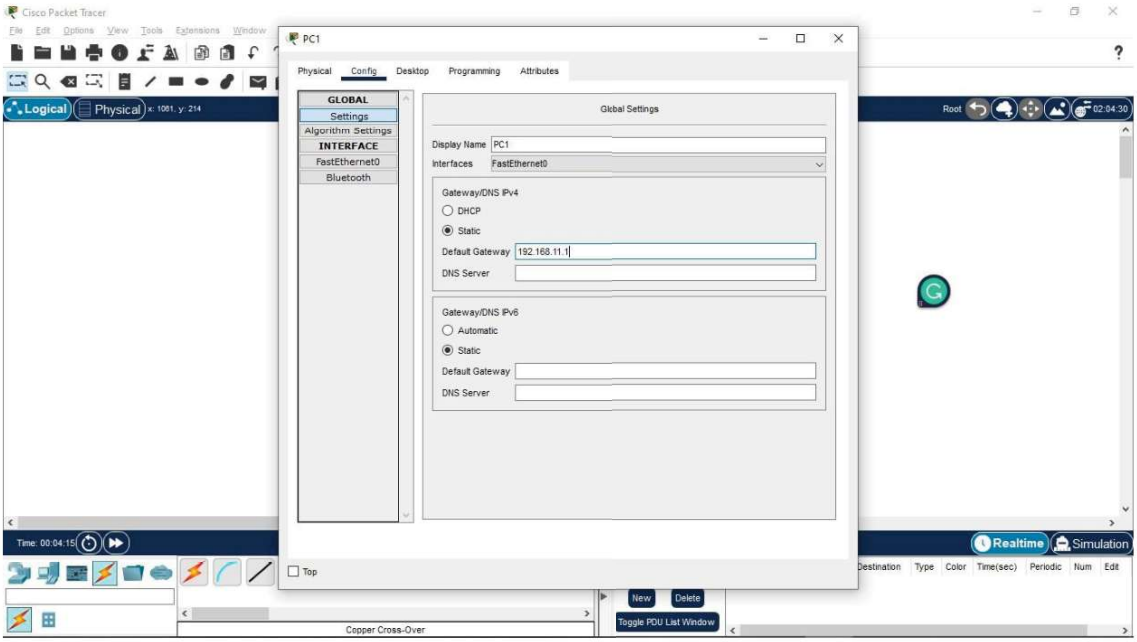
Step 7: Now click on settings and configure the gateway as 192.168.10.1



Step 8: Repeat the same procedure for PC1 and Configure the ip address 192.168.11.2 and subnet mask 255.255.255.0

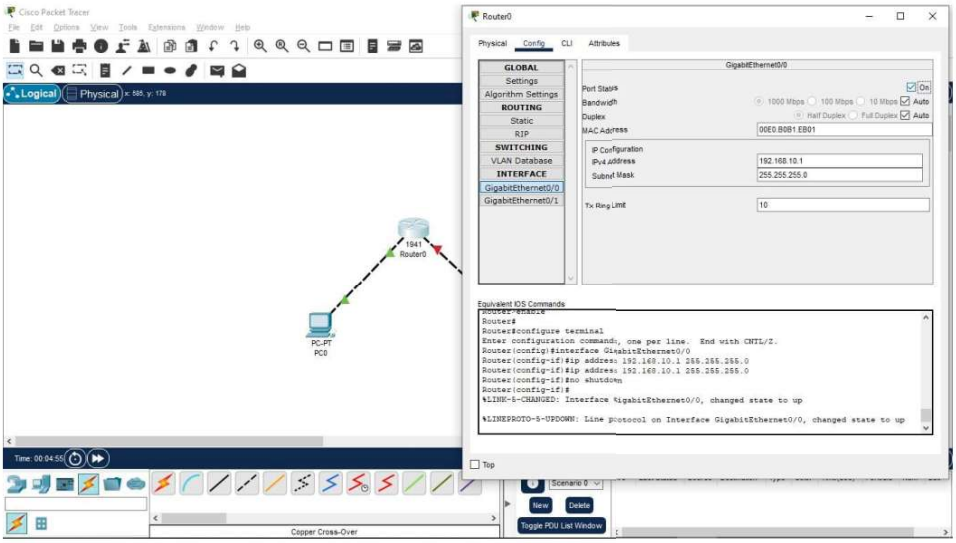


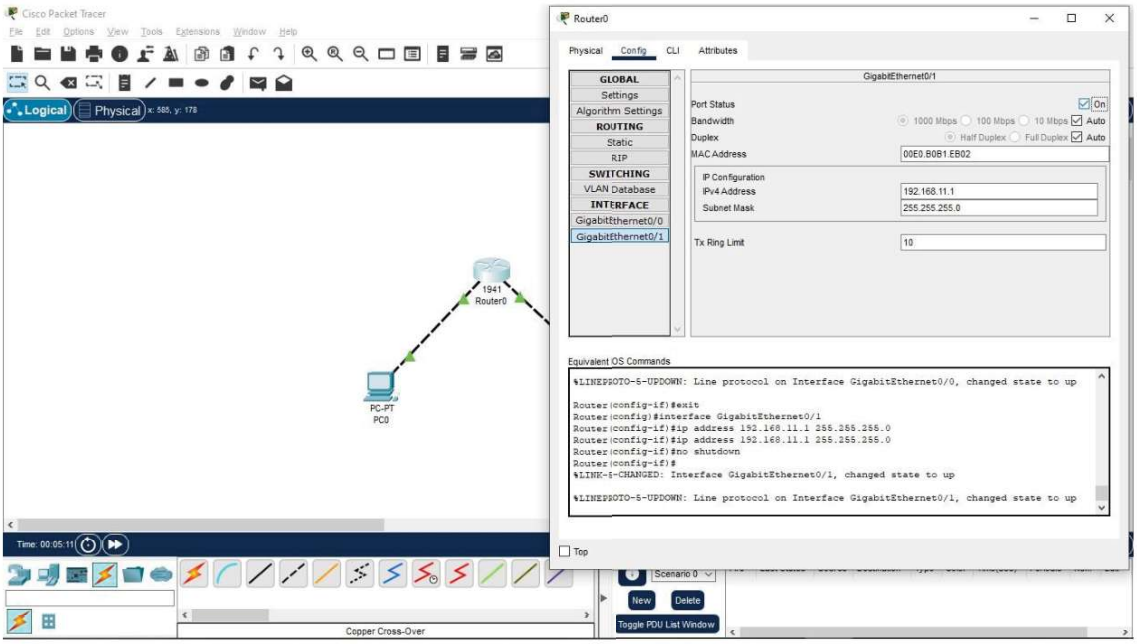
Step 9: Now click on settings and configure the gateway as 192.168.11.1



Step 10: Router configuration

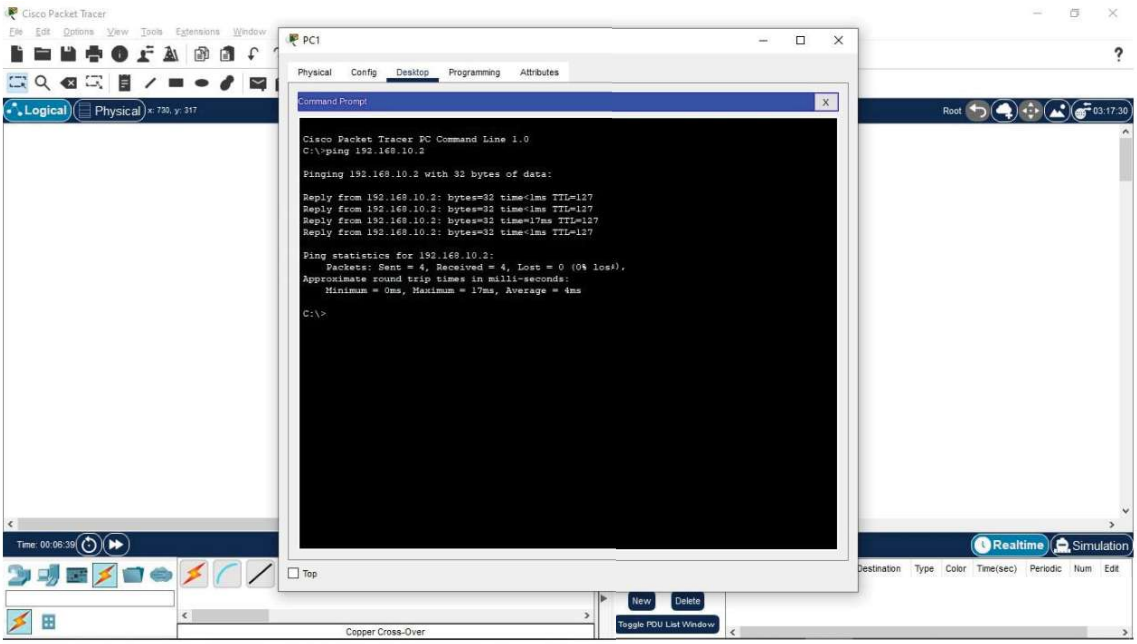
- Click on Router0 and select Config tab.
- Click on GigabitEthernet0/0 in the left pane and configure the ip address 192.168.10.1 and subnet mask 255.255.255.0
- Click on GigabitEthernet0/1 in the left pane and configure the ip address 192.168.11.1 and subnet mask 255.255.255.0





Step 11: Now both the PCs are physically and logically connected. To check the logical connectivity,

- Click on PC1.
- Select Desktop tab.
- Click on Command Prompt icon.
- Type ping 192.168.10.2 to fetch the output as follows



EXERCISE 4 - B: SUBNETTING IN WAN CONFIGURATION (DTE AND DCE)

Objective: To demonstrate the configuration of IP Addressing with Subnetting in WAN Configuration

Pre-requisite: IP Address, Range of IP Address, Classes of IP Address, Subnetting

Components:

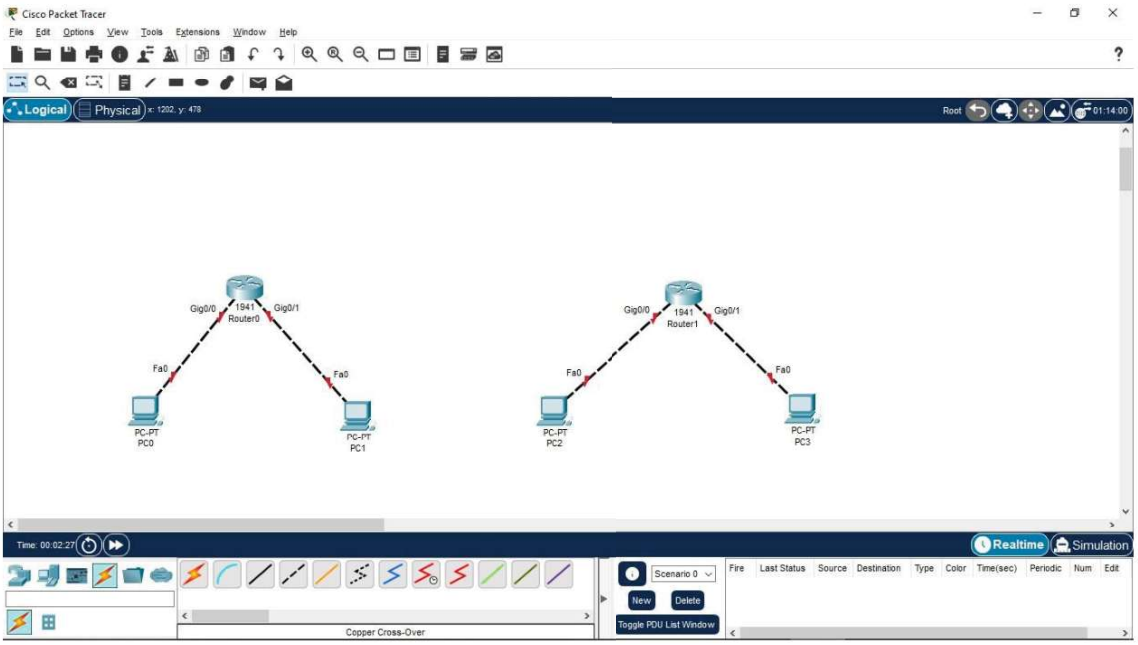
Devices	Required Nos
PCs	4
Copper cross-over Cables	4
Routers	2
Serial DCE	1

Addressing Table:

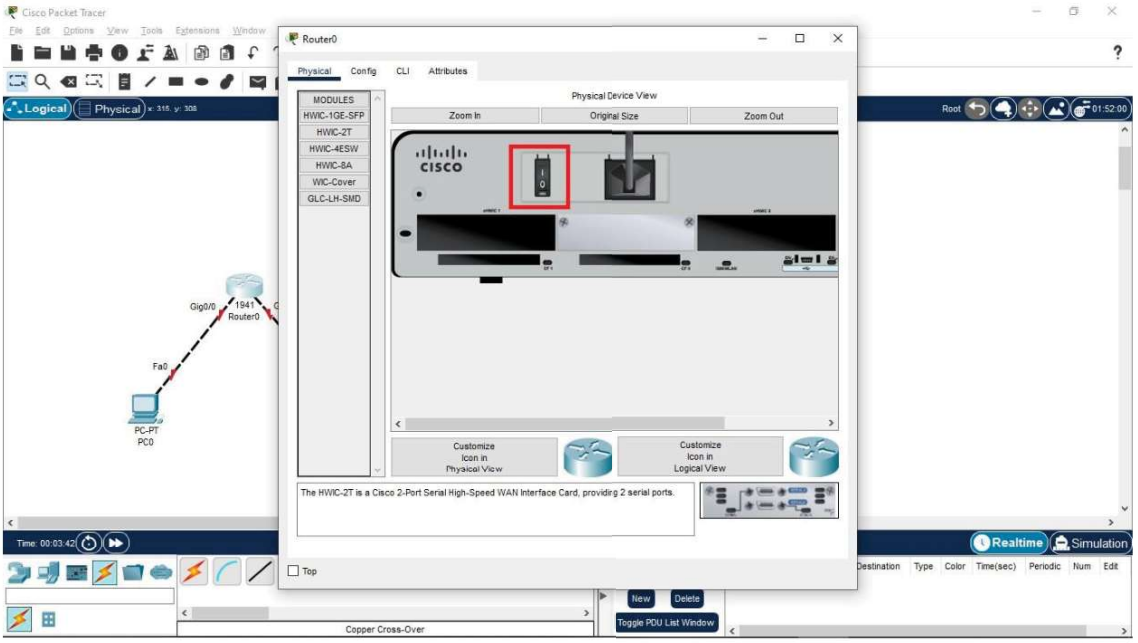
Device	Interface	IP Address	Subnet Mask	Gateway
PC0	Fa0/0	192.168.10.2	255.255.255.224	192.168.10.1
PC1	Fa0/0	192.168.10.34	255.255.255.224	192.168.10.33
PC2	Fa0/0	192.168.10.98	255.255.255.224	192.168.10.97
PC3	Fa0/0	192.168.10.130	255.255.255.224	192.168.10.129
Router0	Gigabit 0/0	192.168.10.1	255.255.255.224	-
Router0	Gigabit 0/1	192.168.10.33	255.255.255.224	-
Router0	Se0/1/0	192.168.10.65	255.255.255.224	-
Router1	Gigabit 0/0	192.168.10.97	255.255.255.224	-
Router1	Gigabit 0/1	192.168.10.129	255.255.255.224	-
Router1	Se0/1/0	192.168.10.66	255.255.255.224	-

Procedure:

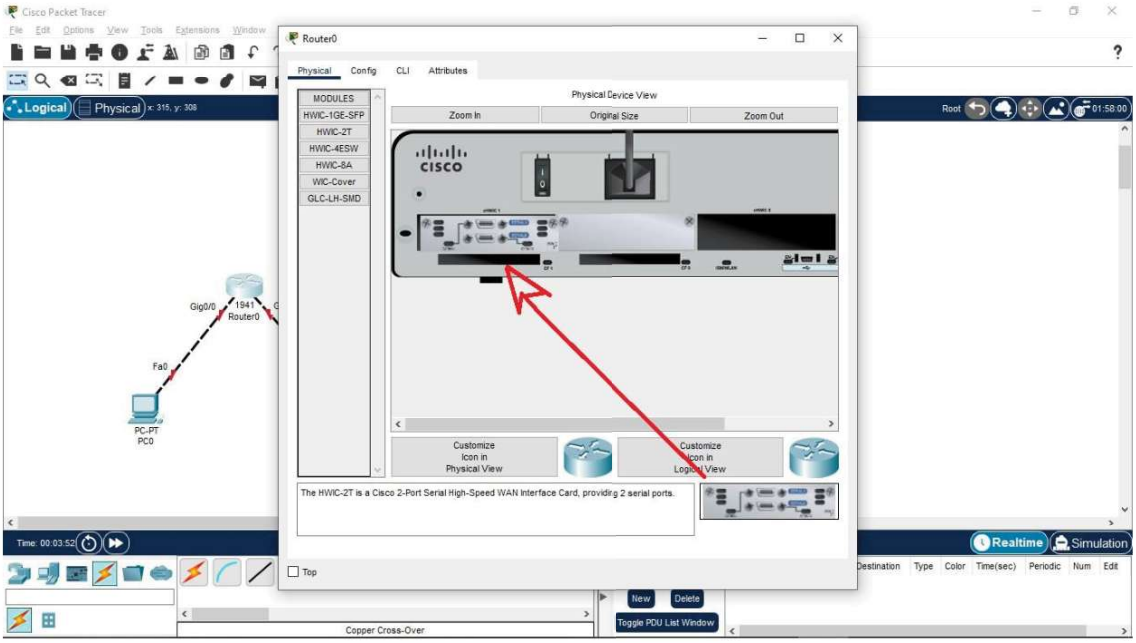
Step 1: Drag 4 PCs and 2 routers in the console area as shown in figure



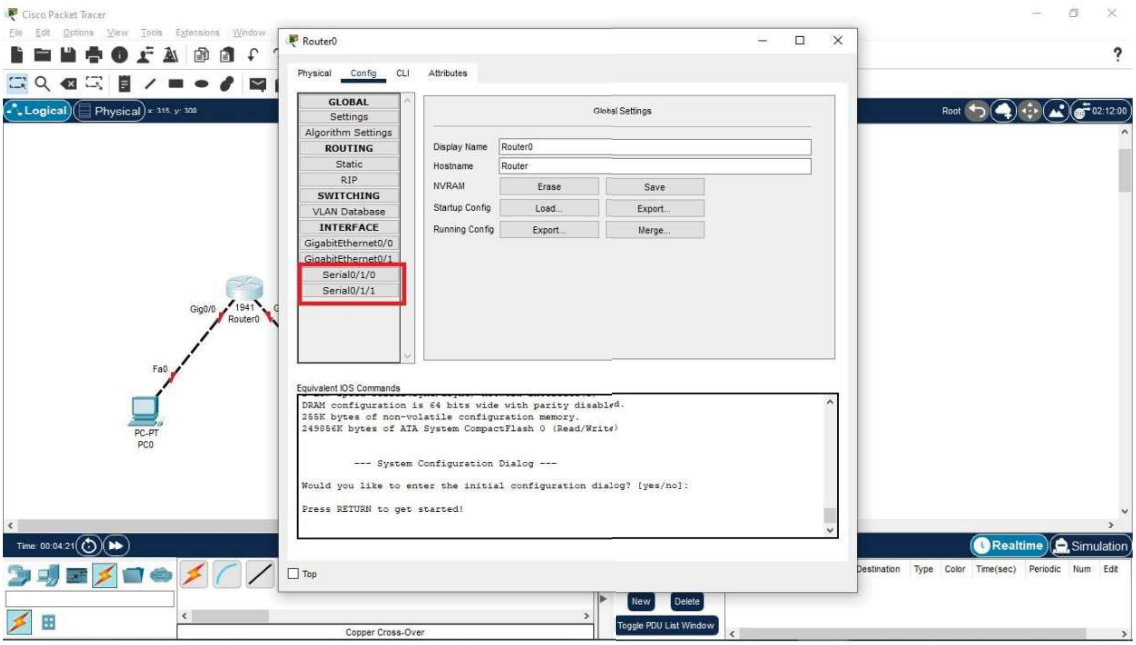
Step 2: Click on Router0 and go to Physical tab. Click HWIC2T in the left pane and Click on zoom in. Switch off the Hardware



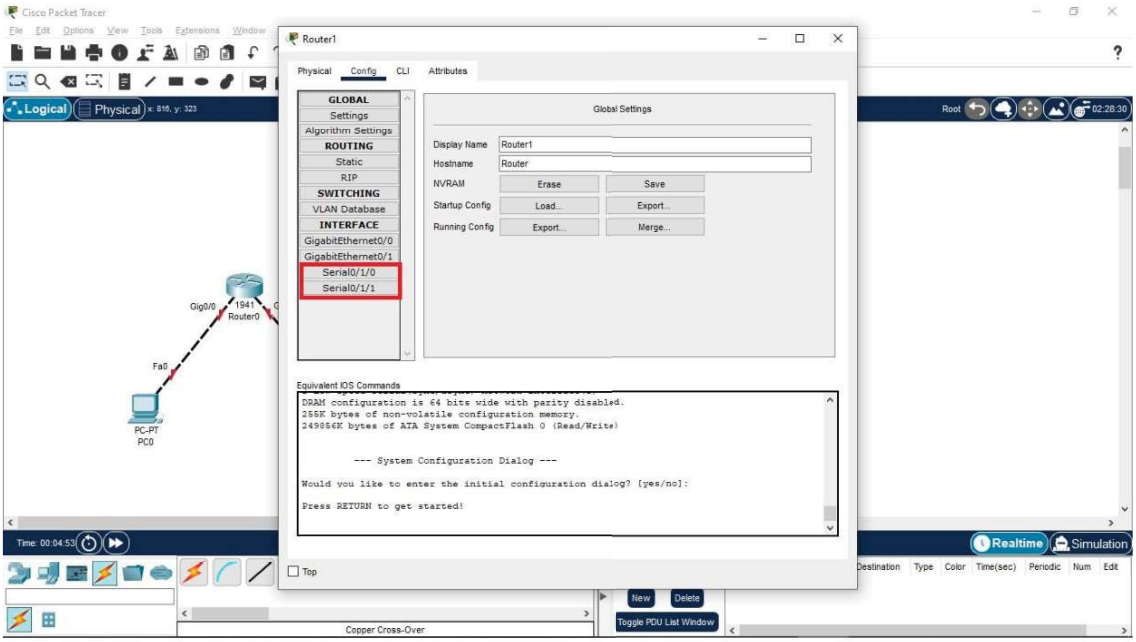
Step 3: Find the console from lower right corner. Drag and drop the console in the empty area as shown in the figure.



Step 4: Now again switch on the hardware and check in config tab for 2 serial ports added.

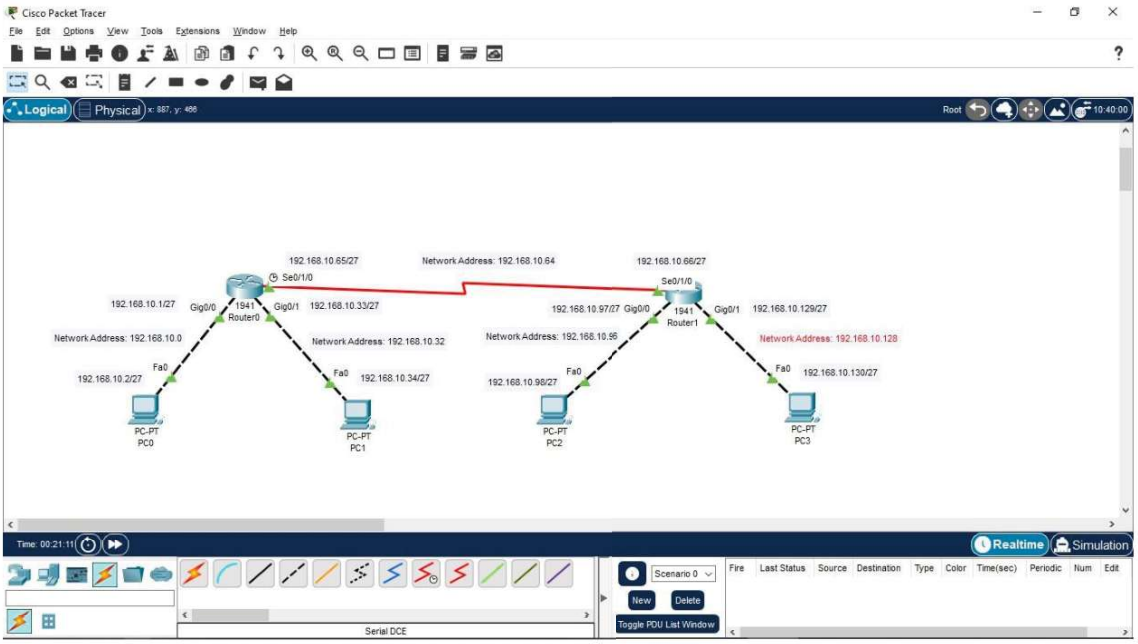


Step 5: Repeat the same procedure (Step 3 and Step 4) for Router1



Step 6: Now the PCs are physically connected through Router. To establish logical connectivity, assign IP addresses for 4 PCs (each 1 interface and corresponding router interface as gateway) and 2 Routers (each 3 ip addresses for 3 interfaces) as shown in the following table.

Device	Interface	IP Address	Subnet Mask	Gateway
PC0	Fa0/0	192.168.10.2	255.255.255.224	192.168.10.1
PC1	Fa0/0	192.168.10.34	255.255.255.224	192.168.10.33
PC2	Fa0/0	192.168.10.98	255.255.255.224	192.168.10.97
PC3	Fa0/0	192.168.10.130	255.255.255.224	192.168.10.129
Router0	Gigabit 0/0	192.168.10.1	255.255.255.224	-
Router0	Gigabit 0/1	192.168.10.33	255.255.255.224	-
Router0	Se0/1/0	192.168.10.65	255.255.255.224	-
Router1	Gigabit 0/0	192.168.10.97	255.255.255.224	-
Router1	Gigabit 0/1	192.168.10.129	255.255.255.224	-
Router1	Se0/1/0	192.168.10.66	255.255.255.224	-



Scenario with Network Address for each link

Step 7:

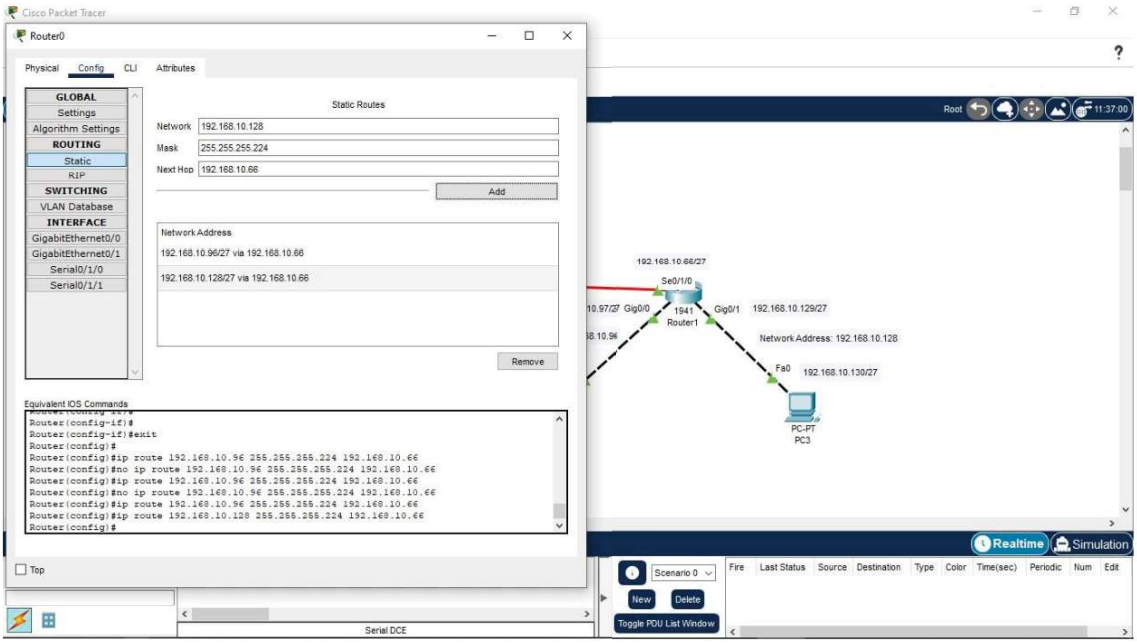
- **To enable packet transmission among the devices in the scenario, Static Routing has to be configured.**
- **To configure static routing, unknown networks and next-hop IP address to reach the unknown network for Router0 and Router1 has to be determined.**
- **Note: While specifying devices we should use IP-Address and while specifying network we should use Network Address.**
- **Unknown networks for the routers are derived in the following table**

Device	Known Networks	Subnet Mask	Unknown Networks	Subnet Mask	Next-hop Address
Router0	192.168.10.0	255.255.255.224	192.168.10.96	255.255.255.224	192.168.10.66
Router0	192.168.10.32	255.255.255.224	192.168.10.128	255.255.255.224	192.168.10.66
Router0	192.168.10.64	255.255.255.224	-	-	-
Router1	192.168.10.96	255.255.255.224	192.168.10.0	255.255.255.224	192.168.10.65
Router1	192.168.10.128	255.255.255.224	192.168.10.32	255.255.255.224	192.168.10.65
Router1	192.168.10.64	255.255.255.224	-	-	-

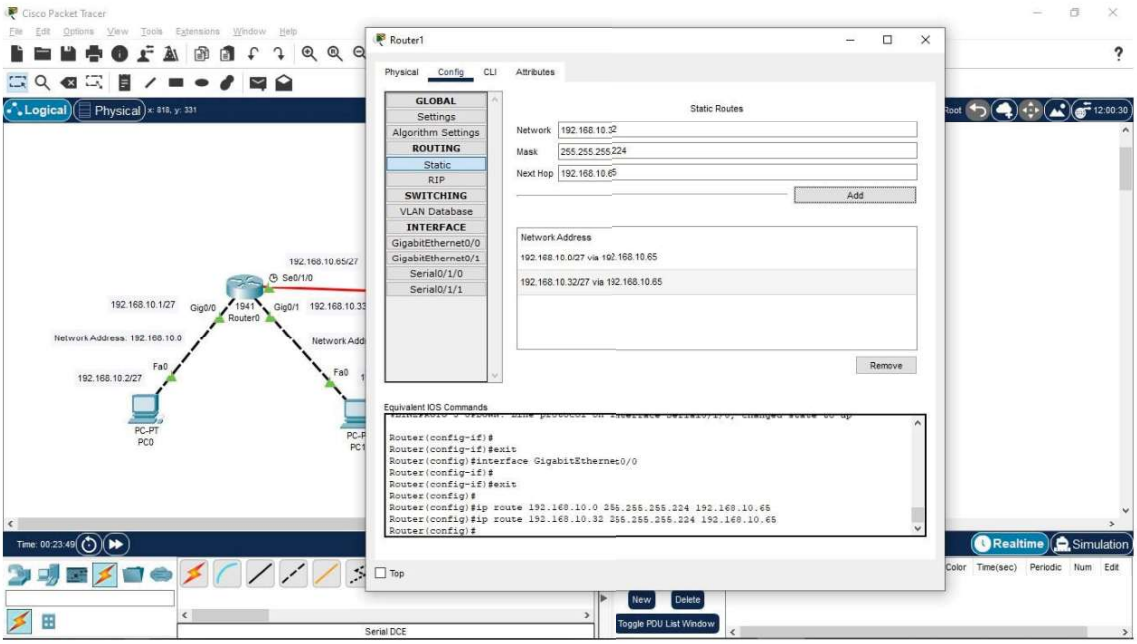
Only unknown networks should be configured for static routing

Step 8:

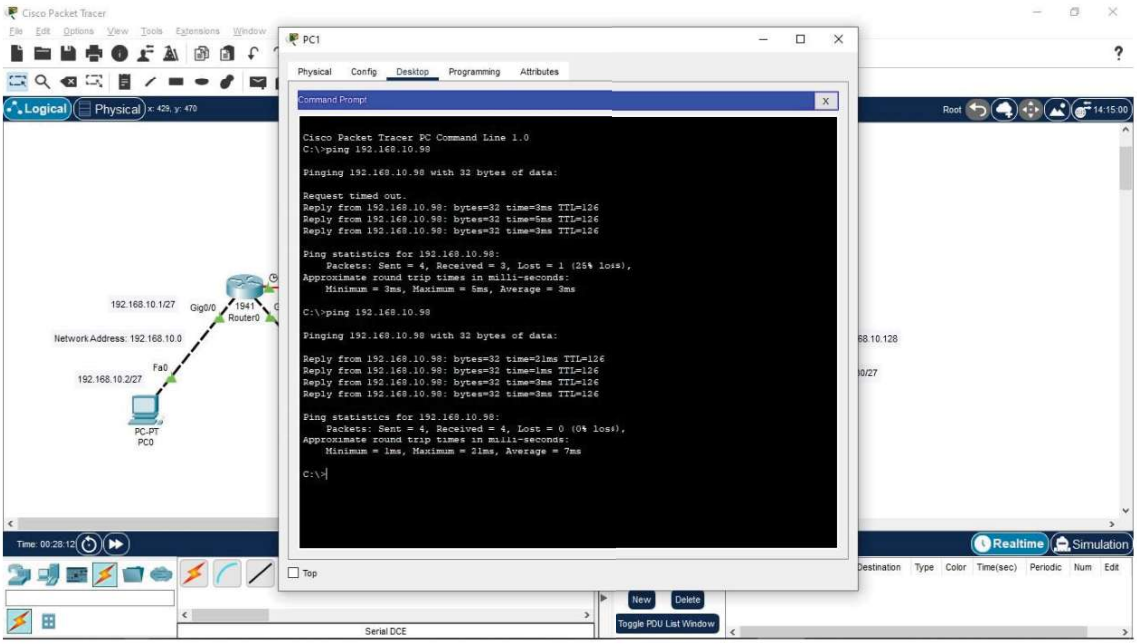
- To configure static routing, click on
    - Router0 => Config tab => Static =>Enter network address (refer unknown network for router0) => Subnet Mask for network address => Next-hop address => Add network
- Repeat the same to add the next network.
- Two networks should be added for the given scenario



Step 9: Repeat the same procedure to configure for Router1



Step 10: After configuration of Static routing in both Routers, Check the connectivity among any two devices using Ping Command



Step 11:

- To check routing table, go to CLI tab in Router and press enter to get the router prompt.
    - Router>
- Now type enable or en and press enter
    - Router>en
    - Router#

Follow the command "show ip route" to get the routing table of a router

```
Router>en
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

192.168.10.0/24 is variably subnetted, 8 subnets, 2 masks
C 192.168.10.0/27 is directly connected, GigabitEthernet0/0
L 192.168.10.1/32 is directly connected, GigabitEthernet0/0
C 192.168.10.32/27 is directly connected, GigabitEthernet0/1
L 192.168.10.33/32 is directly connected, GigabitEthernet0/1
C 192.168.10.64/27 is directly connected, Serial0/1/0
L 192.168.10.65/32 is directly connected, Serial0/1/0
S 192.168.10.96/27 [1/0] via 192.168.10.66
S 192.168.10.128/27 [1/0] via 192.168.10.66
```

Result:

Thus, the implementation of IP addressing & subnetting is done and verified in packet tracer.

EXERCISE 5: DEMONSTRATION OF STATIC AND DEFAULT ROUTING

Exercise 5. a: Demonstration of Static Routing

Objective: To demonstrate the configuration of IP Addressing with Subnetting in WAN Configuration

Pre-requisite: IP Address, Range of IP Address, Classes of IP Address, Subnetting

Components:

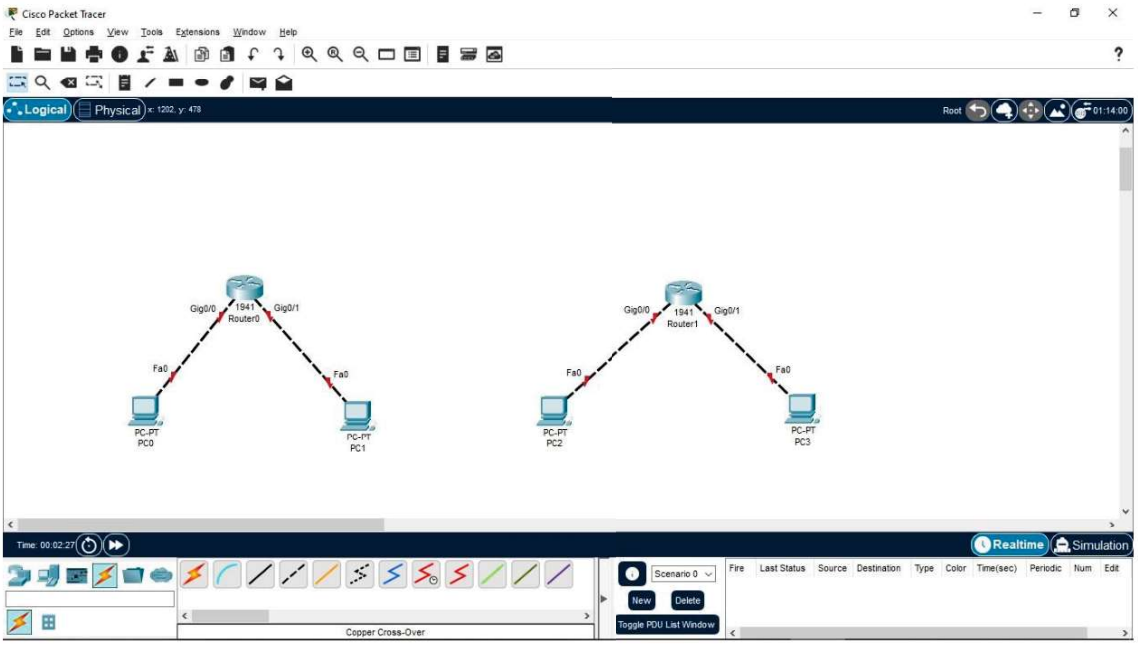
Devices	Required Nos
PCs	4
Copper cross-over Cables	4
Routers	2
Serial DCE	1

Addressing Table:

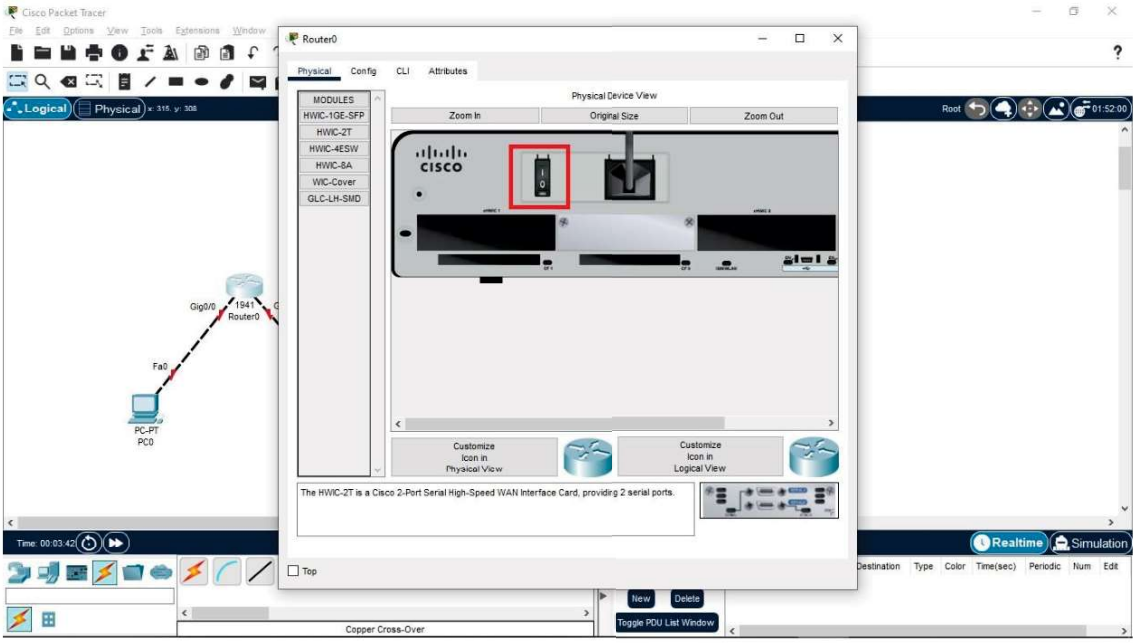
Device	Interface	IP Address	Subnet Mask	Gateway
PC0	Fa0/0	192.168.10.2	255.255.255.224	192.168.10.1
PC1	Fa0/0	192.168.10.34	255.255.255.224	192.168.10.33
PC2	Fa0/0	192.168.10.98	255.255.255.224	192.168.10.97
PC3	Fa0/0	192.168.10.130	255.255.255.224	192.168.10.129
Router0	Gigabit 0/0	192.168.10.1	255.255.255.224	-
Router0	Gigabit 0/1	192.168.10.33	255.255.255.224	-
Router0	Se0/1/0	192.168.10.65	255.255.255.224	-
Router1	Gigabit 0/0	192.168.10.97	255.255.255.224	-
Router1	Gigabit 0/1	192.168.10.129	255.255.255.224	-
Router1	Se0/1/0	192.168.10.66	255.255.255.224	-

Procedure:

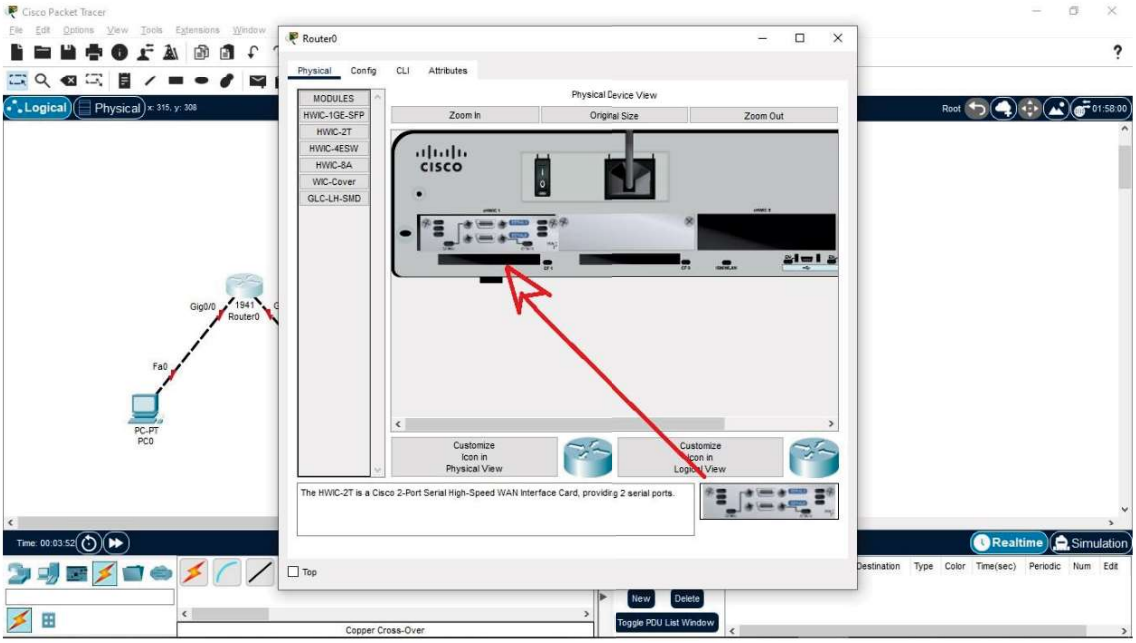
Step 1: Drag 4 PCs and 2 routers in the console area as shown in figure



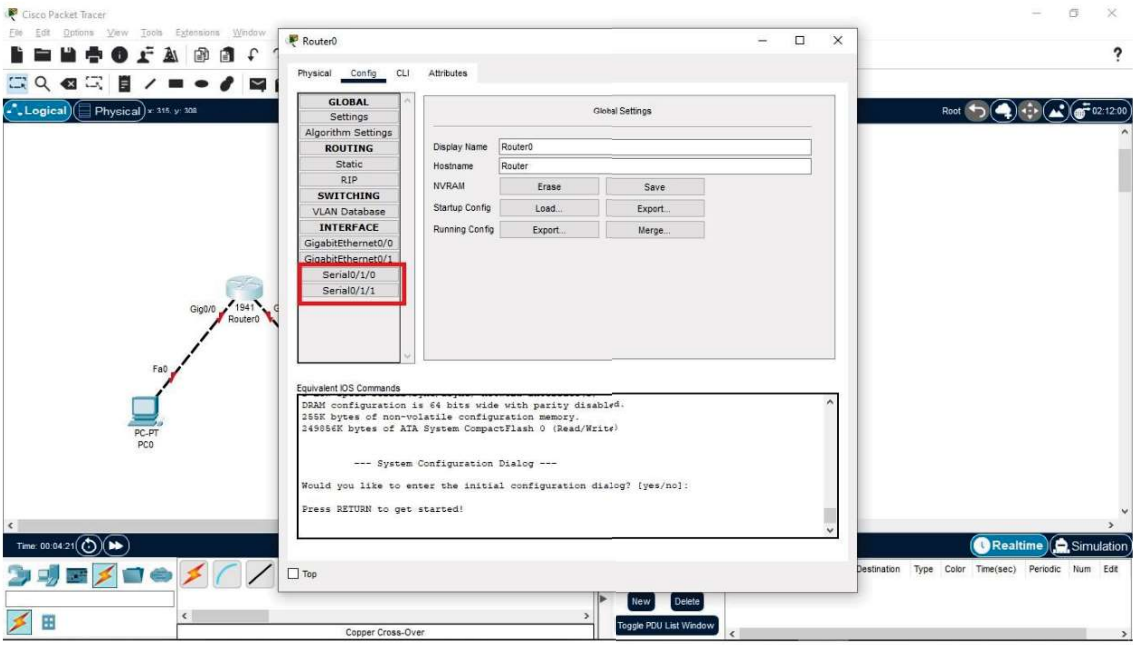
Step 2: Click on Router0 and go to Physical tab. Click HWIC2T in the left pane and Click on zoom in. Switch off the Hardware



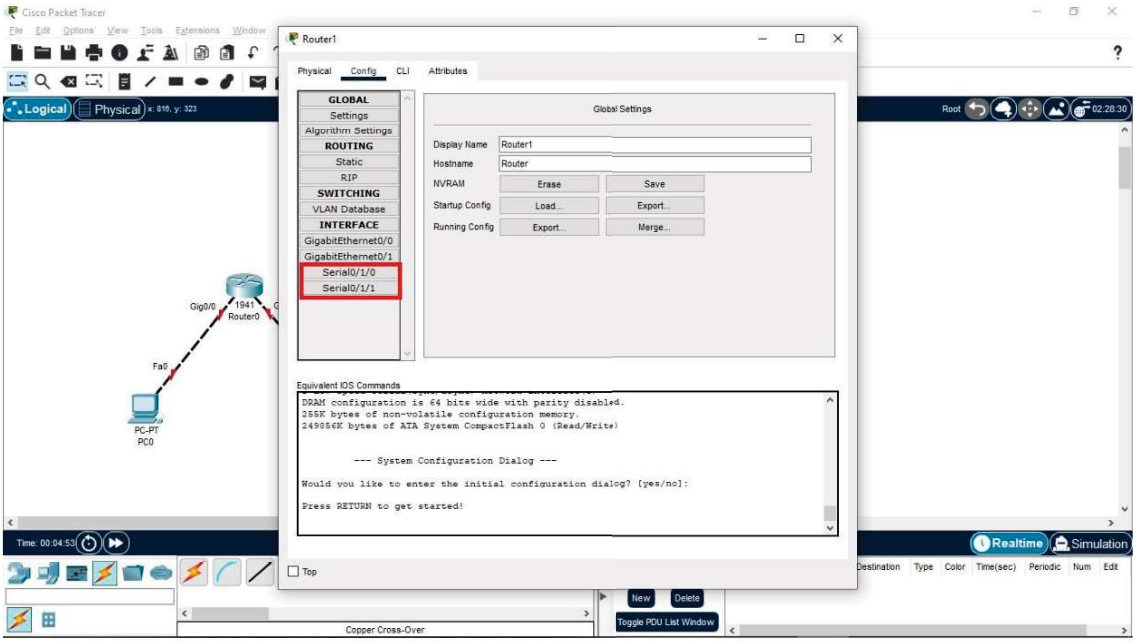
Step 3: Find the console from lower right corner. Drag and drop the console in the empty area as shown in the figure.



Step 4: Now again switch on the hardware and check in config tab for 2 serial ports added.

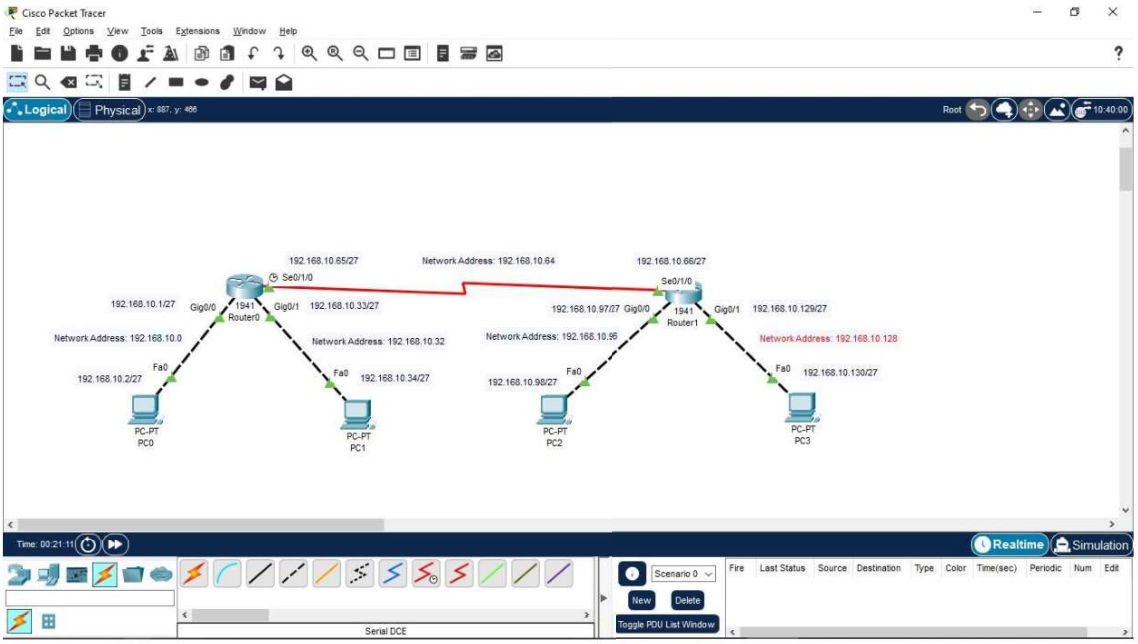


Step 5: Repeat the same procedure (Step 3 and Step 4) for Router1



Step 6: Now the PCs are physically connected through Router. To establish logical connectivity, assign IP addresses for 4 PCs (each 1 interface and corresponding router interface as gateway) and 2 Routers (each 3 ip addresses for 3 interfaces) as shown in the following table.

Device	Interface	IP Address	Subnet Mask	Gateway
PC0	Fa0/0	192.168.10.2	255.255.255.224	192.168.10.1
PC1	Fa0/0	192.168.10.34	255.255.255.224	192.168.10.33
PC2	Fa0/0	192.168.10.98	255.255.255.224	192.168.10.97
PC3	Fa0/0	192.168.10.130	255.255.255.224	192.168.10.129
Router0	Gigabit 0/0	192.168.10.1	255.255.255.224	-
Router0	Gigabit 0/1	192.168.10.33	255.255.255.224	-
Router0	Se0/1/0	192.168.10.65	255.255.255.224	-
Router1	Gigabit 0/0	192.168.10.97	255.255.255.224	-
Router1	Gigabit 0/1	192.168.10.129	255.255.255.224	-
Router1	Se0/1/0	192.168.10.66	255.255.255.224	-



Scenario with Network Address for each link

Step 7:

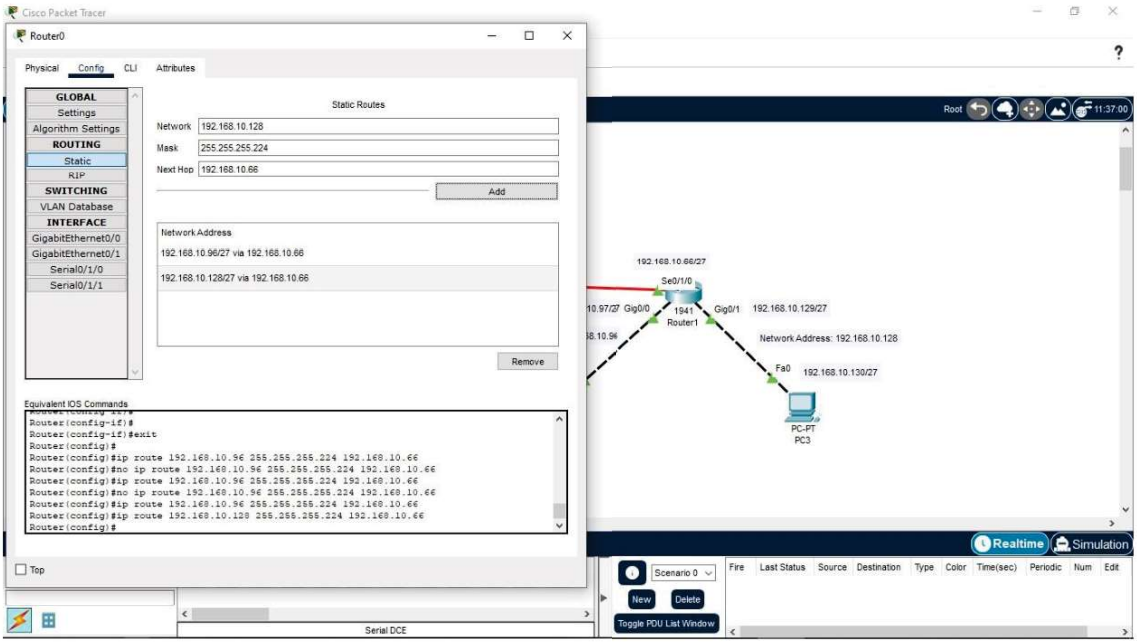
- **To enable packet transmission among the devices in the scenario, Static Routing has to be configured.**
- **To configure static routing, unknown networks and next-hop IP address to reach the unknown network for Router0 and Router1 has to be determined.**
- **Note: While specifying devices we should use IP-Address and while specifying network we should use Network Address.**
- **Unknown networks for the routers are derived in the following table**

Device	Known Networks	Subnet Mask	Unknown Networks	Subnet Mask	Next-hop Address
Router0	192.168.10.0	255.255.255.224	192.168.10.96	255.255.255.224	192.168.10.66
Router0	192.168.10.32	255.255.255.224	192.168.10.128	255.255.255.224	192.168.10.66
Router0	192.168.10.64	255.255.255.224	-	-	-
Router1	192.168.10.96	255.255.255.224	192.168.10.0	255.255.255.224	192.168.10.65
Router1	192.168.10.128	255.255.255.224	192.168.10.32	255.255.255.224	192.168.10.65
Router1	192.168.10.64	255.255.255.224	-	-	-

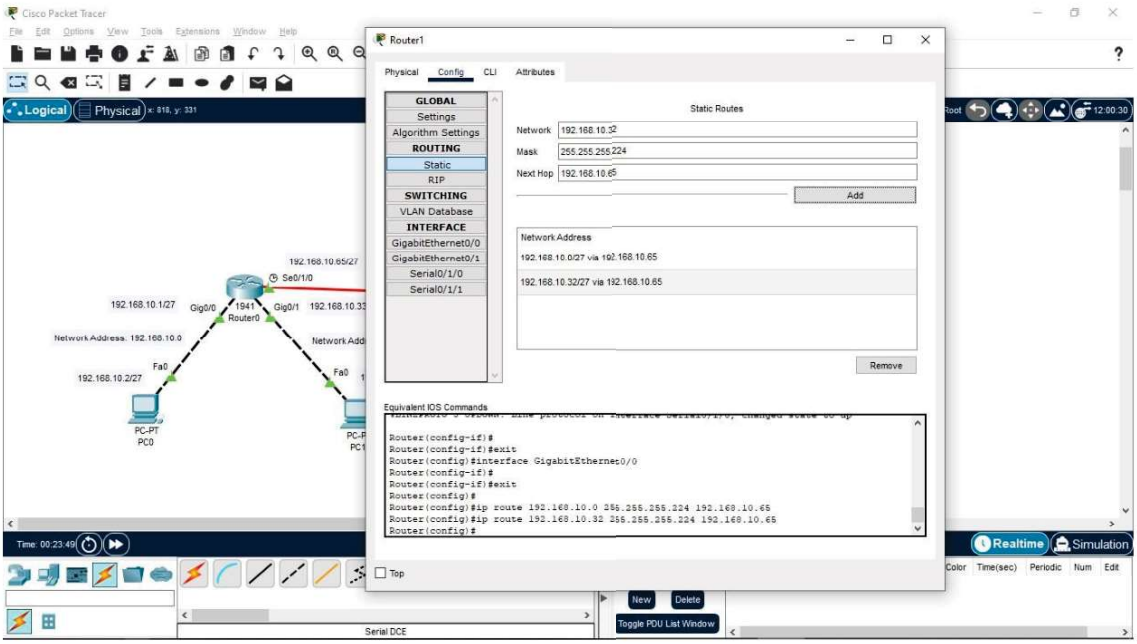
Only unknown networks should be configured for static routing

Step 8:

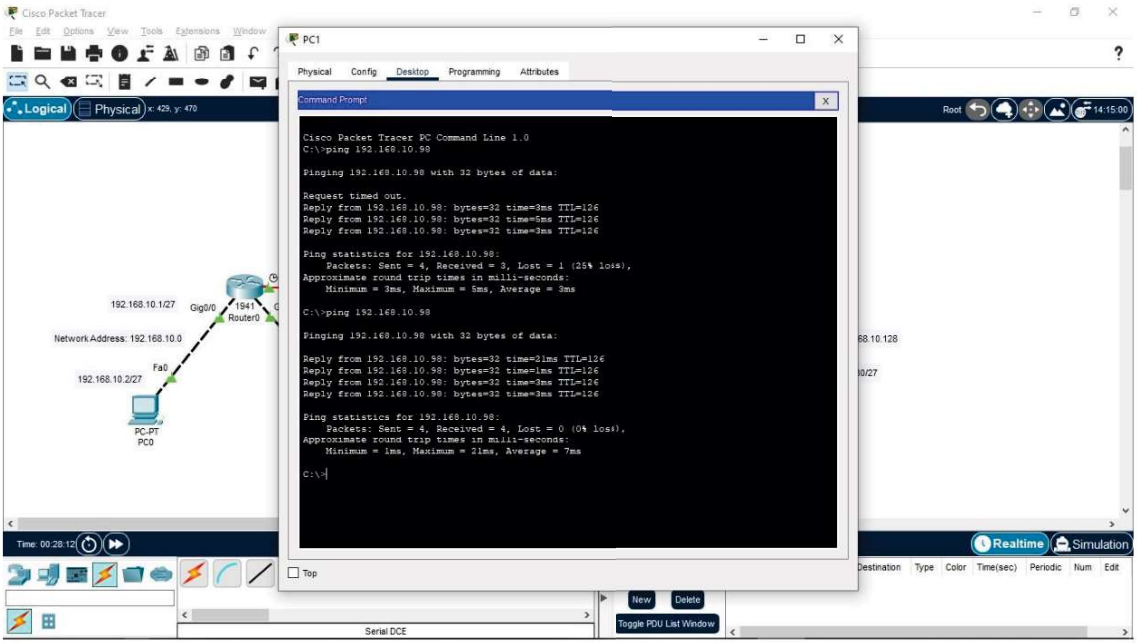
- To configure static routing, click on
    - Router0 => Config tab => Static =>Enter network address (refer unknown network for router0) => Subnet Mask for network address => Next-hop address => Add network
- Repeat the same to add the next network.
- Two networks should be added for the given scenario



Step 9: Repeat the same procedure to configure for Router1



Step 10: After configuration of Static routing in both Routers, Check the connectivity among any two devices using Ping Command



Step 11:

- To check routing table, go to CLI tab in Router and press enter to get the router prompt.
    - Router>
- Now type enable or en and press enter
    - Router>en
    - Router#

Follow the command "show ip route" to get the routing table of a router

```
Router>en
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

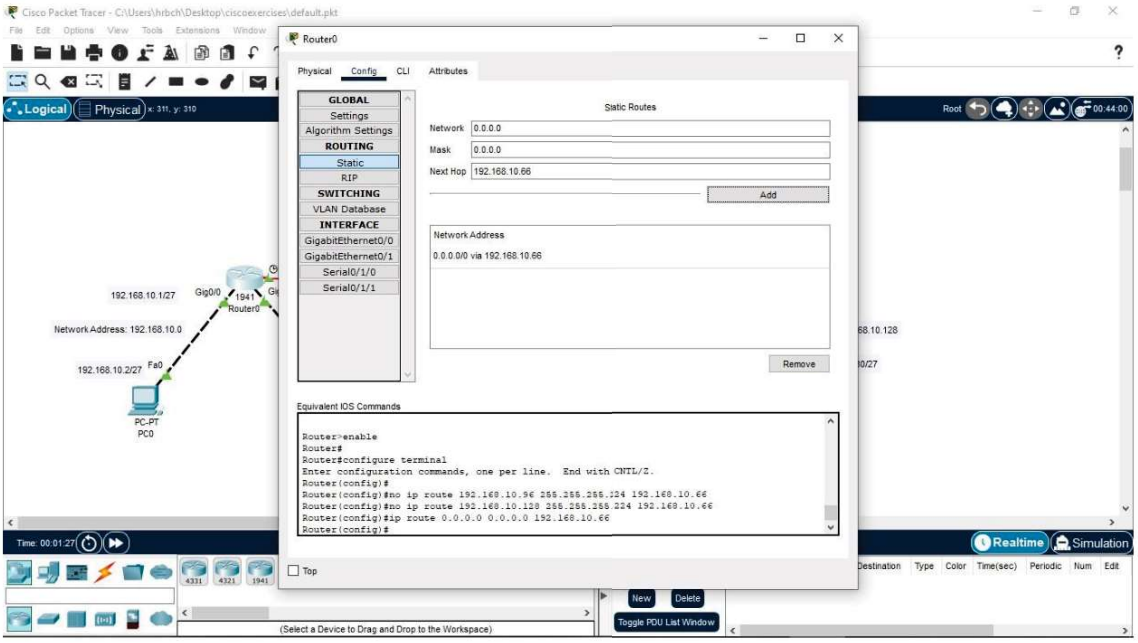
Gateway of last resort is not set

192.168.10.0/24 is variably subnetted, 8 subnets, 2 masks
C 192.168.10.0/27 is directly connected, GigabitEthernet0/0
L 192.168.10.1/32 is directly connected, GigabitEthernet0/0
C 192.168.10.32/27 is directly connected, GigabitEthernet0/1
L 192.168.10.33/32 is directly connected, GigabitEthernet0/1
C 192.168.10.64/27 is directly connected, Serial0/1/0
L 192.168.10.65/32 is directly connected, Serial0/1/0
S 192.168.10.96/27 [1/0] via 192.168.10.66
S 192.168.10.128/27 [1/0] via 192.168.10.66
```

5 b: For Default Routing replace Step 8 and Step 9 with Step 12 and Step 13

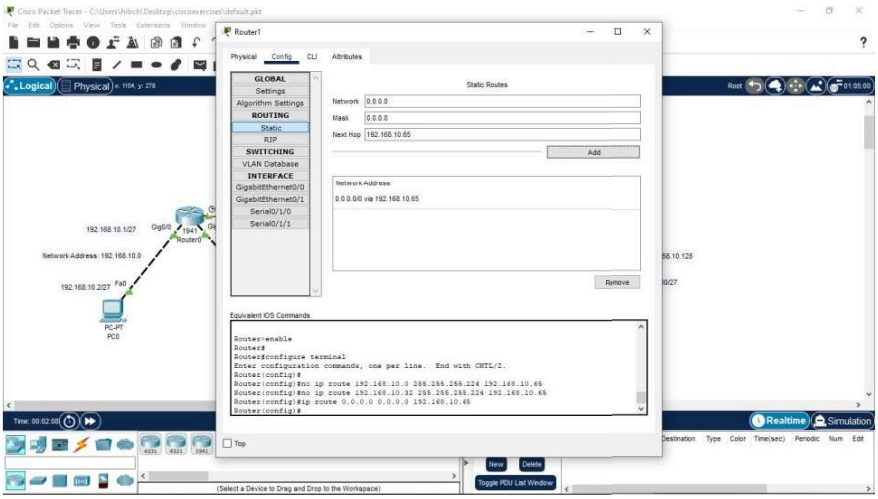
Step 12:

- To configure static routing, click on
    - Router0 => Config tab => Static =>Enter network address (as 0.0.0.0) => Subnet Mask (as 0.0.0.0) => Next-hop address (192.168.10.66) => Add network
- One entry should be added for the given scenario

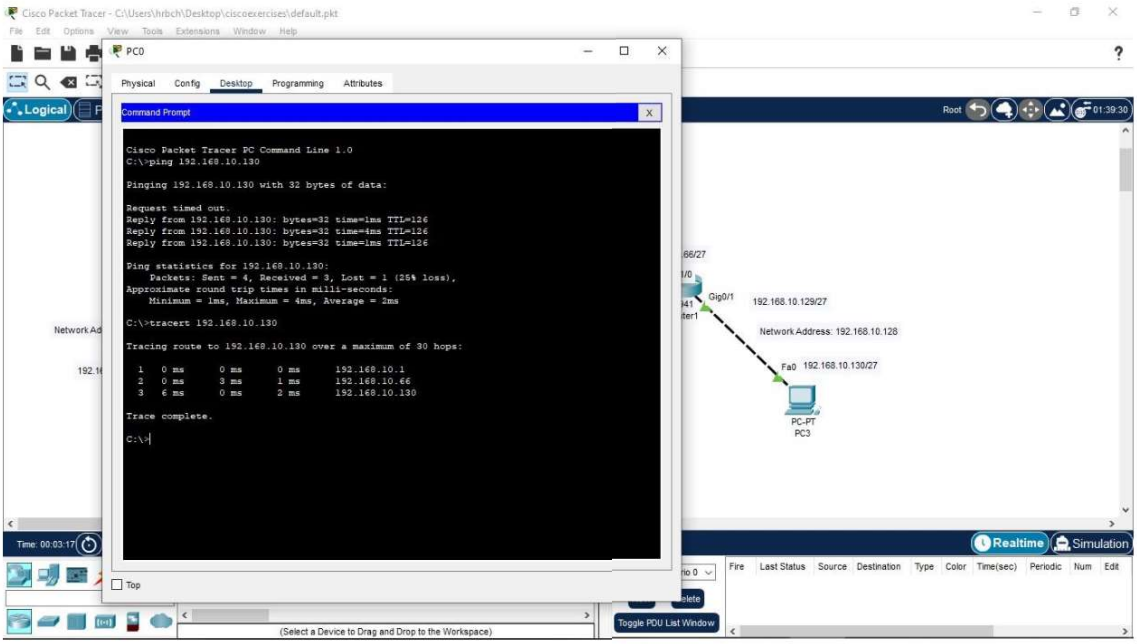


Step 13: Repeat the same procedure to configure for Router1.

Router1 => Config tab => Static =>Enter network address (as 0.0.0.0) => Subnet Mask (as 0.0.0.0) => Next-hop address (192.168.10.65) => Add network



Output for Default Routing: Click on PC0 => Desktop => Command Prompt and check **"tracert 192.168.10.130"** command



EXERCISE-6: NAT CONFIGURATION

AIM:

To configure the Network Address Translation on Cisco Packet Tracer.

Ex. 6A: Static NAT CONFIGURATION

Components:

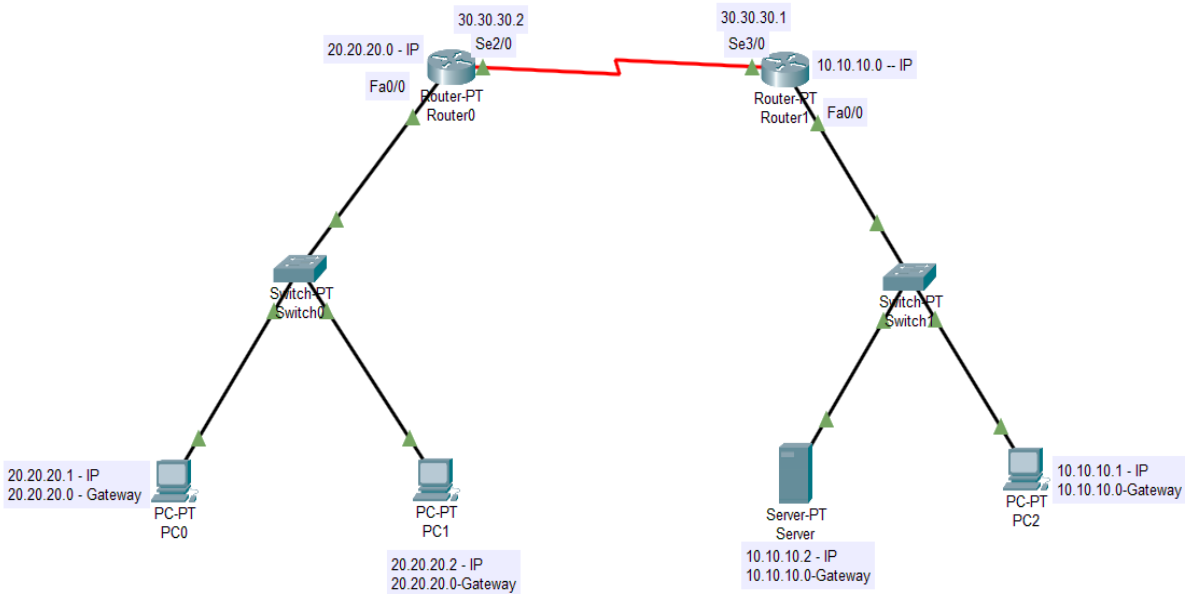
Components/Devices	Required No's
PC's	3
Server	1
Copper Straight through cable	6
Serial Cable (DCE)	1
Routers	2
Switches	2

Addressing Table:

Device	Interface	IP address	Subnet Mask	Gateway
PC0	Fa0/0	20.20.20.1	255.0.0.0	20.20.20.0
PC1	Fa0/0	20.20.20.2	255.0.0.0	20.20.20.0
Router 0	FastEthernet0/0	20.20.20.0	255.0.0.0	
PC2	Fa0/0	10.10.10.1	255.0.0.0	10.10.10.0
Server-PT	Fa0/0	10.10.10.2	255.0.0.0	10.10.10.0
Router 1	FastEthernet0/0	10.10.10.0	255.0.0.0	

Procedure:

Step 1: Drag and drop the required components and connect them using the appropriate cables as shown in the figure.



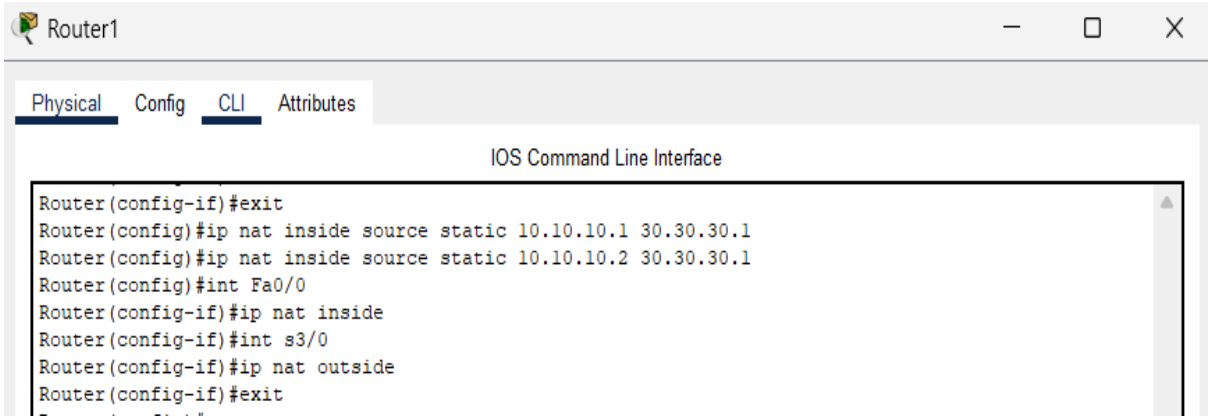
Step 2: Assign the IP address, subnet mask and Gateway for all the PC's and Server as per the addressing table.

Step 3: Connect the two routers using the serial DCE cable.

Step 4: Open Router1 and click on "config" tab and go to FastEthernet0/0 to assign the IP address and subnet mask, Serial port Se3/0 addresses.

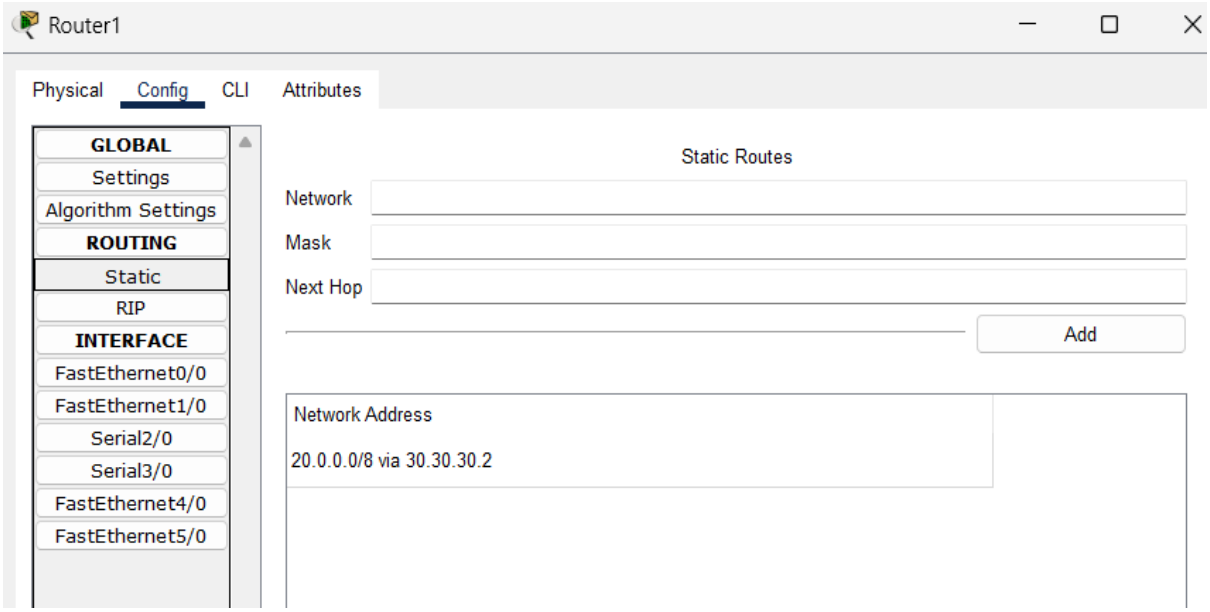
Step 5: Repeat the same procedure for Router0 as well.

Step 6: Click on Router1, go to CLI tab and type the following commands to configure the NAT (static).

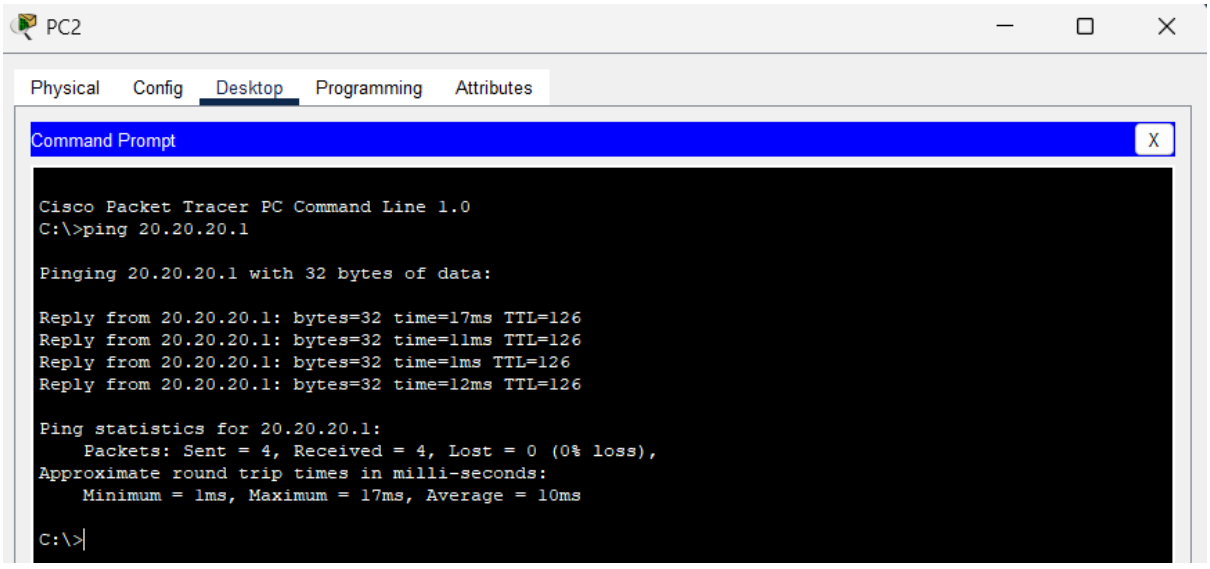


Step 7: In Router1, go to config tab and click the Static menu and type the following

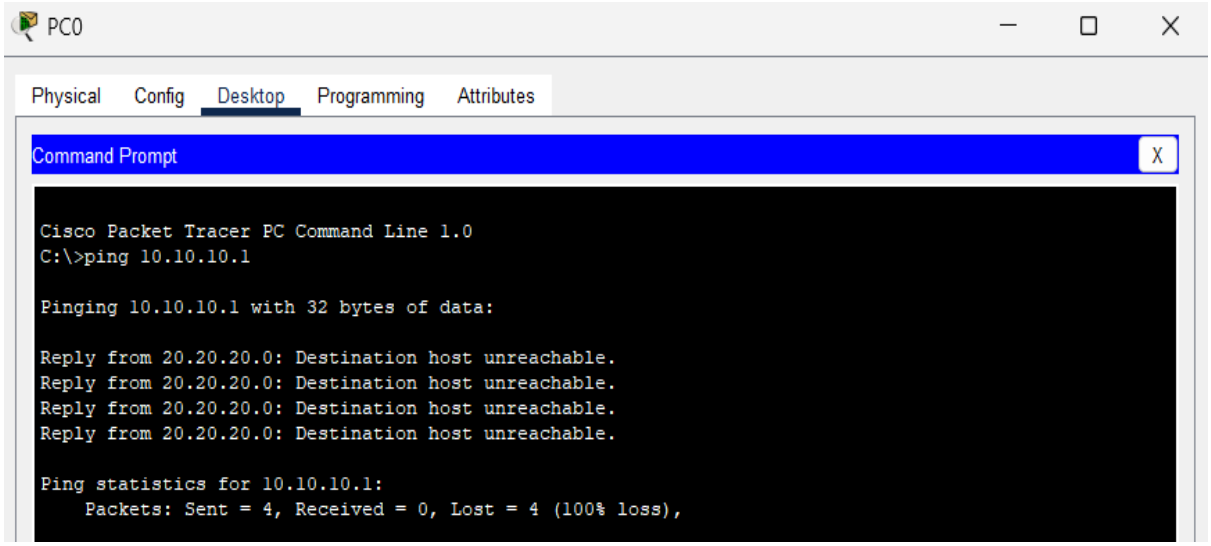
IP address: 20.0.0.0 → Subnet mask: 255.0.0.0 → Next hop: 30.30.30.2 → Add



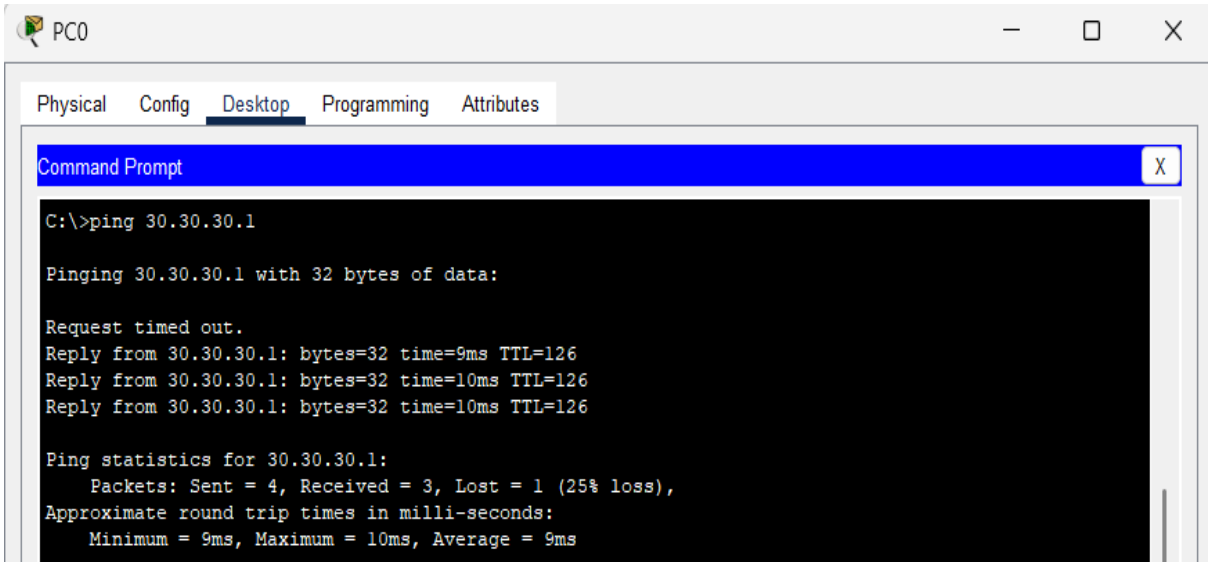
Step 8: After the static configuration of NAT routing, check the connectivity between the Private (PC2 or Server) and public network (PC0 or PC1) devices using ping command.



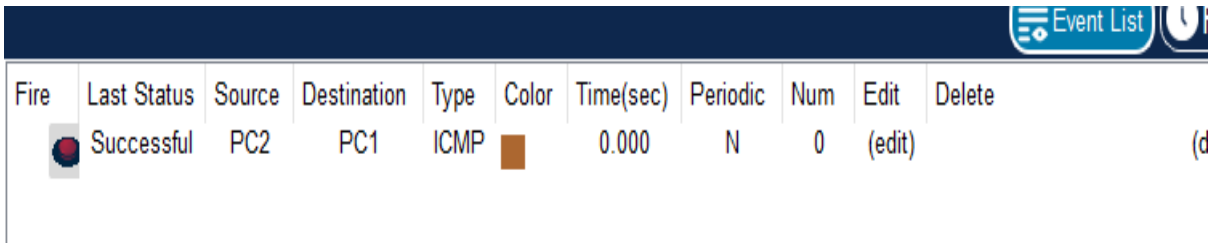
Public network device could not communicate to any private network.



Instead, Public network device can communicate to the NAT router.



Step 9: check the connectivity between the Private (PC2 or Server) and public network (PC0 or PC1) devices using packet simulation method.



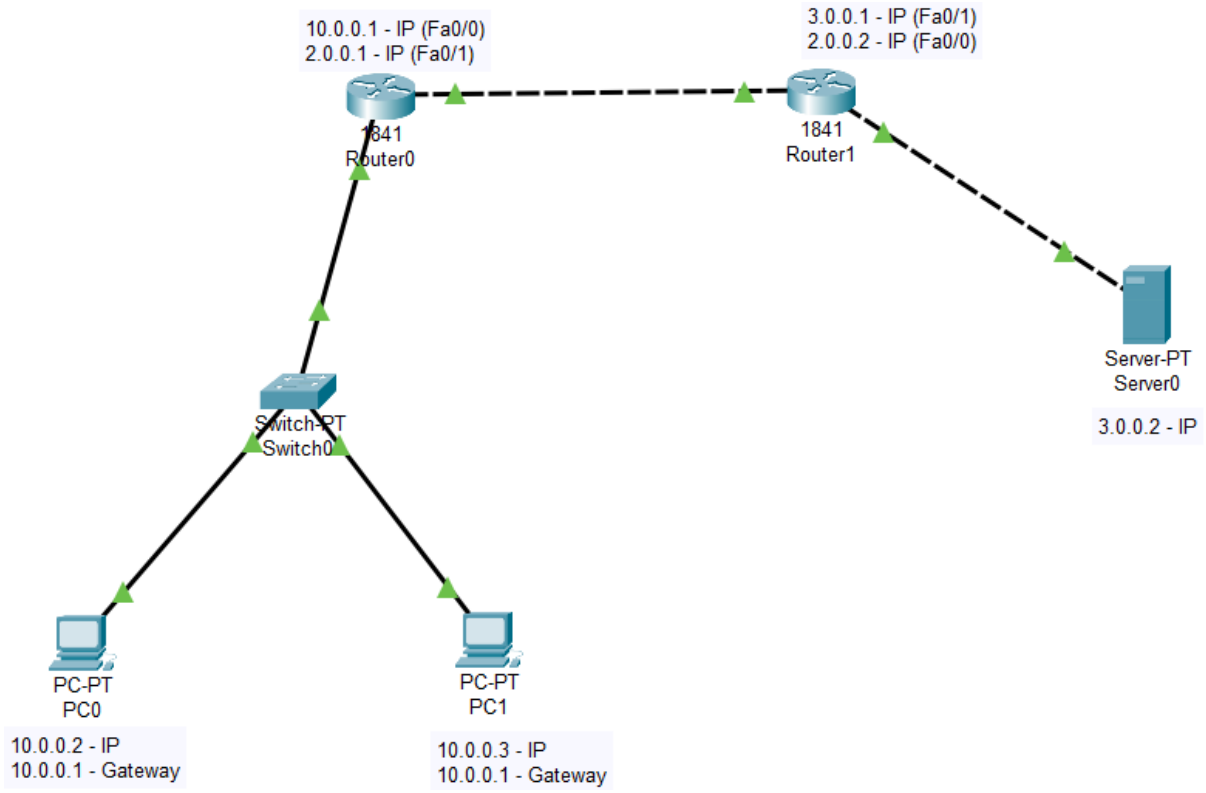
Step 10: To view the NAT table, click on the inspect icon and go to the Router1 and select the NAT table.



NAT Table for Router1

Protocol	Inside Global	Inside Local	Outside Local	Outside Global
icmp	30.30.30.1:14	10.10.10.1:14	20.20.20.2:14	20.20.20.2:14
---	30.30.30.1	10.10.10.1	---	---

6B: Dynamic NAT CONFIGURATION



Components:

Components/Devices	Required No's
PC's	2
Server	1
Copper Straight through cable	3
Serial Cable (DCE)	1
Routers	2
Switches	2

Addressing Table:

Device	Interface	IP address	Subnet Mask	Gateway
PC0	Fa0/0	10.0.0.2	255.0.0.0	10.0.0.1
PC1	Fa0/0	10.0.0.3	255.0.0.0	10.0.0.1
Router 0	FastEthernet0/0	10.0.0.1	255.0.0.0	
Router 0	FastEthernet0/1	2.0.0.1	255.0.0.0	
Server-PT	Fa0/0	3.0.0.2	255.0.0.0	
Router 1	FastEthernet0/0	2.0.0.1	255.0.0.0	
Router 1	FastEthernet0/1	3.0.0.1	255.0.0.0	

Procedure:

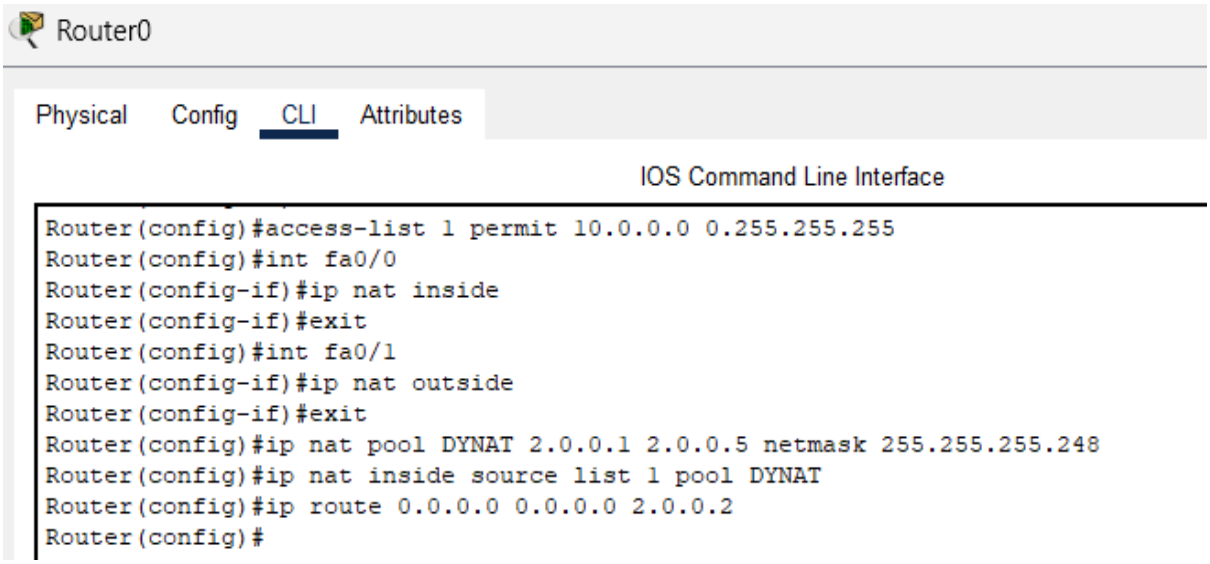
Step 1: Draw the network topology Diagram using cisco packet tracer software. There are two routers.

Router R0(Home Router)

Router R1(ISP Router)

Step 2: Assign the IP Address on HOME ROUTER R0 and ISP ROUTER R1 as per the addressing table.

Step 3: Dynamic NAT configuration on HOME ROUTER R0 and assign the address pool.



Step 4: Route Configuration on Home Router and ISP Router.

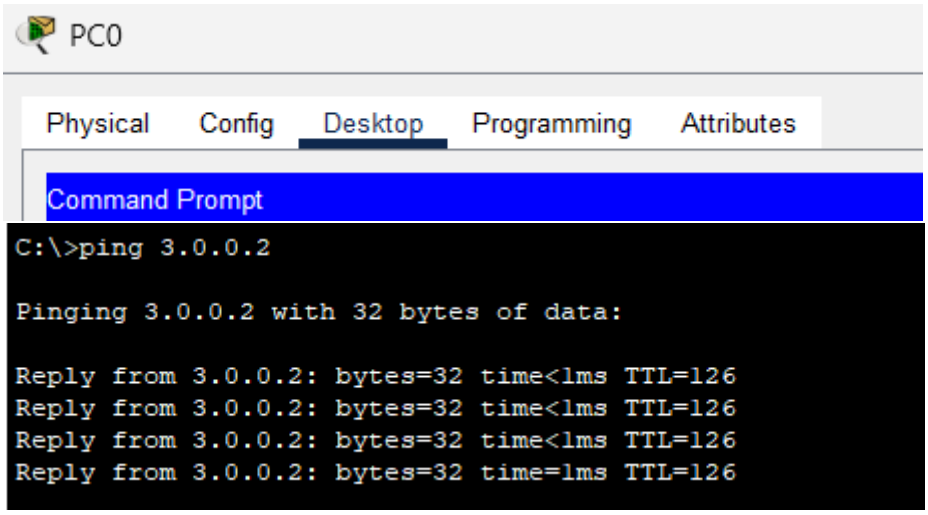
First Do routing configuration on R0 Router:

R0(config)#ip route 0.0.0.0 0.0.0.0 2.0.0.2

Then Configure Routing on Router R1

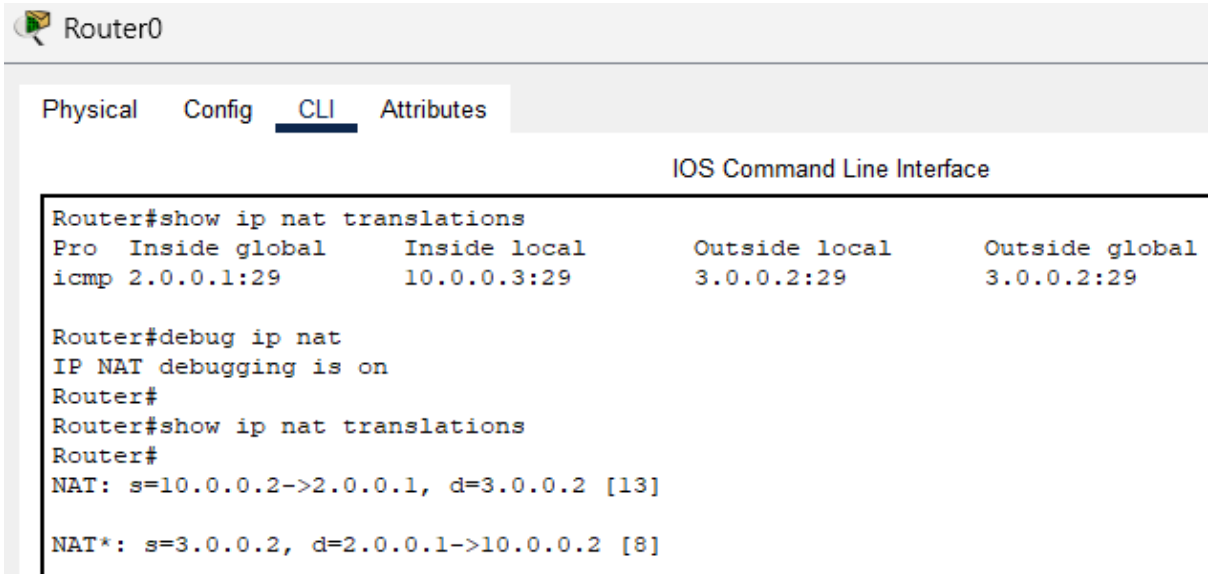
R1(config)#ip route 0.0.0.0 0.0.0.0 2.0.0.1

Step 5: Now Check NAT configuration using ping command



Just when Ping Run type the below command.

This command will help to see the translation process in Router0.



Result:

The implementation of NAT configuration is done successfully using Cisco Packet tracer.

EXERCISE 7. A: DEMONSTRATION OF RIP V1

[Note: RIP v1 does not support classless addressing mode and RIP v2 must be used in Subnet enabled Scenario]

Objective: To demonstrate the configuration of IP Addressing with Subnetting in WAN Configuration

Pre-requisite: IP Address, Range of IP Address, Classes of IP Address, Subnetting

Components:

Devices	Required Nos
PCs	4
Copper cross-over Cables	4
Routers [1941]	2
Serial DCE	1

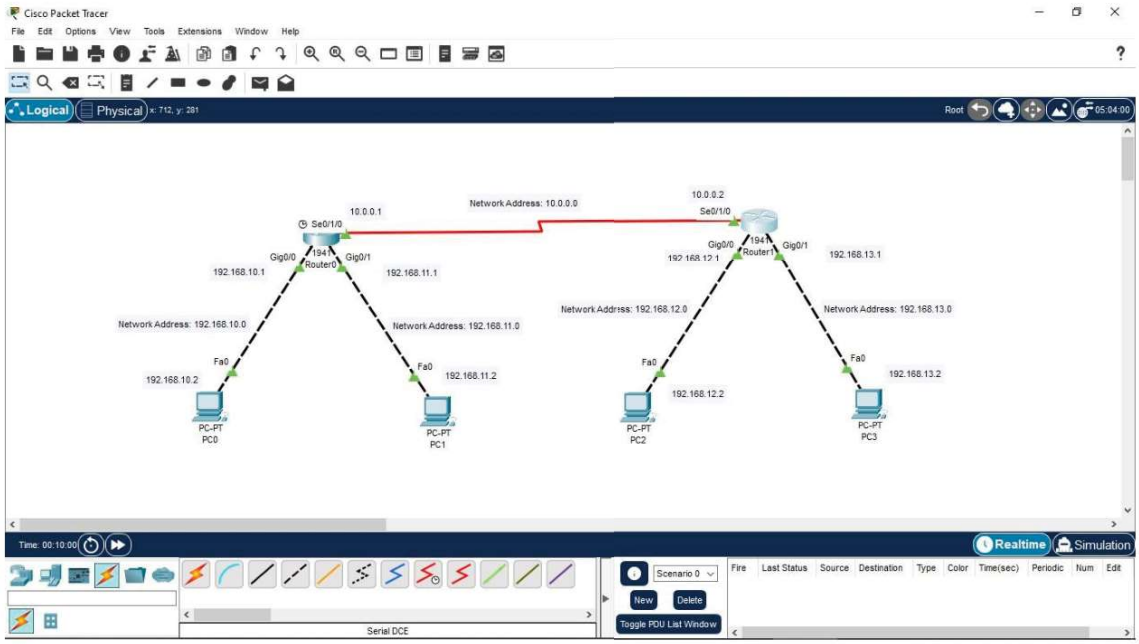
Addressing Table:

Device	Interface	IP Address	Subnet Mask	Gateway
PC0	Fa0/0	192.168.10.2	255.255.255.0	192.168.10.1
PC1	Fa0/0	192.168.11.2	255.255.255.0	192.168.11.1
PC2	Fa0/0	192.168.12.2	255.255.255.0	192.168.12.1
PC3	Fa0/0	192.168.13.2	255.255.255.0	192.168.13.1
Router0	Gigabit 0/0	192.168.10.1	255.255.255.0	-
Router0	Gigabit 0/1	192.168.11.1	255.255.255.0	-
Router0	Se0/1/0	10.0.0.1	255.0.0.0	-
Router1	Gigabit 0/0	192.168.12.1	255.255.255.0	-
Router1	Gigabit 0/1	192.168.13.1	255.255.255.0	-
Router1	Se0/1/0	10.0.0.2	255.0.0.0	-

Procedure:

Step 1:

- Drag 4 PCs and 2 routers in the console area as shown in the figure.
- Follow the procedure for connecting **Serial DCE cable from Exercise 4.**
- Assign IP addresses for 4 PCs (each 1 interface and corresponding router interface as gateway) and 2 Routers (each 3 ip addresses for 3 interfaces) as shown in the Addressing Table



Step 2:

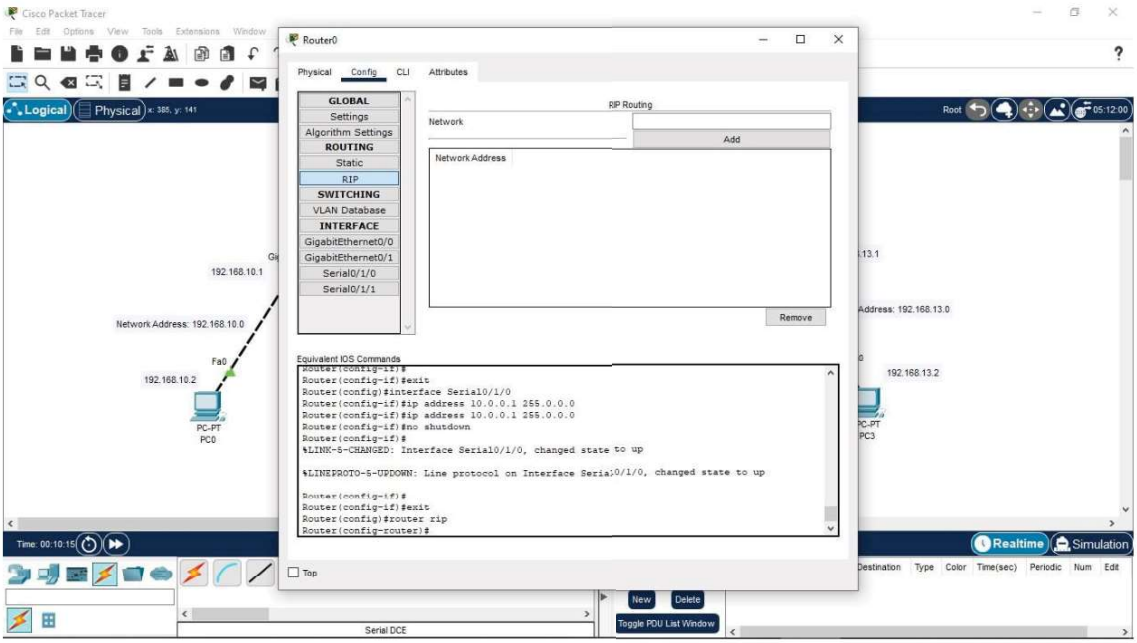
- To enable packet transmission among the devices in the scenario, RIP Routing has to be configured.
- To configure RIP routing, **known networks** for Router0 and Router1 has to be determined.
- Note: While specifying devices we should use IP-Address and while specifying network we should use Network Address.
- The **known networks** for the routers are derived in the following table

Device	Known Networks	Subnet Mask
Router0	192.168.10.0	255.255.255.0
Router0	192.168.11.0	255.255.255.0
Router0	10.0.0.0	255.0.0.0
Router1	192.168.12.0	255.255.255.0
Router1	192.168.13.0	255.255.255.0
Router1	10.0.0.0	255.0.0.0

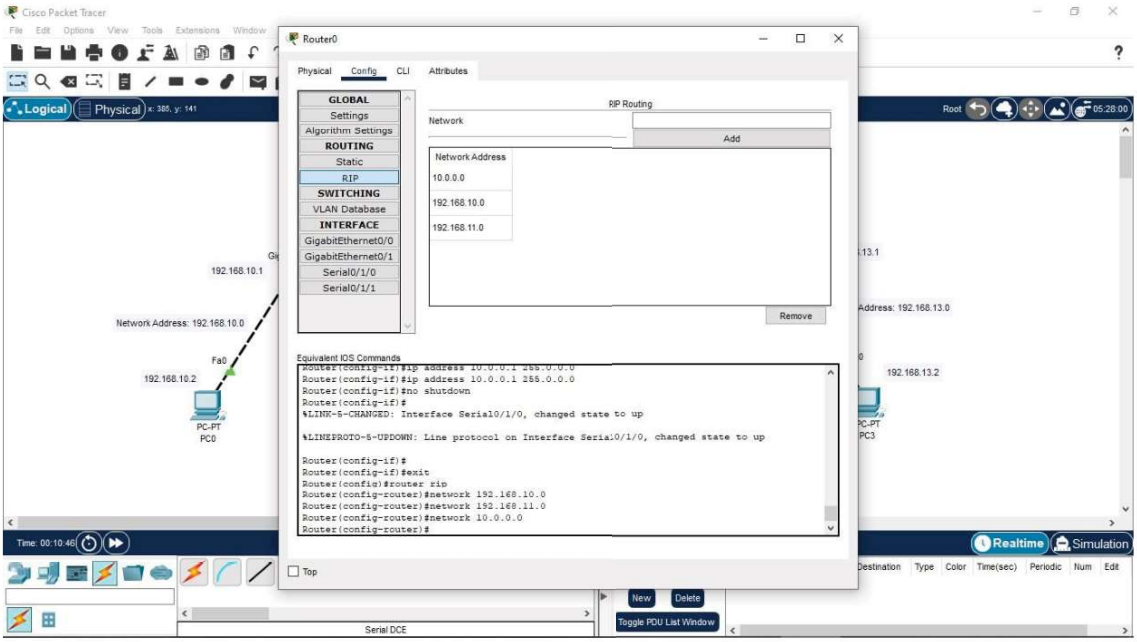
Only known networks should be configured for RIP routing

Step 3:

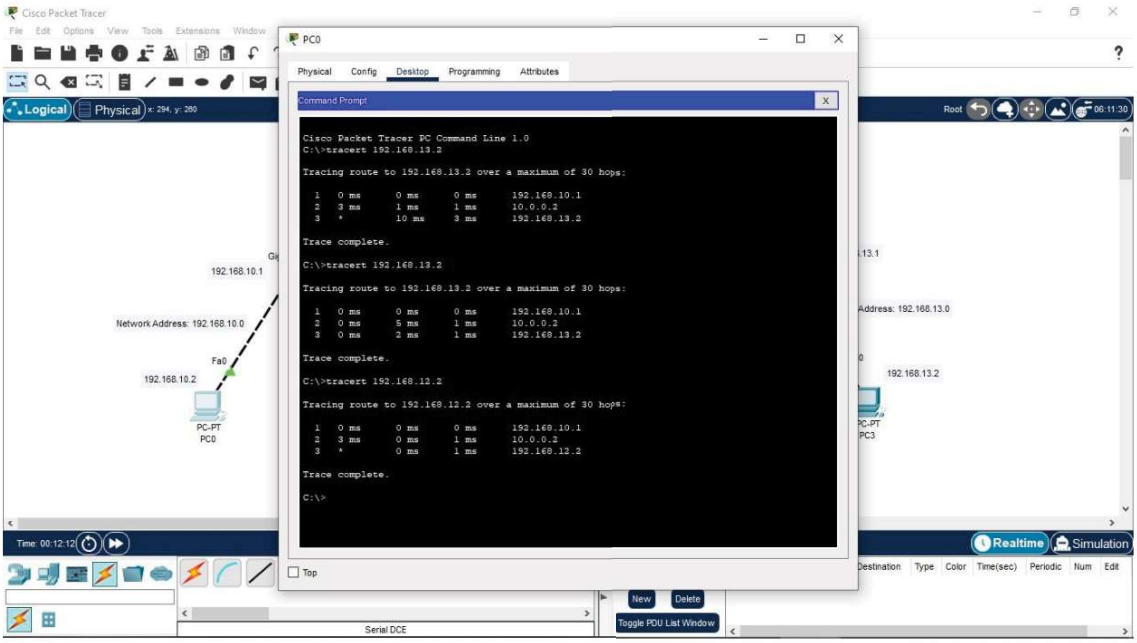
- To configure RIP routing, click on
  - Router0 => Config tab => RIP =>Enter network address (refer table) => Add network
- Repeat the same to add the next network.
- Three networks should be added for the given scenario



Step 4: Repeat the same procedure to configure for Router1

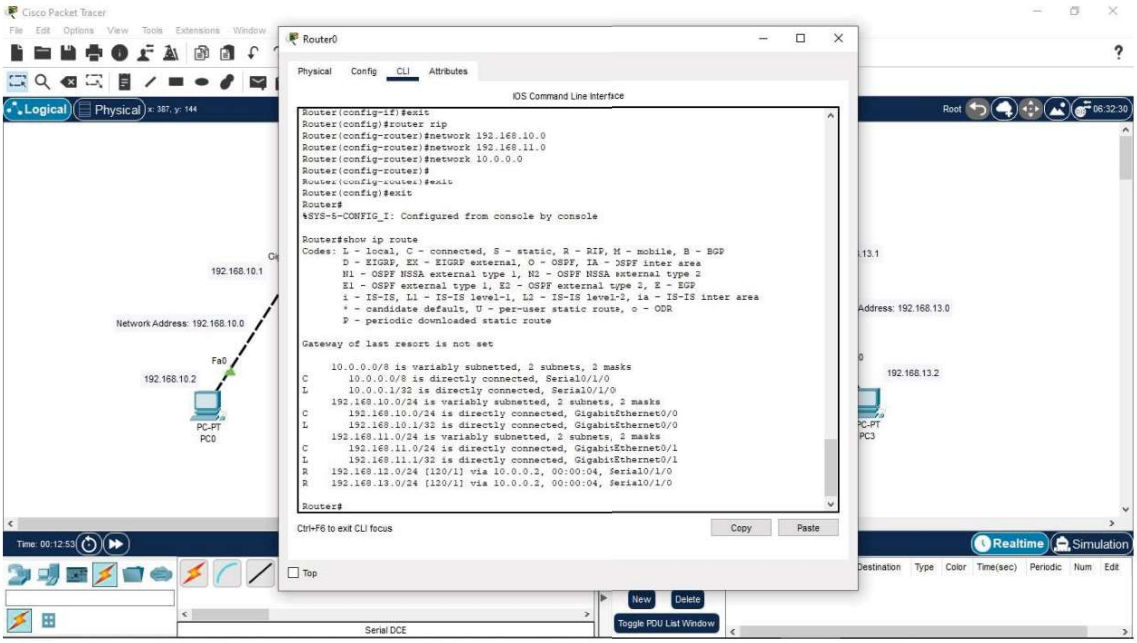


Step 5: After configuration of RIP routing in both Routers, Check the connectivity among any two devices using Ping Command or tracertcommand



Step 6:

- To check routing table, go to CLI tab in Router and press enter to get the router prompt.
    - Router>
- Now type enable or en and press enter
    - Router>en
    - Router# show ip route



EXERCISE 7. B: DEMONSTRATION OF RIP V2

Objective: To demonstrate the configuration of RIP v2

Pre-requisite: IP Address, Range of IP Address, Classes of IP Address, Subnetting

Components:

Devices	Required Nos
PCs	4
Copper cross-over Cables	4
Routers	2
Serial DCE	1

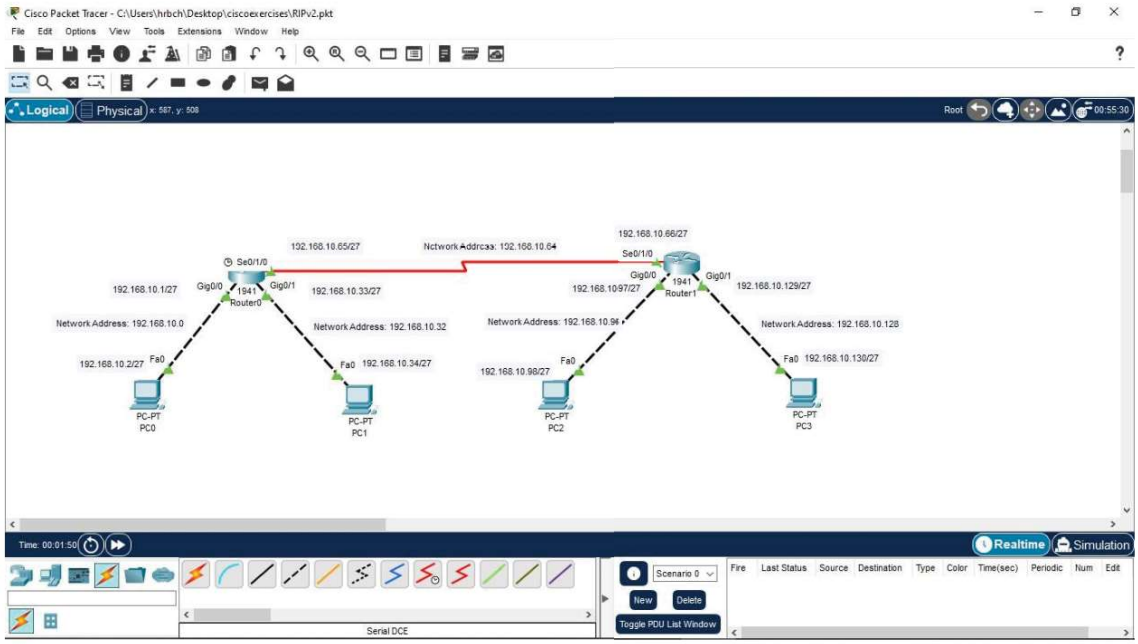
Addressing Table:

Device	Interface	IP Address	Subnet Mask	Gateway
PC0	Fa0/0	192.168.10.2	255.255.255.224	192.168.10.1
PC1	Fa0/0	192.168.10.34	255.255.255.224	192.168.10.33
PC2	Fa0/0	192.168.10.98	255.255.255.224	192.168.10.97
PC3	Fa0/0	192.168.10.130	255.255.255.224	192.168.10.129
Router0	Gigabit 0/0	192.168.10.1	255.255.255.224	-
Router0	Gigabit 0/1	192.168.10.33	255.255.255.224	-
Router0	Se0/1/0	192.168.10.65	255.255.255.224	-
Router1	Gigabit 0/0	192.168.10.97	255.255.255.224	-
Router1	Gigabit 0/1	192.168.10.129	255.255.255.224	-
Router1	Se0/1/0	192.168.10.66	255.255.255.224	-

Procedure:

Step 1:

- Drag 4 PCs and 2 routers in the console area as shown in the figure.
- Follow the procedure for connecting **Serial DCE cable from Exercise 4.**
- Assign IP addresses for 4 PCs (each 1 interface and corresponding router interface as gateway) and 2 Routers (each 3 ip addresses for 3 interfaces) as shown in the Addressing Table



Step 2:

- To enable packet transmission among the devices in the scenario, RIP v2 Routing has to be configured.
- To configure RIP v2 routing, **known networks** for Router0 and Router1 has to be determined.
- Note: While specifying devices we should use IP-Address and while specifying network we should use Network Address.
- The **known networks** for the routers are derived in the following table

Device	Known Networks	Subnet Mask	Unknown Networks	Subnet Mask	Next-hop Address
Router0	192.168.10.0	255.255.255.224	192.168.10.96	255.255.255.224	192.168.10.66
Router0	192.168.10.32	255.255.255.224	192.168.10.128	255.255.255.224	192.168.10.66
Router0	192.168.10.64	255.255.255.224	-	-	-
Router1	192.168.10.96	255.255.255.224	192.168.10.0	255.255.255.224	192.168.10.65
Router1	192.168.10.128	255.255.255.224	192.168.10.32	255.255.255.224	192.168.10.65
Router1	192.168.10.64	255.255.255.224	-	-	-

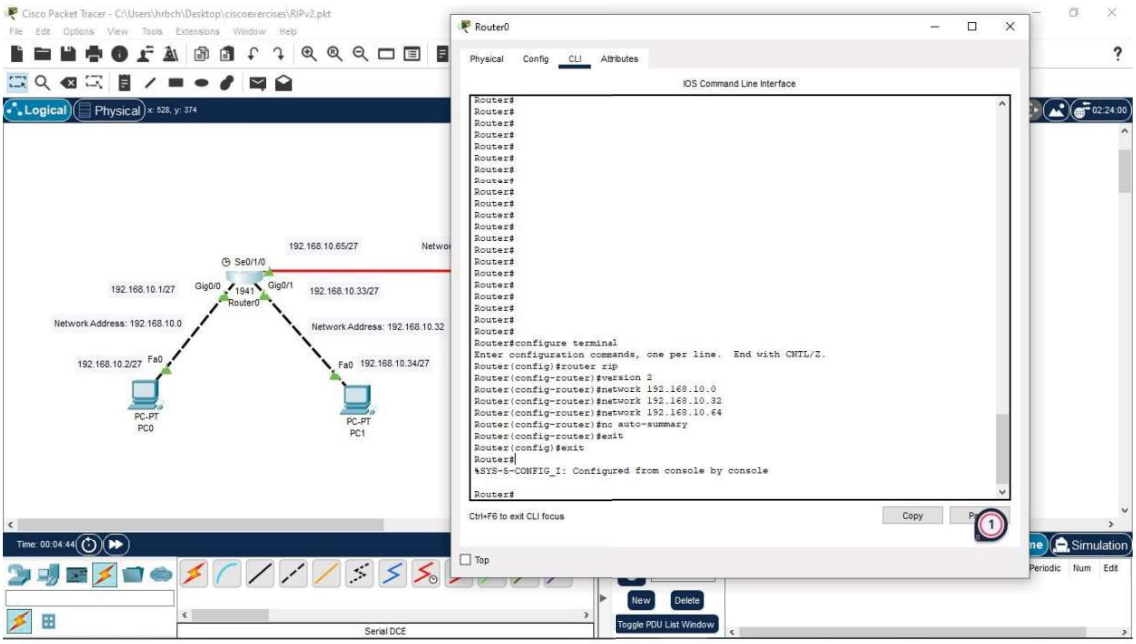
Only unknown networks should be configured for static routing

Step 3:

- To configure static routing, click on
  - Router0 => Config tab => CLI
- Type "enable" in "Router>" prompt and type the following commands to configure RIP v2 in Router0

```
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#network 192.168.10.0
Router(config-router)#network 192.168.10.32
Router(config-router)#network 192.168.10.64
Router(config-router)#no auto-summary
Router(config-router)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

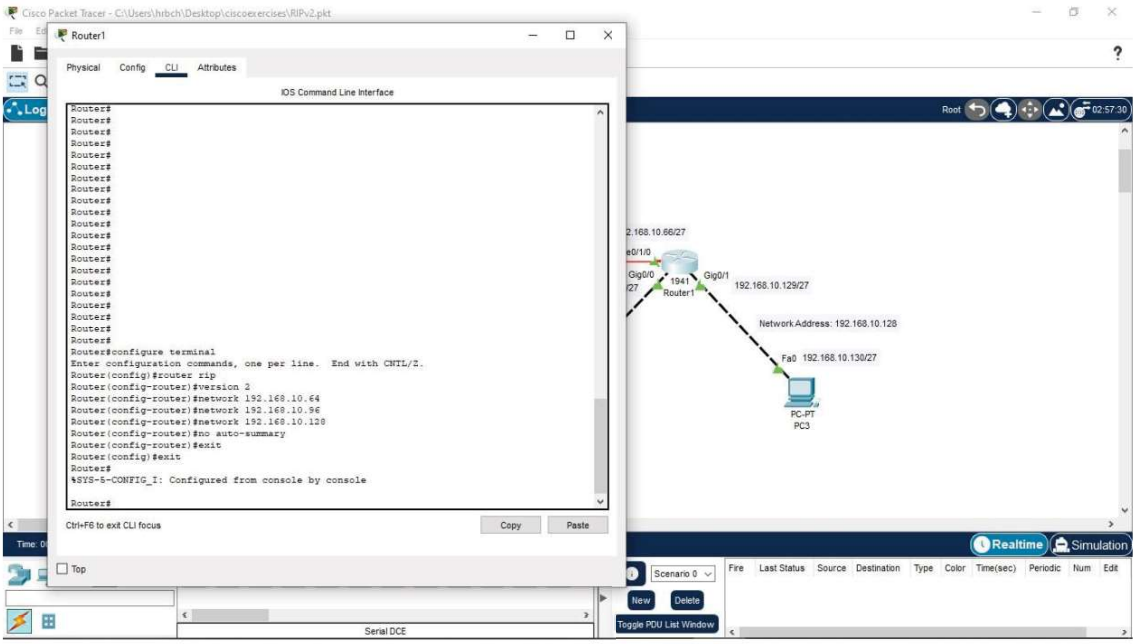
Router#
```



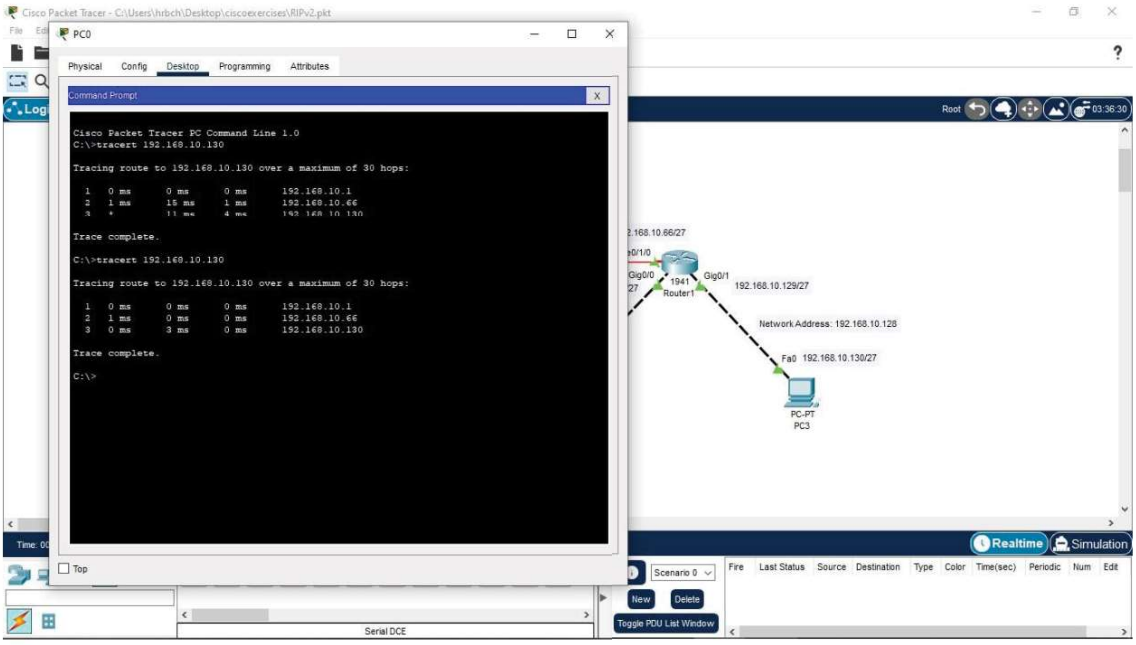
Step 4: Repeat the same procedure to configure for Router1

```
Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#network 192.168.10.64
Router(config-router)#network 192.168.10.96
Router(config-router)#network 192.168.10.128
Router(config-router)#no auto-summary
Router(config-router)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console


Router#
```



Step 5: After configuration of RIP v2 routing in both Routers, Check the connectivity among any two devices using Ping Command/ tracert command



Result:

The demonstration of RIP v1 & RIP v2 is done successfully in the packet tracer.

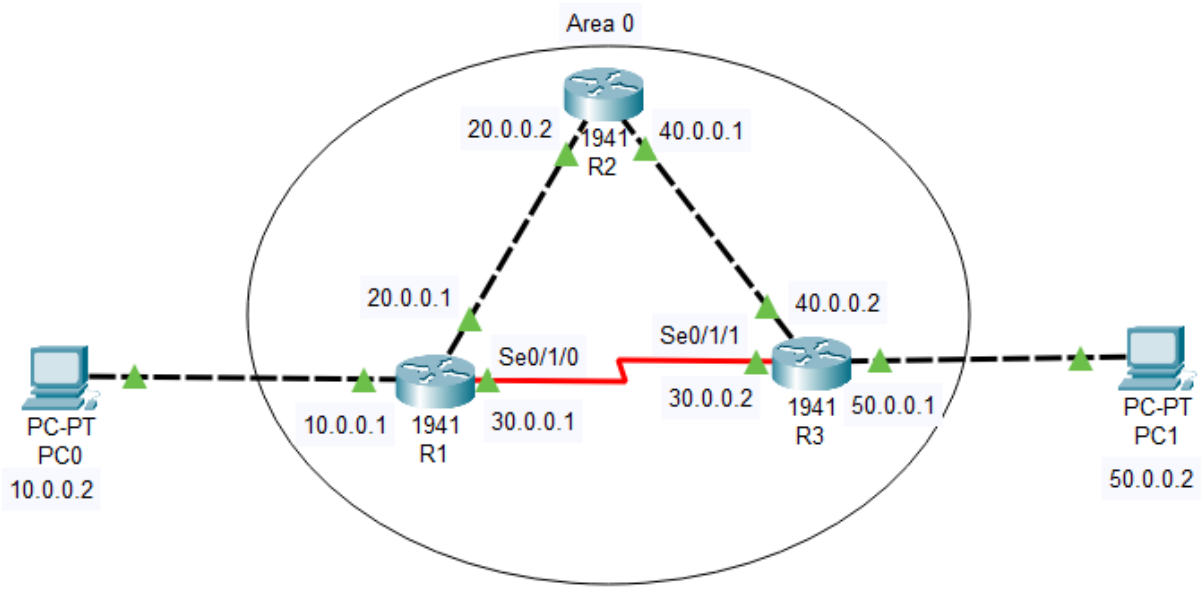
EXERCISE-8-A: CONFIGURATION OF SINGLE AREA OSPF

AIM:

To configure the single area OSPF protocol on Cisco Packet Tracer.

Components:

Components/Devices	Required No's
PC's	2
Copper crossover cable	4
Serial Cable (DCE)	1
Routers	3



Addressing Table:

Device	Interface	IP address	Subnet Mask	Gateway
PC0	Fa0/0	10.0.0.2	255.0.0.0	10.0.0.1
PC1	Fa0/0	50.0.0.2	255.0.0.0	50.0.0.1
Router 1	Gig0/0	10.0.0.1	255.0.0.0	--
Router 1	Gig0/1	20.0.0.1	255.0.0.0	--
Router 1	Se0/1/0	30.0.0.1	255.0.0.0	
Router 2	Gig0/0	20.0.0.2	255.0.0.0	
Router 2	Gig0/1	40.0.0.1	255.0.0.0	
Router 3	Gig0/0	50.0.0.1	255.0.0.0	

Router 3	Gig0/1	40.0.0.2	255.0.0.0	
Router 3	Se0/1/1	30.0.0.2	255.0.0.0	

Procedure:

Step 1: Drag and drop the required components and connect them using the appropriate cables as shown in the figure.

Step 2: Assign the IP address, subnet mask and Gateway for all the PC's and Routers as per the addressing table.

Step 3: Connect the two routers R1 and R3 using the serial DCE cable.

Step 4: Open Router1 and click on "config" tab and go to FastEthernet0/0 to assign the IP address and subnet mask, Serial port Se0/1/0 addresses.

Step 5: Repeat the same procedure for Router2 and Router3 as well.

Step 6: Click on Router1, go to CLI tab and type the following commands to configure the OSPF.

```
Router>enable
Router#configure terminal
Router(config)#router ospf 1
Router(config-router)#network 10.0.0.0 0.255.255.255 area 0
Router(config-router)#network 20.0.0.0 0.255.255.255 area 0
Router(config-router)#network 30.0.0.0 0.255.255.255 area 0
Router(config-router)#exit
Router(config)#exit
Router#
```

Step 7: In Router2, go to config tab and type the following commands to configure the OSPF

```
Router>enable
Router#configure terminal
Router(config)#router ospf 2
Router(config-router)#network 20.0.0.0 0.255.255.255 area 0
Router(config-router)#network 40.0.0.0 0.255.255.255 area 0
Router(config-router)#exit
Router(config)#exit
Router#
```

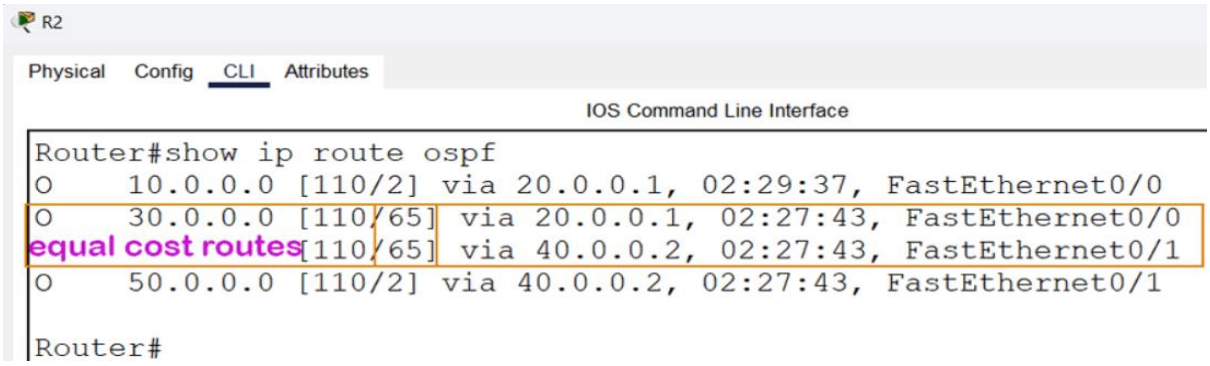
Step 8: In Router3, go to config tab and type the following commands to configure the OSPF

```
Router>enable
Router#configure terminal
Router(config)#router ospf 3
Router(config-router)#network 30.0.0.0 0.255.255.255 area 0
Router(config-router)#network 40.0.0.0 0.255.255.255 area 0
Router(config-router)#network 50.0.0.0 0.255.255.255 area 0
Router(config-router)#exit
Router(config)#exit
Router#
```

Step 9: OSPF shares routing information only with neighbors. Use the **show ip ospf neighbor** command on R1 to verify OSPF neighbors.

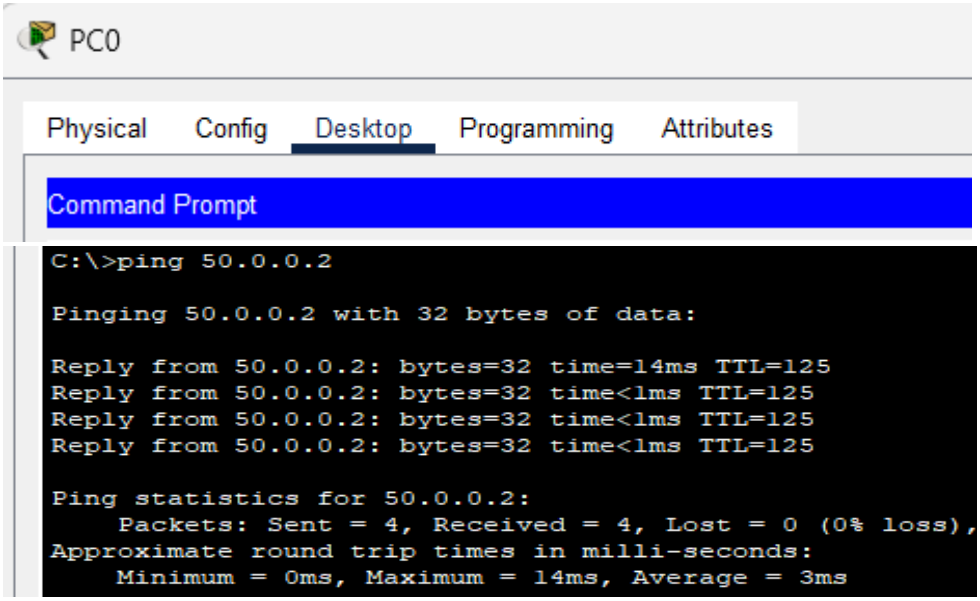


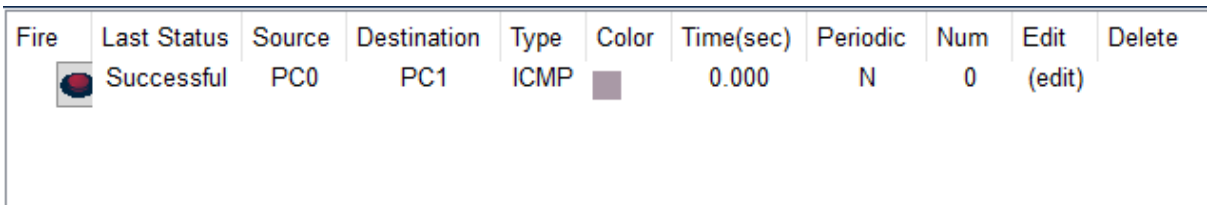
Step 10: The **show ip route ospf** command displays a list of all OSPF routes in the routing table. The following image shows the output of this command on R1.



OUTPUT

Verify the connectivity between End devices by using the ping command and packet simulation.





Result:

The implementation of single area OSPF configuration is done successfully using Cisco Packet tracer.

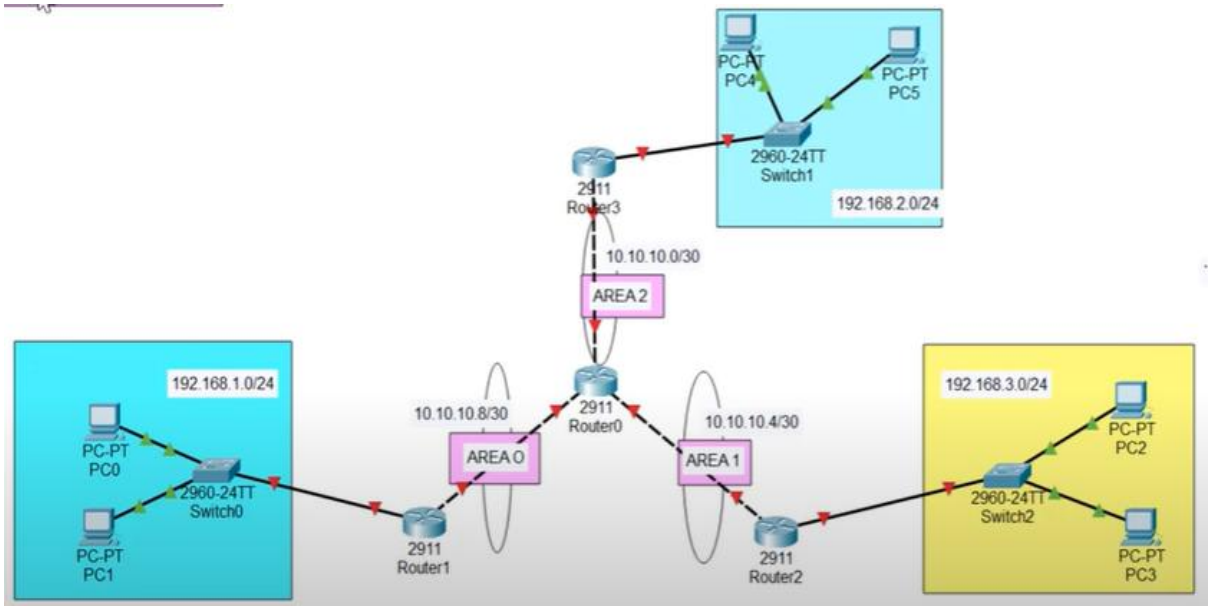
EXERCISE-8-B: CONFIGURATION OF MULTI-AREA OSPF

AIM:

To configure the multi-area OSPF protocol on Cisco Packet Tracer.

Components:

Components/Devices	Required No's
PC's	6
Copper crossover cable	3
Copper straight-through cable	9
Routers	4
Switches	3



Addressing Table:

Device	Interface	IP address	Subnet Mask	Gateway
PC0		192.168.1.2	255.255.255.0	192.168.1.1
PC1		192.168.1.3	255.255.255.0	192.168.1.1
Router 1	Gig0/0	192.168.1.1	255.255.255.0	--
Router 1	Gig0/1	10.10.10.8	255.0.0.0	--
PC2		192.168.3.2	255.255.255.0	192.168.3.1
PC3		192.168.3.3	255.255.255.0	192.168.3.1

Router 2	Gig0/0	192.168.3.1	255.255.255.0	
Router 2	Gig0/1	10.10.10.4	255.0.0.0	
PC4		192.168.2.2	255.255.255.0	192.168.2.1
PC5		192.168.2.3	255.255.255.0	192.168.2.1
Router 3	Gig0/0	192.168.2.1	255.255.255.0	
Router 3	Gig0/1	10.10.10.0	255.0.0.0	

Procedure:

Step 1: Drag and drop the required components and connect them using the appropriate cables as shown in the figure.

Step 2: Assign the IP address, subnet mask and Gateway for all the PC's and Routers as per the addressing table.

Step 3: In Router-1, go to CLI tab and type the following configuration commands

```
Router(config)#router ospf 15
Router(config-router)#rou
Router(config-router)#router-id 1.1.1.1
Router(config-router)#net
Router(config-router)#network 192.168.1.0 0.0.0.255 are
Router(config-router)#network 192.168.1.0 0.0.0.255 area 0
Router(config-router)#network 10.10.10.8 0.0.0.3 area 0
Router(config-router)#exit
Router(config)#
Router(config)#
Router(config)#
Router(config)#do wr
Building configuration...
```

In Router-2, go to CLI tab and type the following configuration commands

```
Router(config)#router ospf 15
Router(config-router)#rou
Router(config-router)#router-id 2.2.2.2
Router(config-router)#
Router(config-router)#
Router(config-router)#net
Router(config-router)#network 192.168.3.0 0.0.0.255 ar
Router(config-router)#network 192.168.3.0 0.0.0.255 area 1
Router(config-router)#network 10.10.10.4 0.0.0.3 area 1
Router(config-router)#ex
Router(config)#
Router(config)#
Router(config)#
Router(config)#do wr
```

In Router-2, go to CLI tab and type the following configuration commands

```
Router(config)#router ospf 15
Router(config-router)#rou
Router(config-router)#router-id 3.3.3.3
Router(config-router)#
Router(config-router)#net
Router(config-router)#network 192.168.2.0 0.0.0.255 ar
Router(config-router)#network 192.168.2.0 0.0.0.255 area 2
Router(config-router)#network 10.10.10.0 0.0.0.3 area 2
Router(config-router)#ex
Router(config)#
Router(config)#
Router(config)#
Router(config)#do wr
```

Finally in Router-0, configure all the areas that are adjacent to it.

```
Router(config)#router ospf 15
Router(config-router)#
Router(config-router)#
Router(config-router)#r
Router(config-router)#rou
Router(config-router)#router-id 4.4.4.4
Router(config-router)#
Router(config-router)#
Router(config-router)#ne
Router(config-router)#net
Router(config-router)#network 10.10.10.8 0.0.0.3 are
Router(config-router)#network 10.10.10.8 0.0.0.3 area 0
Router(config-router)#
05:52:39: %OSPF-5-ADJCHG: Process 15, Nbr 1.1.1.1 on GigabitEthernet0/0 from LOADING to FULL,
Loading Done

Router(config-router)#
Router(config-router)#network 10.10.10.4 0.0.0.3 area 1
Router(config-router)#
Router(config-router)#
Router(config-router)#
Router(config-router)#
05:53:16: %OSPF-5-ADJCHG: Process 15, Nbr 2.2.2.2 on GigabitEthernet0/1 from LOADING to FULL,
Loading Done

Router(config-router)#
Router(config-router)#
Router(config-router)#network 10.10.10.0 0.0.0.3 area 2
Router(config-router)#
Router(config-router)#
Router(config-router)#ex
Router(config)#
Router(config)#
Router(config)#do wr
Building configuration...
```

Step 4: OSPF shares routing information only with neighbors. Use the **show ip ospf neighbor** command on R0 to verify OSPF neighbors.

```
Router#sh ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.1           1   FULL/DR         00:00:38    10.10.10.10     GigabitEthernet0/0
2.2.2.2           1   FULL/DR         00:00:35    10.10.10.6      GigabitEthernet0/1
3.3.3.3           1   FULL/DR         00:00:30    10.10.10.2      GigabitEthernet0/2
```

```
Router#sh ip ospf database
            OSPF Router with ID (4.4.4.4) (Process ID 15)

                Router Link States (Area 0)

Link ID         ADV Router      Age         Seq#       Checksum Link count
1.1.1.1         1.1.1.1         102         0x80000003 0x00bfbf 2
4.4.4.4         4.4.4.4         78          0x80000003 0x00f6eb 1

                Net Link States (Area 0)
Link ID         ADV Router      Age         Seq#       Checksum
10.10.10.10     1.1.1.1         102         0x80000001 0x00840f

                Summary Net Link States (Area 0)
Link ID         ADV Router      Age         Seq#       Checksum
10.10.10.4      4.4.4.4         63          0x80000001 0x0077b7
192.168.3.0     4.4.4.4         58          0x80000002 0x00508f
10.10.10.0      4.4.4.4         39          0x80000003 0x009b95
192.168.2.0     4.4.4.4         25          0x80000004 0x005787

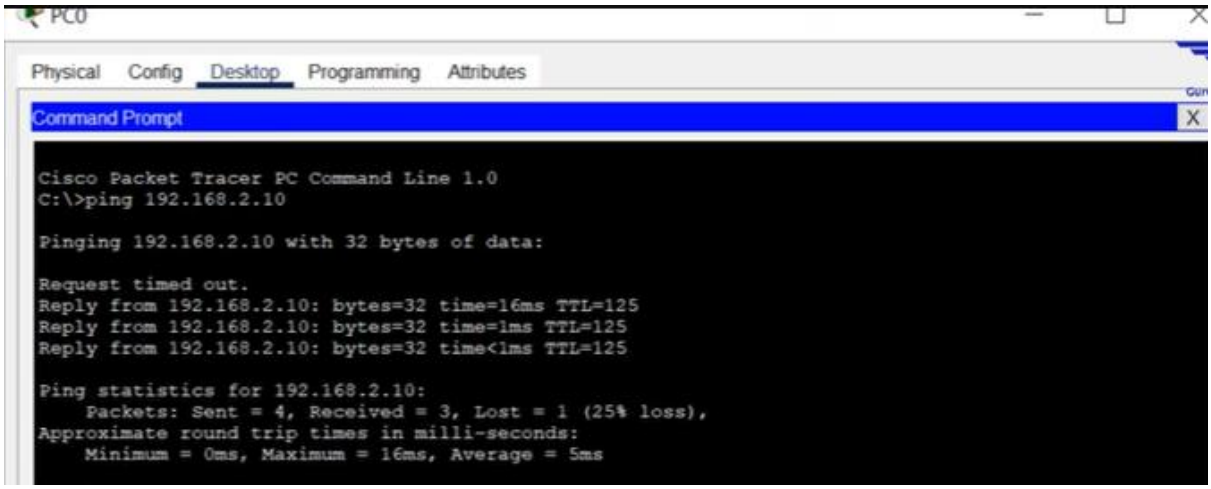
                Router Link States (Area 1)

Link ID         ADV Router      Age         Seq#       Checksum Link count
4.4.4.4         4.4.4.4         65          0x80000002 0x009853 1
2.2.2.2         2.2.2.2         65          0x80000003 0x00c4b8 2

                Net Link States (Area 1)
Link ID         ADV Router      Age         Seq#       Checksum
10.10.10.6      2.2.2.2         65          0x80000001 0x00efc4
```

OUTPUT

Verify the connectivity between End devices by using the ping command and packet simulation.



Result:

The implementation of multi-area OSPF configuration is done successfully using Cisco Packet tracer.

EXERCISE-9-A: CONFIGURATION OF POINT-TO-POINT PROTOCOL

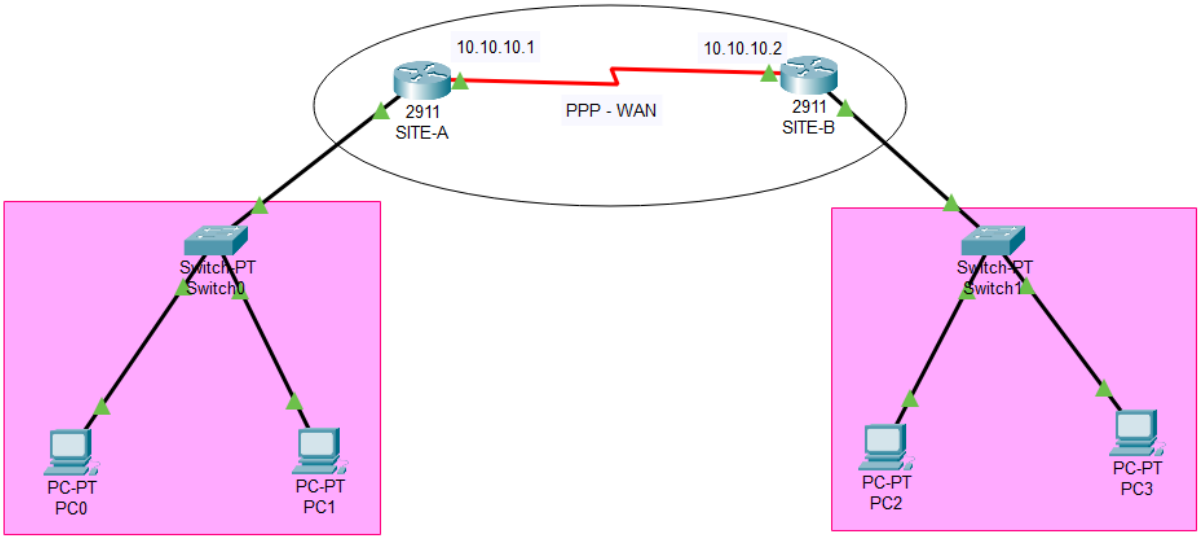
AIM:

To configure the point-to-point protocol (PPP) on Cisco Packet Tracer.

Components:

Components/Devices	Required No's
PC's	4
Copper straight-through cable	4
Serial Cable (DCE)	1
Routers	2
Switches	2

Topology Diagram:



Addressing Table:

Device	Interface	IP address	Subnet Mask	Gateway
PC0	Fa0/0	192.168.10.2	255.255.255.0	192.168.10.1
PC1	Fa0/0	192.168.10.3	255.255.255.0	192.168.10.1
Router 1	Gig0/0	192.168.10.1	255.255.255.0	--
Router 1	Se0/3/0	10.10.10.1	255.255.255.252	
Router 2	Gig0/0	192.168.20.1	255.255.255.0	
Router 2	Se0/3/1	10.10.10.2	255.255.255.252	

PC2	Fa0/0	192.168.20.2	255.255.255.0	192.168.20.1
PC3	Fa0/0	192.168.20.3	255.255.255.0	192.168.20.1

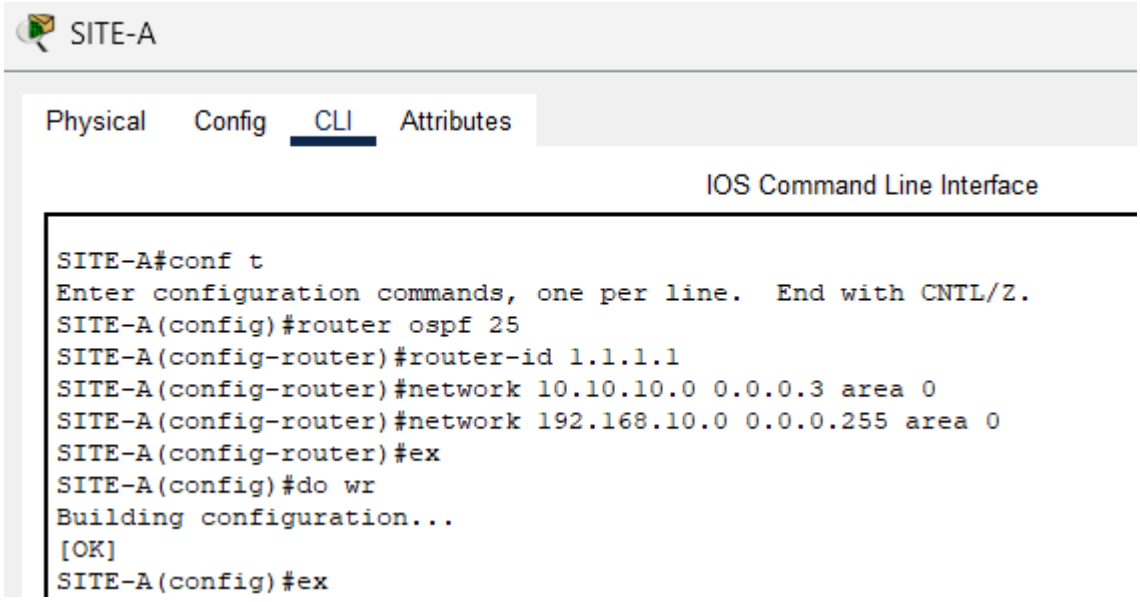
Procedure:

Step 1: Drag and drop the required components and connect them using the appropriate cables as shown in the figure.

Step 2: Assign the IP address, subnet mask and Gateway for all the PC's and IP address for Routers as per the addressing table.

Step 3: Connect the two routers R1 and R3 using the serial DCE cable.

Step 4: Open Router1 and click on "CLI" tab and configure the OSPF protocol as follows.



Step 5: Repeat the same procedure for Router2 as well.

SITE-B

Physical Config CLI Attributes

IOS Command Line Interface

```
SITE-B#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
SITE-B(config)#router ospf 25
SITE-B(config-router)#router-id 1.1.2.2
SITE-B(config-router)#network 10.10.10.0 0.0.0.3 area 0
SITE-B(config-router)#network 192.168.20.0 0.0.0.255 area 0
SITE-B(config-router)#ex
SITE-B(config)#do wr
Building configuration...
[OK]
```

Step 6: Configure the point-to point protocol for both the sites as per the commands.

SITE-A

Physical Config CLI Attributes

IOS Command Line Interface

```
Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname SITE-A
SITE-A(config)#username SITE-B password cisco
SITE-A(config)#int se0/3/0
SITE-A(config-if)#encapsulation ppp
SITE-A(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to down

SITE-A(config-if)#ppp authentication chap
SITE-A(config-if)#ex
SITE-A(config)#do wr
```

SITE-B

Physical Config CLI Attributes

IOS Command Line Interface

```
Router(config)#hostname SITE-B
SITE-B(config)#username SITE-A password cisco
SITE-B(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/1, changed state to down

SITE-B(config)#encapsulation ppp
                ^
% Invalid input detected at '^' marker.

SITE-B(config)#int se0/3/1
SITE-B(config-if)#encapsulation ppp
SITE-B(config-if)#ppp authentication chap
SITE-B(config-if)#ex
SITE-B(config)#ex
```

Step 7: check the configuration of PPP and ospf neighbor.

```
SITE-A#sh ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.2.2           0   FULL/  -        00:00:32    10.10.10.2      Serial0/3/0
```

```
SITE-B#sh ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.1           0   FULL/  -        00:00:36    10.10.10.1      Serial0/3/1
```

Step 8: Check the configuration of the topology on both the routers.

SITE-B

Physical Config CLI Attributes

IOS Command Line Interface

```
SITE-B#sh int se0/3/1
Serial0/3/1 is up, line protocol is up (connected)
  Hardware is HD64570
  Internet address is 10.10.10.2/30
  MTU 1500 bytes, BW 1544 Kbit, DLY 20000 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation PPP, loopback not set, keepalive set (10 sec)
  LCP Open
  Open: IPCP, CDPCP
  Last input never, output never, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0 (size/max/drops); Total output drops: 0
  Queueing strategy: weighted fair
  Output queue: 0/1000/64/0 (size/max total/threshold/drops)
     Conversations  0/0/256 (active/max active/max total)
     Reserved Conversations 0/0 (allocated/max allocated)
     Available Bandwidth 1158 kilobits/sec
  5 minute input rate 74 bits/sec, 0 packets/sec
  5 minute output rate 60 bits/sec, 0 packets/sec
     43 packets input, 3324 bytes, 0 no buffer
     Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
     0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
     35 packets output, 2612 bytes, 0 underruns
 --More--
```

OUTPUT

Verify the connectivity between End devices by using the ping command and packet simulation.

Result:

The implementation of PPP configuration is done successfully using Cisco Packet tracer.

EXERCISE-9-B: CONFIGURATION OF HDLC PROTOCOL

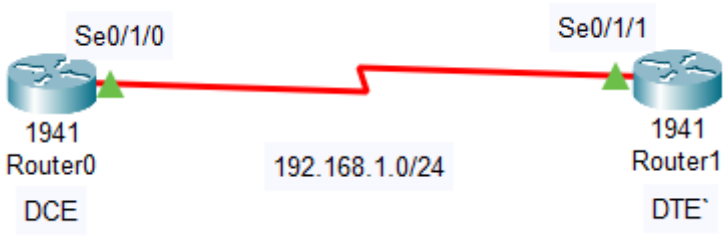
AIM:

To configure the point-to-point protocol (PPP) on Cisco Packet Tracer.

Components:

Components/Devices	Required No's
Serial Cable (DCE)	1
Routers	2

Topology Diagram:



Procedure:

Step 1: Drag and drop the required components and connect them using the appropriate cable as shown in the diagram.

Step 2: Find the DCE and DTE routers by using the command "show controllers serial0/1/0" on both the routers.

```
Router#show controllers serial 0/0/0
Interface Serial0/0/0
Hardware is PowerQUICC MPC860
DTE V.35 TX and RX clocks detected
idb at 0x81081AC4, driver data structure at 0x81084AC0
SCC Registers:
General [GSMR]=0x2:0x00000000, Protocol-specific [PSMR]=0x8
Events [SCCE]=0x0000, Mask [SCCM]=0x0000, Status [SCCS]=0x00
Transmit on Demand [TODR]=0x0, Data Sync [DSR]=0x7E7E
Interrupt Registers:
Config [CICR]=0x00367F80, Pending [CIPR]=0x0000C000
Mask   [CIMR]=0x00200000, In-srv  [CISR]=0x00000000
Command register [CR]=0x580
```

Step 3: Configure the HDLC protocol by using the following command on both the routers

Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int se0/1/0
Router(config-if)#ip ad 192.168.1.1 255.255.255.0
Router(config-if)#encapsulation hdlc
Router(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/1/0, changed state to down
Router(config-if)#ex
Router(config)#ex
```

Router1

Physical Config CLI Attributes

IOS Command Line Interface

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int se0/1/1
Router(config-if)#ip ad 192.168.1.2 255.255.255.0
Router(config-if)#encapsulation hdlc
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/1/1, changed state to up

Router(config-if)#ex
Router(config)#ex
```

Step 4: Check the IP connectivity between the routers using the ping command.

Router0

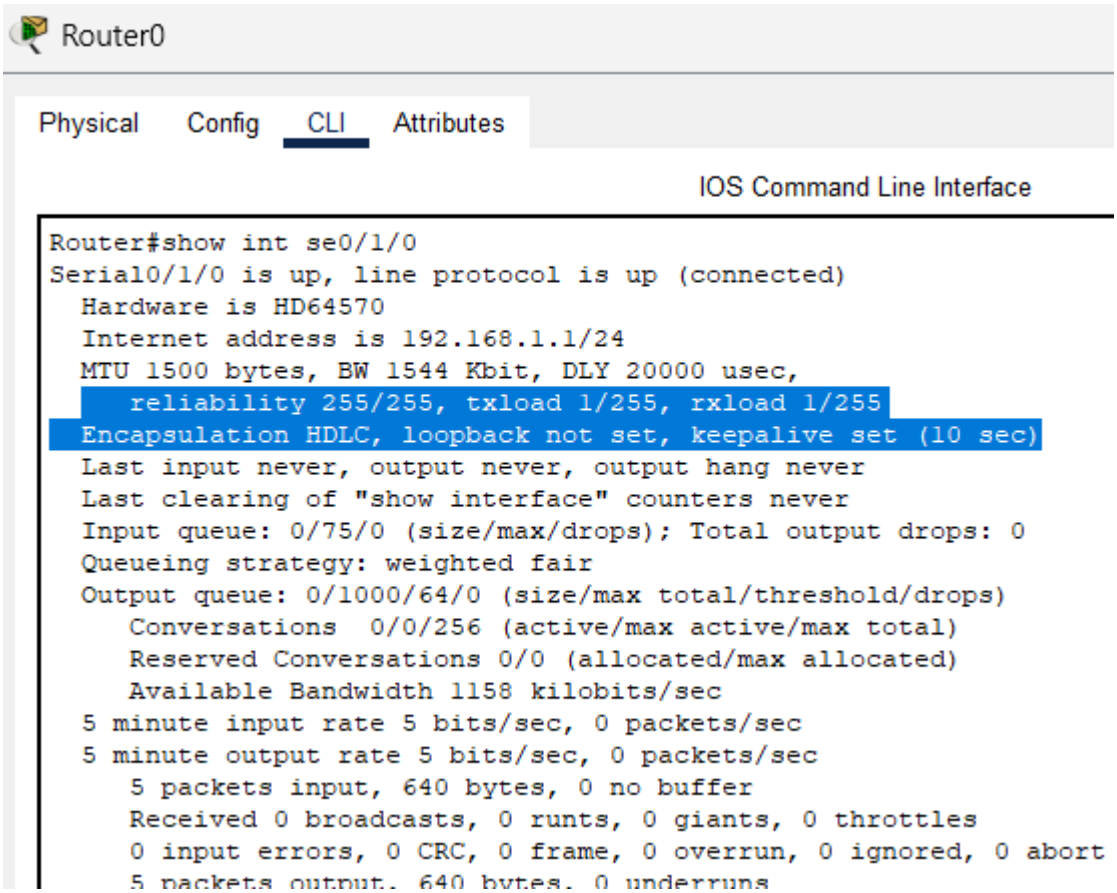
Physical Config CLI Attributes

IOS Command Line Interface

```
Router#ping 192.168.1.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/24/91 ms
```

Step 5: Check the HDLC configuration on both the routers



Result:

The implementation of HDLC configuration is done successfully using Cisco Packet tracer.

EXERCISE-10: CONFIGURATION OF BORDER GATEWAY PROTOCOL

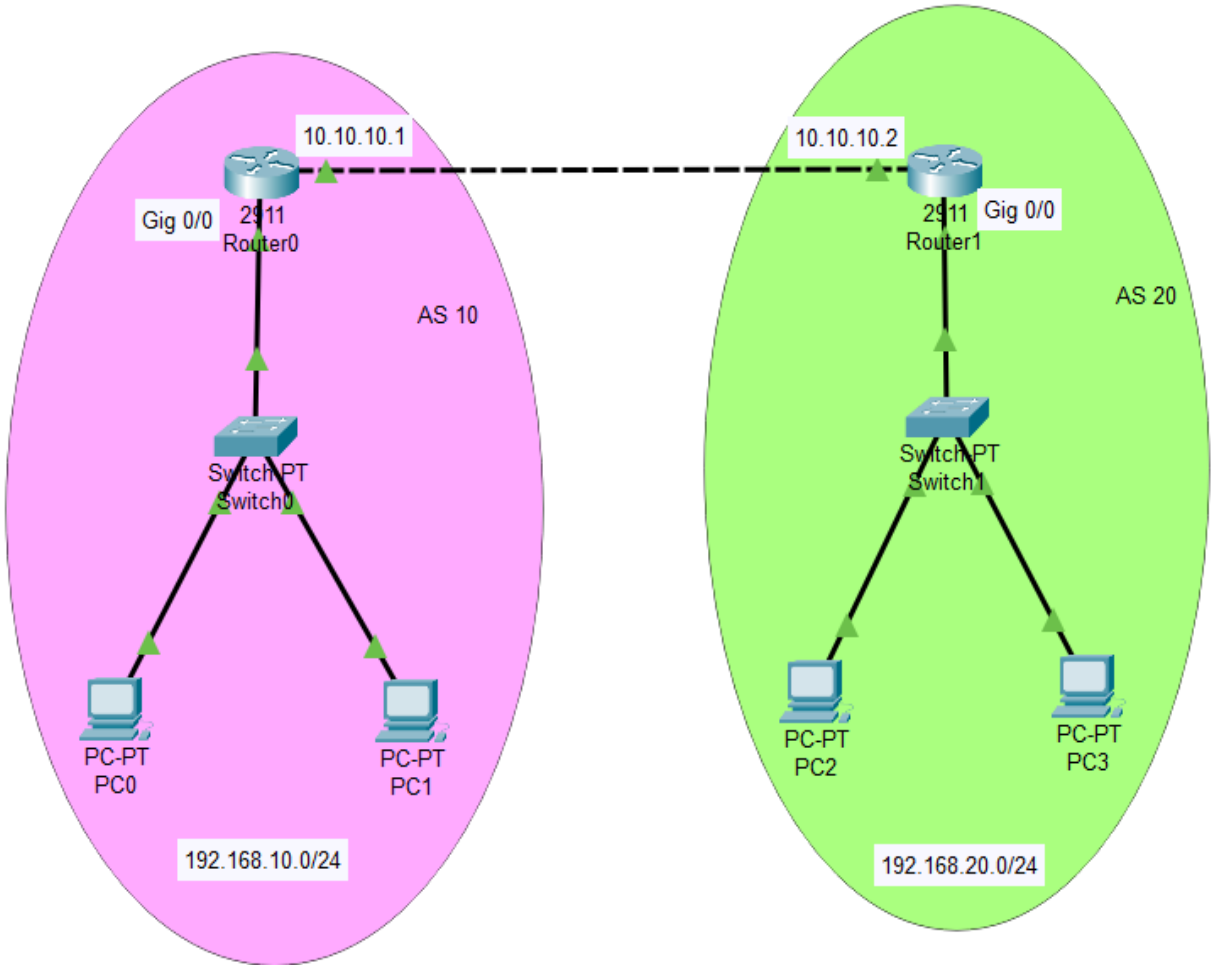
AIM:

To configure the Border Gateway Protocol (BGP) on Cisco Packet Tracer.

Components:

Components/Devices	Required No's
PC's	4
Copper straight-through cable	6
Cross-over cable	1
Routers	2
Switches	2

Topology Diagram:



Addressing Table:

Device	Interface	IP address	Subnet Mask	Gateway
PC0	Fa0	192.168.10.2	255.255.255.0	192.168.10.1
PC1	Fa0	192.168.10.3	255.255.255.0	192.168.10.1
Router 0	Gig0/0	192.168.10.1	255.255.255.0	--
Router 0	Gig0/1	10.10.10.1	255.0.0.0	
Router 1	Gig0/0	192.168.20.1	255.255.255.0	
Router 1	Gig0/1	10.10.10.2	255.0.0.0	
PC2	Fa0	192.168.20.2	255.255.255.0	192.168.20.1
PC3	Fa0	192.168.20.3	255.255.255.0	192.168.20.1

Procedure:

Step 1: Drag and drop the required components and connect them using the appropriate cables as shown in the figure.

Step 2: Assign the IP address, subnet mask and Gateway for all the PC's and IP address for Routers as per the addressing table.

Step 3: Connect the two routers R0 and R1 using the copper cross-over cable.

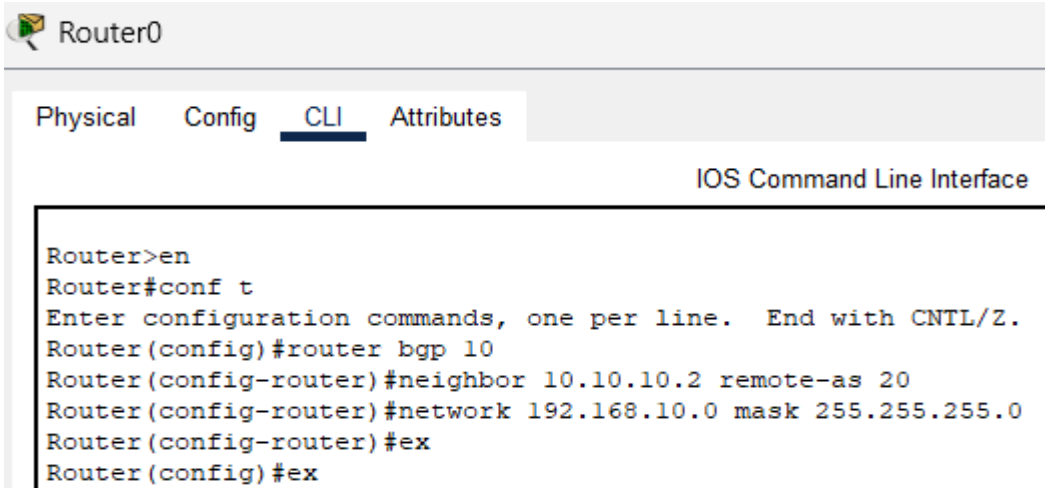
Basic configuration Syntax of BGP

```
Router(config)# router bgp <Autonomous number>

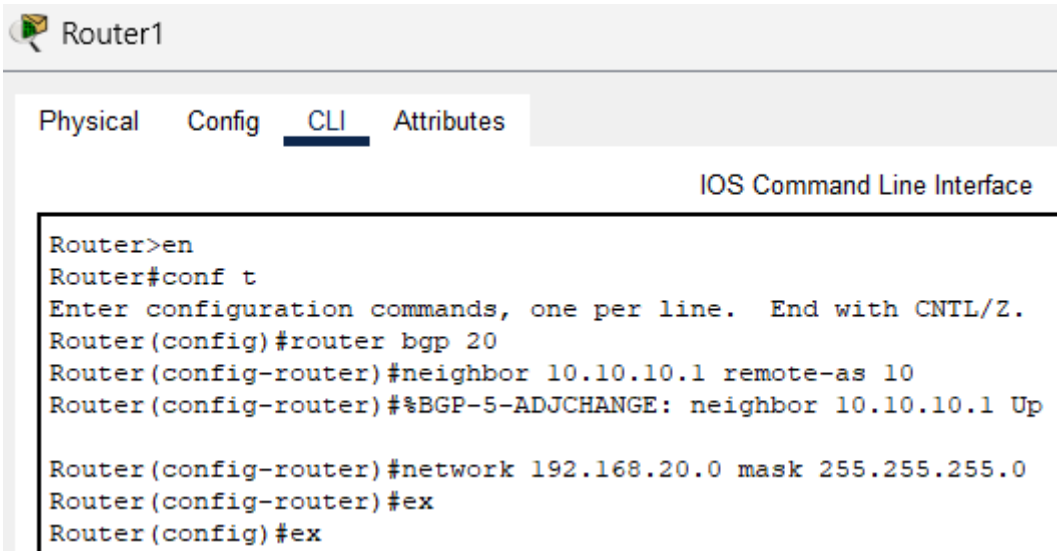
Router(config-router)# neighbour <IP-Address> remote-as <AS number>

Router(config-router)# network <network ID> [mask <subnet mask>]
```

Step 4: Open Router0 and click on "CLI" tab and configure the border gateway protocol as follows.



Step 5: Repeat the same procedure for Router1 as well.



Step 6: check the configuration of BGP address summary & neighbour using the following commands.

Router#show ip bgp summary

Router#show ip bgp neighbor

OUTPUT

Verify the connectivity between End devices by using the ping command and packet simulation.

Result:

The implementation of BGP configuration is done successfully using Cisco Packet tracer.

EXERCISE-11: CONFIGURATION OF ENHANCED INTERIOR GATEWAY ROUTING PROTOCOL

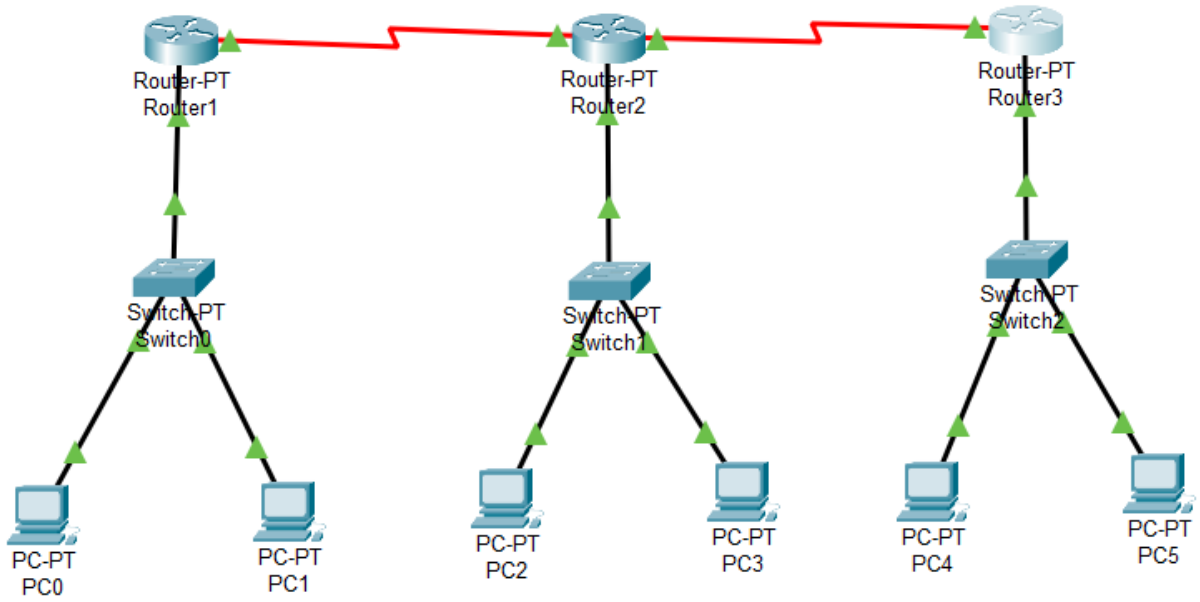
AIM:

To configure the Enhanced Interior Gateway Routing Protocol (EIGRP) on Cisco Packet Tracer.

Components:

Components/Devices	Required No's
PC's	6
Copper straight-through cable	9
Serial Cable (DCE)	2
Routers	3
Switches	3

Topology Diagram:



Addressing Table:

Device	Interface	IP address	Subnet Mask	Gateway
PC0	Fa0/0	192.168.2.2	255.255.255.0	192.168.2.1
PC1	Fa0/0	192.168.2.3	255.255.255.0	192.168.2.1

PC2	Fa0/0	192.168.3.2	255.255.255.0	192.168.3.1
PC3	Fa0/0	192.168.3.3	255.255.255.0	192.168.3.1
PC4	Fa0/0	192.168.4.2	255.255.255.0	192.168.4.1
PC5	Fa0/0	192.168.4.3	255.255.255.0	192.168.4.1
Router 1	Gig0/0	192.168.2.1	255.255.255.0	--
Router 1	Se2/0	10.10.10.1	255.0.0.0	
Router 2	Gig0/0	192.168.3.1	255.255.255.0	
Router 2	Se2/0	10.10.10.2	255.0.0.0	
Router 2	Se3/0	11.11.11.1	255.0.0.0	
Router 3	Gig0/0	192.168.4.1	255.255.255.0	
Router 3	Se2/0	11.11.11.2	255.0.0.0	

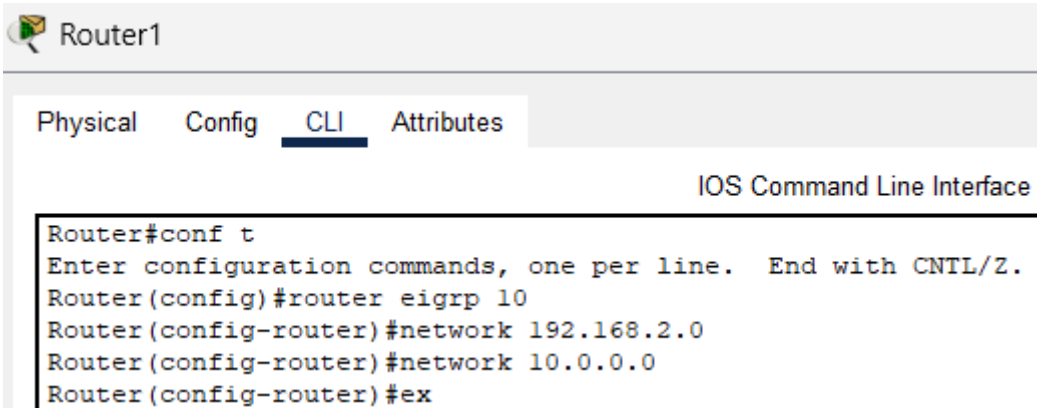
Procedure:

Step 1: Drag and drop the required components and connect them using the appropriate cables as shown in the figure.

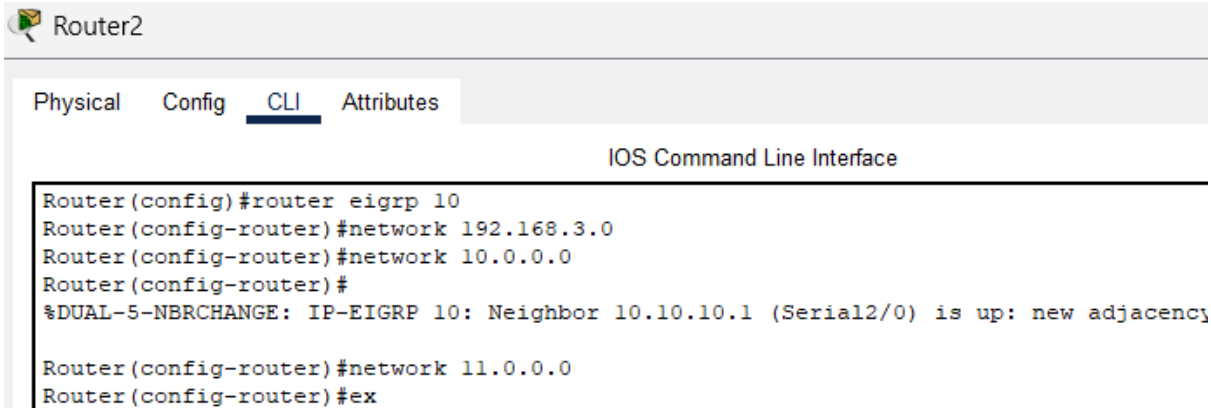
Step 2: Assign the IP address, subnet mask and Gateway for all the PC's and IP address for Routers as per the addressing table.

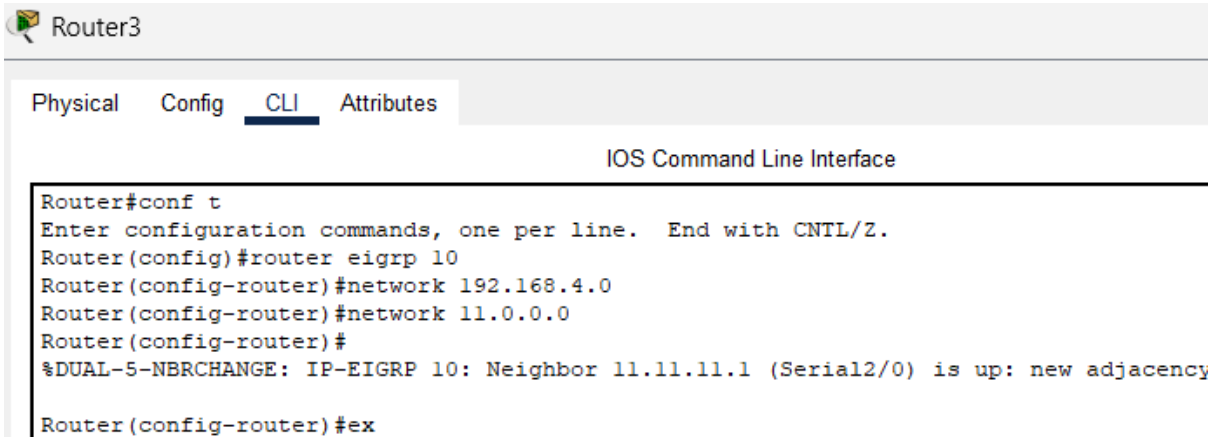
Step 3: Connect all the three routers R1, R2 and R3 using the serial DCE cable.

Step 4: Open Router1 and click on "CLI" tab and configure the EIGRP protocol as follows.



Step 5: Repeat the same procedure for Router2 & Router3 as well.





Step 6: check the configuration of EIGRP neighbors using the following command.

Router#show ip eigrp neighbors

Router#show ip eigrp neighbors

Step 7: Check the configuration of the topology on the remaining routers.

OUTPUT

Verify the connectivity between End devices by using the ping command and packet simulation.

Result:

The implementation of EIGRP configuration is done successfully using Cisco Packet tracer.

EXERCISE-12: CONFIGURATION OF TELNET & SSH

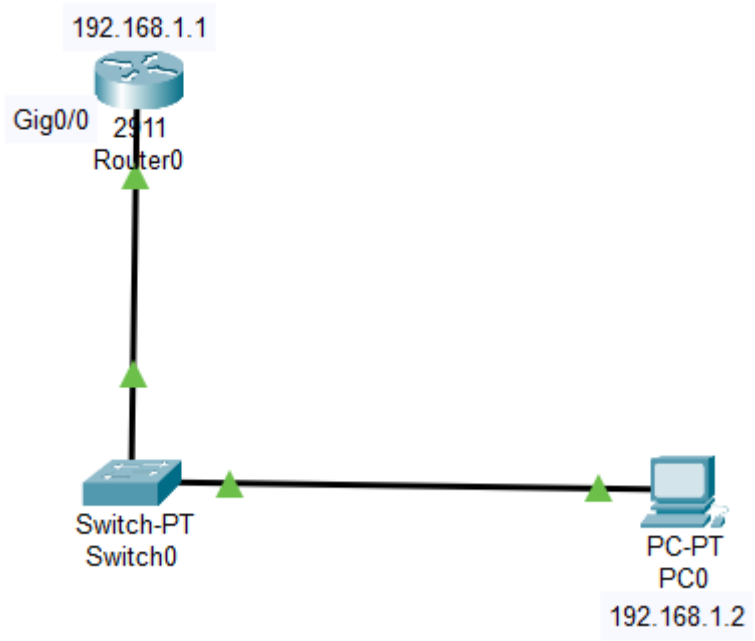
AIM:

To configure the Telnet and SSH protocols on Cisco Packet Tracer.

Components:

Components/Devices	Required No's
PC's	1
Copper straight-through cable	2
Routers	1
Switches	1

Topology Diagram:



Addressing Table:

Device	Interface	IP address	Subnet Mask	Gateway
PC0	Fa0/0	192.168.1.2	255.255.255.0	192.168.1.1
Router0	Gig0/0	192.168.1.1	255.255.255.0	--

Protocols (abbreviation)	Ports Numbers	Transport protocols	Meanings
Secure Shell (SSH)	22	TCP and UDP	It is a cryptographic network protocol used to secure data communication.
Telnet	23	TCP	It is the used for remote management protocol for managing network devices.

Procedure:

Step 1: Drag and drop the required components and connect them using the appropriate cables as shown in the figure.

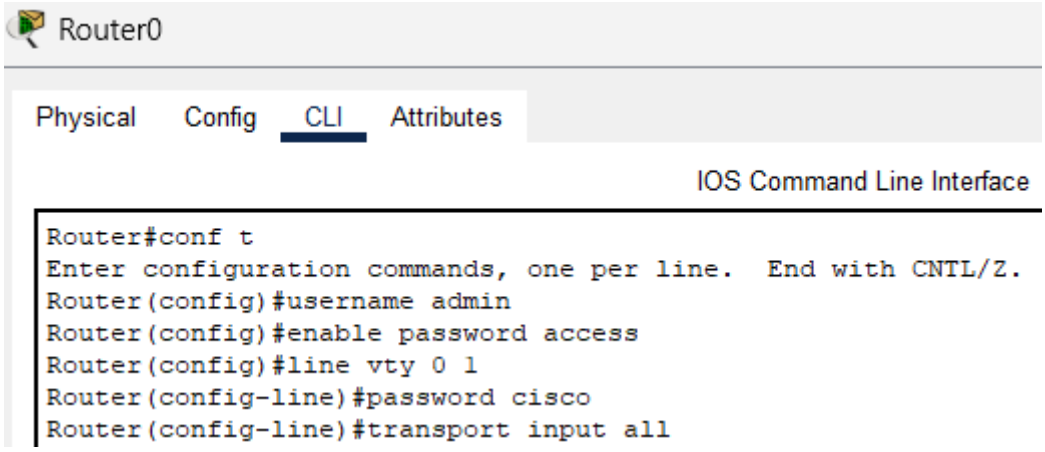
Step 2: Assign the IP address, subnet mask and Gateway for all the PC's and IP address for Routers as per the addressing table.

Step 3: Click on PC0 and go to command prompt & type the following command.

```
C:\>telnet 192.168.1.1
Trying 192.168.1.1 ...Open

[Connection to 192.168.1.1 closed by foreign host]
```

Step 4: Configure the telnet protocol for the Router0 as per the following commands.



Step 5: In PC0, Verify the user access by typing the following command.

```
Router#show user
    Line       User       Host(s)              Idle       Location
   0 con 0                 idle                 00:03:52
*390 vty 0                 idle                 00:00:00 192.168.1.2

  Interface    User               Mode         Idle     Peer Address
```

Step 6: We can encrypt the password by typing the following command

Router(config)#service password-encryption

Step 7: Reset all the configurations and then configure the SSH protocol for the Router0 as per the following commands.

Router(config)#hostname R1

R1(config)#ip domain-name simplenetworking

R1(config)#crypto key generate rsa

How many bits in the modulus [512]: 758

% Generating 758 bit RSA keys, keys will be non-exportable...[OK]

R1(config)#ip ssh version 2

*Mar 1 0:9:47.375: %SSH-5-ENABLED: SSH 1.99 has been enabled

R1(config)#ip ssh time-out 10

R1(config)#ip ssh authentication-retries 2

R1(config)#enable secret access

R1(config)#line vty 0 1

R1(config-line)#transport input ssh

R1(config-line)#password cisco

R1(config-line)#exit

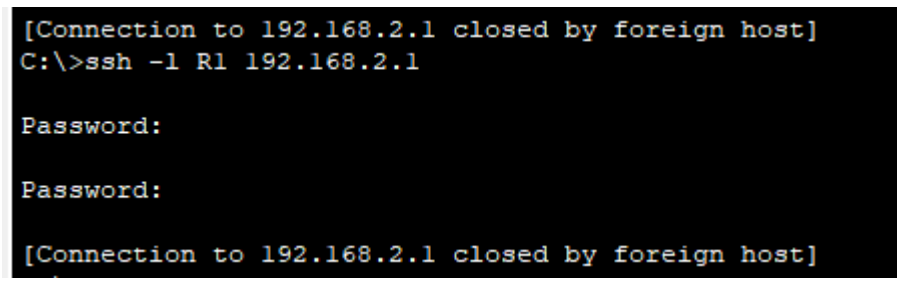
OUTPUT

Verify the remote connectivity between the PC0 and Router0 by using the following command.

For Telnet:



For SSH:



```
R1#show user
    Line       User       Host(s)              Idle       Location
   0 con 0                 idle                 00:07:05
*134 vty 0     R1         idle                 00:00:00

  Interface    User               Mode         Idle     Peer Address
```

```
R1#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 10 secs; Authentication retries: 2
```

```
R1#show ssh
Connection      Version Mode Encryption          Hmac      State             Username
0               1.99    IN   aes128-cbc      hmac-sha1     Session Started         R1
0               1.99    OUT  aes128-cbc      hmac-sha1     Session Started         R1
%No SSHv1 server connections running.
```

Result:

The implementation of Telnet & SSH configuration is done successfully using Cisco Packet tracer.